

A TREATISE ON THE PRINCIPLES OF
FINANCE

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*Being a Plain Account of Money, Capital, Risk and
the Management of Funds*

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*Mistake: retained earnings are
a pile of cash.*

*Mistake: a profitable business
always has money to pay its
bills.*

*Mistake: the financial
manager and the accountant
do the same work.*

*A London bank collapsed in
1995 because one trader's
work was never checked.*

*Leeson held both the trading
and the records of it, so no one
checked him.*

*The inquiry found weak
control and weak
management, as well as one
man's wrongdoing.*

the second asks how to pay. A cheap loan does not make a poor purchase sensible.

Some students picture retained earnings as cash lying in the bank. Retained earnings are profit the business has kept, and it may already have spent that profit on stock or fittings. At Meghna and Sons, Tk 210,000 of retained earnings buys textbook stock.

A business can earn a profit and still lack cash on wage day, because customers may pay late or the profit may be tied up in stock. Profit and liquidity are different things.

The accountant records what has happened. The financial manager plans, raises and controls money to decide what should happen next. In a small shop one person may do both, but the two kinds of work stay different.

IN HISTORY

In 1762 a family founded Barings in London. It was a merchant bank, which means a bank that serves trade and large clients. By 1995 it was 233 years old. On 26 February 1995 it collapsed. The cause lay in Singapore, where a Barings subsidiary employed a trader named Nick Leeson. A trader buys and sells financial contracts for a bank. Leeson dealt in contracts called futures, which are agreements to buy or sell at a fixed price on a later date.

Leeson was allowed to control both the trading and the recording of those trades. Normally two people hold these tasks, so that each checks the other. With one man in both roles, his losses stayed hidden while they grew. Reports put the losses at about £827 million, more than the bank could absorb.

The Bank of England, which is Britain's central bank and supervises banks, had a board inquire into the collapse. Its report in July 1995 found that several people at Barings had failed to do their jobs properly. The Dutch group ING bought the bank for £1, and in December 1995 a Singapore court sentenced Leeson to six and a half years in prison. In this chapter's words, weak financial control, not bad luck, brought down the bank: the money moved without checks that would have shown the problem in time.

REVIEW QUESTIONS

2.1 Describe the roles of a financial manager in a small firm, such as a bookshop run by two brothers. Name the three duties and give one example of each.

2.2 A clinic decides to buy a scanner for Tk 2,000,000 and to pay for it with a bank loan. Name the investment decision and the financing decision in this choice. Which of the three core decisions has not yet been made?

2.3 Explain the difference between financial planning and financial control, using a tailor who expects to pay Tk 30,000 for cloth.

2.4 A shop earns a profit every month but cannot pay its suppliers on time. Explain how weak liquidity disrupts the daily operations of the shop.

*To the one I would avenge from death, if
given the chance; the one with whom I truly
shared love; my first pet, my Cutush and
Lokkhi. Pufi*

*Choices matter because of
what they do to a business
every day.*

Operating efficiency. How little money and effort a business uses to produce each unit of sales.

Operating efficiency measures how little is used for each unit of sales.

Liquidity. How easily a business can pay its bills on time from cash it has or can quickly raise.

Liquidity is the ability to pay bills on time.

The case file shows why stock is not the same as cash.

Profitability. How much profit a business earns compared with the money invested or the sales made.

Profitability compares profit with money invested or with sales.

The case file gives the profit and sales of 2023.

The three qualities can pull in different directions.

The section ends with a short recap.

which of the three questions a given choice answers, and to give one example of each.

* * *

2.3 HOW FINANCE SHAPES BUSINESS SUCCESS

Everything so far has described choices. Choices matter because of their effect on a business each day. A shop that buys what it cannot sell, or sells below what the goods cost, or cannot pay a supplier on time, will struggle however loyal its customers are. Three qualities show whether finance is serving the business well.

Operating efficiency means how little money and effort a business uses to produce each unit of sales. Suppose a tailor spends Tk 70 on cloth, thread and labour for every Tk 100 she sells, while a rival spends Tk 80 for the same Tk 100. The first tailor is more efficient. Size does not decide this: a small shop can be more efficient than a large one.

Liquidity means how easily a business can pay its bills on time from cash it has or can quickly raise. Weak liquidity disrupts daily work: a supplier refuses a delivery, wages are late, a chance to buy cheaply passes.

Case file, illustrative. At 1 January 2024, Meghna and Sons held stock of Tk 1,850,000 and a cash and bank balance of Tk 240,000. Books on the shelves cannot pay a supplier until they are sold, but the balance at the bank can.

Profitability means how much profit a business earns compared with the money invested or the sales made. A profit of Tk 20,000 on Tk 200,000 invested is 10% of the money invested. Compare ratios, not amounts, because a large profit can be a poor result on a very large investment.

Case file, illustrative. Profit after tax for 2023 was Tk 480,000, and sales in 2023 were Tk 7,200,000. The profit is about Tk 6.67 for each Tk 100 of sales.

The three qualities can pull in different directions. Buying more stock may raise sales and profit, yet it spends cash and so lowers liquidity. The financial manager's work is to keep all three in view, and each of the three decisions of section 2.2 affects at least one of them.

Before moving on. Operating efficiency measures how little is used for each unit of sales. Liquidity measures the ability to pay bills on time. Profitability compares profit with the money invested or with sales. *You should now be able to say* which of the three a business lacks if it earns a profit but cannot pay its wages on time.

* * *

WORKED CASE: THREE DECISIONS AT MEGHNA AND SONS

CONTENTS

I. MONEY, FINANCE AND THE FIRM	1
1.1 What Finance Is	2
1.2 Finance Among the Business Functions	4
1.3 Where Funds Come From and Where They Go	5
II. THE FINANCIAL MANAGER'S WORK	11
2.1 The Financial Manager's Role	12
2.2 The Three Core Decisions	13
2.3 How Finance Shapes Business Success	16
III. THE GOAL OF THE FIRM AND AGENCY	21
3.1 Profit Maximisation and Its Limits	22
3.2 Wealth Maximisation	23
3.3 Agency Theory	25
IV. THE TIME VALUE OF MONEY: SINGLE SUMS	31
4.1 Why Time Changes the Value of Money	32
4.2 Future Value of a Single Sum	33
4.3 Present Value of a Single Sum	36
4.4 Compounding Within the Year	38
V. ANNUITIES	43
5.1 What an Annuity Is	44
5.2 Future Value of an Annuity	46
5.3 Present Value of an Annuity	48
5.4 Finding the Payment	50
VI. UNEVEN CASH FLOWS	53
6.1 What a Mixed Stream Is	54
6.2 Present and Future Value of a Mixed Stream	55
6.3 Comparing Streams with a Common Rate	57
VII. CAPITAL BUDGETING: DECISIONS AND CASH FLOWS	63
7.1 What Capital Budgeting Is and Why It Matters	64
7.2 The Steps of the Process	65
7.3 Cash Flow from an Investment	67
VIII. CAPITAL BUDGETING: EVALUATION TECHNIQUES	73
8.1 Payback Period	74
8.2 Net Present Value	76
8.3 Internal Rate of Return and Profitability Index	77
8.4 When the Methods Disagree	79
IX. RETURN AND RISK	83
9.1 What Return Is	84
9.2 What Risk Is	85
9.3 Describing Uncertain Outcomes	86
9.4 Measuring the Spread of Outcomes	87
X. RISK, REQUIRED RETURN AND MARKET EFFICIENCY	93
10.1 Risk Premium	94
10.2 Diversification and Beta	95
10.3 The Capital Asset Pricing Model	98
10.4 Efficient Markets	99
XI. OPERATING LEVERAGE AND BREAK-EVEN ANALYSIS	103
11.1 Fixed and Variable Costs	104
11.2 The Operating Break-even Point	106

Stock. Word in use. In daily life, a supply of something kept for later use. In a shop, the goods held for sale. In finance, stock can also mean ownership parts of a company, taught in Chapter 14, "Valuing Common and Preferred Stock".

The rule is that an investment commits money for a long time, so it is hard to reverse.

The first decision is complete: it asks what to buy.

Financing decision.

Choosing how to raise the money for purchases, from owners or lenders.

The financing decision asks how to pay.

Recall: sources of funds, equity and debt, Chapter 1, section 1.3, "Where Funds Come From and Where They Go".

The case file shows how the corner was paid for.

A preview: the kept profit has a name, which this section gives a little further on.

The two decisions use the same money but ask different questions.

Dividend decision.

Choosing how much profit to pay out to owners and how much to keep.

The dividend decision asks how much profit goes to the owners.

Dividend. A share of profit paid out to the owners of a business.

Retained earnings. Profit kept in the business instead of being paid out to owners.

The part paid out is the dividend; the part kept is

The rule is that an investment decision commits money to assets for a long time, so it is hard to undo. A corner, once built, cannot be returned to the supplier. There are two limits to the meaning. Here, investment means spending the business's own funds on its own assets, not saving in a bank. And a shorter-lived use of funds, such as the textbook stock that the shop buys before each school term, is treated in Chapter 15, "Working Capital and the Cash Cycle". The method for judging whether a project is worth buying is taught in Chapter 7, "Capital Budgeting: Decisions and Cash Flows".

So far, one idea only: the investment decision asks what to buy.

The second is the financing decision. The **financing decision** is choosing how to raise the money for purchases, from owners or lenders. Recall that equity is money from owners and debt is borrowed money. A bakery that buys an oven may pay from its owners' funds, take a bank loan, or use both. The purchase is the same in each case, but the way of paying is different.

Case file, illustrative. The corner of Tk 350,000 is funded by Tk 150,000 of profit the business had kept and a new bank loan of Tk 200,000 at 10% a year. That choice is the financing decision. The yearly charge on the new loan, called interest, is Tk 20,000. After the new borrowing, the bank loan stands at Tk 800,000.

The kept profit has a name that this section gives a little further on. For now, read it as owners' funds, which are a source of funds, while the loan is debt.

PAUSE AND CHECK 2.P1 A bakery buys an oven for Tk 200,000. It pays Tk 80,000 from the owners' funds and borrows Tk 120,000. Which part of this is the investment decision, and which part is the financing decision?

Notice how the two decisions differ. Investment asks what to buy; financing asks how to pay. A loan with a low charge does not make a poor purchase wise, and a good purchase can still fail if it is paid for in a way the business cannot repay. A common slip is to answer "should we buy?" with "the loan is cheap". That answers the second question, not the first.

The third is the dividend decision. The **dividend decision** is choosing how much profit to pay out to owners and how much to keep. A shopkeeper who earns a profit of Tk 90,000 may take Tk 30,000 home and leave Tk 60,000 in the business. Neither choice is wrong in itself. A payout rewards the owners now. Keeping profit leaves more to fund growth later, without asking a lender.

The part of profit paid out is called a **dividend**, and the part kept is called **retained earnings**. In the shopkeeper's example, the dividend is Tk 30,000 and the retained earnings are Tk 60,000. A business owned by one family may call its payout a withdrawal. This book uses the word dividend for any share of profit paid to owners.

² Tax is money that a business pays to the government on its profit. Profit after tax is the profit left once that payment has been made.

TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches A Treatise on the Principles of Finance one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

One sum of money has three fair claims on it, and only one can win.

Every business meets this puzzle on a larger scale, and someone must answer it.

You need the idea of sources and uses of funds from Chapter 1.

The chapter has three sections, a worked case and a short history.

Recall: business function, Chapter 1, section 1.2, "Finance Among the Business Functions".

A small shop shows that a role can exist before it has a title.

Tanvir holds the money function, and a name exists for the person who does so.

Financial manager. The person who plans, raises and controls the money of a business.

The financial manager is whoever plans, raises and controls the money.

Each of the three verbs describes a separate part of the duty.

Suppose you hold Tk 100,000 that you do not need to spend this month. Three people ask for it. A friend wants to start a juice stall and needs equipment. Your parents would like a share, because you could not have saved it without them. A bank is waiting for a loan to be repaid, and every month of waiting adds a charge. Each request is reasonable, and the money cannot go to all three. Who should receive it?

A business meets the same puzzle in larger form. It earns a profit, and its owners, its lenders and its own plans for growth all ask for the money. Somebody must hear the claims, weigh them and choose. That person does not guess. The person follows a method, and the method has three parts. This chapter names the person, names the three parts, and shows how the choices appear in the daily life of a shop.

You will need: Chapter 1, section 1.3, "Where Funds Come From and Where They Go": sources and uses of funds, equity and debt. If those words feel uncertain, read that section again before starting.

Section 2.1 describes the person who does finance in a business. Section 2.2 explains the three decisions this person makes, and it goes slowly, because the three decisions sound alike. Section 2.3 shows how finance affects the daily success of a business. A worked case then follows one day of decisions at Meghna and Sons, and a history shows what can happen when control fails.

* * *

2.1 THE FINANCIAL MANAGER'S ROLE

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhaka. Since 2019 it has been run by Imran, who is responsible for the shop and buying, and by Tanvir, who is responsible for accounts and money.

In Chapter 1 we called finance one business function. Someone must perform it, even in a small shop. In the case file above, Tanvir does so, beside his work on accounts. The person who holds this duty has a name.

A **financial manager** is the person who plans, raises and controls the money of a business. In a small shop this may be one brother. In a large business it may be a team led by a senior person with a grander title, such as chief financial officer.¹ The title varies from business to business. The duties are what count.

Look closely at the three verbs. To plan is to decide in advance what money will be needed and when. To raise is to find the money, from owners or from lenders. To control is to check that money is used as intended, and to correct matters when it is not. A tailor who foresees a large order, borrows to buy cloth and later

¹ Chief financial officer is the title of the most senior person in charge of finance in a large business. This book uses "financial manager" for the role in every size of business.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.



CHAPTER I

Money, Finance and the Firm

Where does a business find its money, and what does it do with it?



This chapter introduces finance: what it is, where it sits among the main kinds of work in a business, and how money moves between owners, lenders and the business. A family bookshop in Dhaka supplies the figures, and a banking house in Florence supplies the history.

Finance has limits worth knowing. It is not only for banks or very large businesses; the stall owner uses it every day. It is also not the same as having money: a business with no money may need finance more than a rich one. And finance is a method for choosing, not a promise that every choice will be right, because a loan used badly must still be repaid.

The viewpoint changes with the person. The owner wants the business to grow. A lender wants to be repaid on the agreed day. An employee wants wages paid on time. A customer wants the shop to be open next month. All four depend on how well the money has been found and used.

Funds are money that a business has available to spend, save or invest. To invest means to use money now in the hope of earning more later. Funds may be notes in a purse or a balance held at a bank. On a morning when the stall owner's purse holds Tk 1,200, her funds are Tk 1,200. The tomatoes on her table are not funds, because she cannot spend a tomato at the wholesale market until it has been sold.

Cash flow is money moving into or out of a business during a period. A period is any stretch of time, such as a day, a month or a year. On Monday customers pay the stall owner Tk 3,000, which is a cash flow into the business, and she pays Tk 2,000 to the wholesaler, which is a cash flow out. Funds are measured at one moment; cash flow is measured over a stretch of time.

The table below compares the two ideas. Read each row from left to right: the first column names the idea, the second says how it is measured, and the third gives an example from the stall.

FUNDS AND CASH FLOW COMPARED

IDEA	HOW IT IS MEASURED	EXAMPLE FROM THE STALL
Funds	At one moment	Tk 1,200 in her purse at five in the morning
Cash flow	Over a period	Tk 3,000 received and Tk 2,000 paid during Monday

A sale and a cash flow are different events. Suppose a customer takes goods worth Tk 5,000 on 28 June and promises to pay on 28 July. The sale happens in June, but the cash flow into the business happens in July. In the business sense of the word, the profit from that sale is counted in June, while the money reaches the business in July. A business can therefore show a good profit and still lack money to pay wages on Friday. This is why finance watches cash flow closely.

PAUSE AND CHECK 1.P1 A tailor accepts an order worth Tk 3,000 today. The customer promises to pay when the suit is ready next week. Is there a cash flow into the business today? Give a reason.

Wealth is the total value of what a person or business owns, after taking away what is owed. Suppose the stall owner owns a table, scales and Tk 1,200 in cash, worth Tk 4,000 together, and owes her cousin Tk 800. Her wealth is Tk 4,000 less Tk 800, which is Tk 3,200. Wealth can be large while funds are small: her table is part of

Finance is not only for banks, not the same as having money, and not a promise of good choices.

Each person connected to a business sees finance from a different side.

GO SLOWLY HERE.

Funds. Money that a business has available to spend, save or invest.

Funds are money that is available to use.

Cash flow. Money moving into or out of a business during a period.

Cash flow is money in motion over a period of time.

The table sets the two ideas side by side.

Profit. Word in use. In daily life, the money you gain from a sale. In business, what a sale earns after the costs of making it, counted when the sale is made, even if the customer pays later.

A sale and its cash flow can fall in different months.

Wealth. The total value of what a person or business owns, after taking away what is owed.

Wealth is what is owned after taking away what is owed.

The conclusion states what the sorting shows.

The check confirms that nothing is missing or counted twice.

The chapter in brief.

Mistake: finance belongs only to banks and large businesses.

Mistake: goods on the shelves count as funds.

Mistake: profit and cash flow are the same thing.

Mistake: equity is free money, or debt is always bad.

Conclusion. Two sources supplied the shop: debt of Tk 600,000, which is one quarter of the total, and equity of Tk 1,800,000, which is three quarters. Three assets hold the funds: stock, the cash and bank balance, and fittings. The family's wealth in the shop is Tk 1,800,000.

Check. The totals agree at Tk 2,400,000, and the wealth equals the equity. If the totals had differed, a figure would be missing or counted twice. Note also that the cash and bank balance sits among the assets and not among the sources: it is money held, not money obtained.

* * *

SUMMARY

Finance is the work of finding money and deciding how to use it. Funds are money available at one moment, cash flow is money moving over a period, and wealth is what is owned after taking away what is owed. Finance is one business function, beside selling, buying and managing people, and it differs from accounting and economics. A business obtains funds from sources, equity from owners and debt from lenders, and uses them on assets. On any one date, the sources equal the assets.

THINK IT THROUGH

1.T1 A cinema owner takes Tk 40,000 in ticket sales, owes Tk 25,000 for film hire next month and pays a Tk 10,000 loan instalment. Classify each of the three movements, and say which are cash flows today.

1.T2 What is the net cash flow for the cinema owner today? Which of the three movements is a liability, and what will it become next month?

WHERE PEOPLE GO WRONG

Some students believe that finance belongs only to banks and large businesses. In fact, every stall owner who finds money and decides how to use it practises finance. Size changes the amounts, not the questions.

Some students count the books on the shelves as funds. Books are assets; they become funds only when they are sold and paid for. Meghna and Sons holds stock of Tk 1,850,000 but a cash and bank balance of only Tk 240,000.

A sale made in June and paid for in July creates profit in June, but its cash flow arrives in July. The two measure different things, and a business needs to watch both.

Owners supply equity, but they expect a share of any profit, so equity is not free. Debt carries an agreed charge and must be repaid, but it lets a business obtain funds that its owners cannot supply. Neither is good or bad in itself; each comes with terms.

The three overlap. The tailor uses accounting records to see how much money was received, uses economics to understand why cloth costs more, and uses finance to decide about the machine. A common confusion is to think that finance is only keeping records. Records belong to accounting; finance uses them to make decisions about the future.

Before moving on. A business function is one main kind of work in a business. Finance is one function: it supplies funds to the others and judges the money side of their decisions. It differs from accounting, which records what has happened, and from economics, which studies choice in society.

* * *

1.3 WHERE FUNDS COME FROM AND WHERE THEY GO

Walk into a bookshop and look along the shelves. Every book there was bought with money before it was sold. Somebody supplied that money; the shop did not create it. In any business, then, two questions come before all others: who supplied the money, and what was it used for?

Case file, illustrative. At 1 January 2024, Meghna and Sons held stock of Tk 1,850,000, a cash and bank balance of Tk 240,000 and fittings of Tk 310,000, which together come to Tk 2,400,000. The funds came from a bank loan of Tk 600,000 at 10% a year and from the family's own funds of Tk 1,800,000. Sales in 2023 were Tk 7,200,000.

Two words in the case file need care. Stock here means the books held for sale. Fittings means the shelves and counters fixed in the shop. Both belong to the shop, and both were bought with money.

A **source of funds** is a place a business gets money from, such as its owners or a lender. Meghna and Sons has two: the family and a bank. Each supplied part of the Tk 2,400,000.

A **use of funds** is a purpose a business spends money on, such as stock or equipment. The shop used funds for books to sell and for shelves and counters. Keeping money at the bank is also a use, because the business chose to hold it for later.

An **asset** is something of value that a business owns and uses to earn money. The shop's books, its shelves and counters, and its cash and bank balance are assets. Each asset is the result of a use of funds.

Now the rule that joins the two halves. On any one date, the total from all sources of funds equals the total of the assets that those funds bought or kept. For Meghna and Sons, add the sources: Tk 600,000 plus Tk 1,800,000 is Tk 2,400,000. Add the

The three subjects overlap, so a real decision often uses all three.

The section ends with a short recap.

Every book on a shelf was bought with someone's money, so whose money was it?

The case file gives the figures that the rest of the section sorts.

Two words in the case have special meanings in the shop.

GO SLOWLY HERE.

Source of funds. A place a business gets money from, such as its owners or a lender.

A source of funds is where the money came from.

Use of funds. A purpose a business spends money on, such as stock or equipment.

A use of funds is what the money was spent on.

Asset. Something of value that a business owns and uses to earn money.

An asset is something the business owns that helps it earn money.

The rule: on any one date, the money obtained equals the assets held.

Equity. Money put in by the owners, who share the business profit or loss.

Capital. Word in use. In daily life, the main city of a country, or a large letter. In finance, money put into a business to start or run it.

Equity is the owners' money, and the owners share profit or loss.

Debt. Money borrowed that must be repaid, with an extra charge for using it.

Debt is borrowed money that must be repaid with an extra charge.

The table compares equity and debt on four points.

Equity is money put in by the owners, who share the business profit or loss. The family's Tk 1,800,000 is equity. Money from owners is often called capital. Owners are not promised a fixed payment: if the shop earns a profit they share it, and if it makes a loss they bear it.

Debt is money borrowed that must be repaid, with an extra charge for using it. The bank loan of Tk 600,000 is debt, and its charge is 10% a year.² The lender is promised repayment on agreed dates, whether the shop earns a profit or not.

The table below compares equity and debt. Read down a column to see one kind of source; read across a row to compare the two on one point.

EQUITY AND DEBT COMPARED

POINT	EQUITY	DEBT
Who supplies it	The owners	A lender
Repayment	No fixed repayment; owners share profit or loss	Must be repaid on agreed dates
Extra charge	None promised; a share of any profit	Agreed in advance: 10% a year
At Meghna and Sons	Tk 1,800,000 from the family	Tk 600,000 from the bank

PAUSE AND CHECK 1.P2 A new bakery begins with Tk 300,000 of the owner's savings and a Tk 100,000 loan from a bank. Which amount is equity and which is debt? Who must be repaid on agreed dates?

A **liability** is an amount a business owes to someone else. The bank loan of Tk 600,000 is a liability. Debt is one kind of liability; an unpaid bill to a supplier is another. Now the wealth of the family in the shop can be found: assets of Tk 2,400,000 less the liability of Tk 600,000 leave Tk 1,800,000, which equals the family's equity.

A **financial market** is a place or network where buyers and sellers trade money, loans and ownership parts of businesses. A bank branch where one borrower meets one lender is a very small example. Large markets let a business find many sources at once. Chapter 13, Valuing Bonds, and Chapter 14, Valuing Common and Preferred Stock, show how such trades set prices.

The figure below shows the same flows as a picture. Read it from left to right. A solid arrow shows money going into the shop from its owners or its lender. A dashed arrow shows what goes back to them. The arrow on the right shows the shop spending funds on assets. Each arrow carries a label in plain words, and the

² A charge of 10% a year means Tk 10 for each Tk 100 borrowed, for each year the money is used. Later chapters show how the charge is calculated.

THE PROBLEM OF TIMING

Profit is earned in a particular period, and a plan that raises profit in one period may lower it in another. The following case file shows the difficulty with figures.

Case file, illustrative. Date: 15 May 2024. Business: Meghna and Sons, a bookshop in Dhaka. Choice: cut restocking, meaning the buying of new books to replace those sold. Effect on this year's profit: an increase of Tk 80,000. Effect on next year's profit: a decrease of Tk 150,000.

A rule that says "earn the largest profit this year" favours the cut, because Tk 80,000 is a gain. Look at both years together and the picture changes: Tk 80,000 gained, then Tk 150,000 lost, which is a net fall of Tk 70,000. There is a second point. Money received this year can be used at once, for example to repay a loan or to buy stock, and money received next year cannot. So the two amounts are not equal in usefulness. Chapter 4 teaches how to compare money at different times with exact arithmetic. For now, remember only that *when* a profit arrives matters.

THE PROBLEM OF RISK

The second problem is risk, meaning the chance that the result will differ from what was expected. Imagine two plans for a small tailoring shop. Plan one promises Tk 50,000 from a school uniform order that has been signed and will be paid. Plan two promises Tk 50,000, but only if a large crowd comes to a festival. The promised profit is the same. Nobody who understood the difference would treat the plans as equal. The rule "largest profit" cannot see this difference, because it looks only at the number promised.

There is a third and smaller limit. Profit for a year is a figure produced by accounting for that period. Choices about when to spend, such as delaying the restocking of the shelves, change the figure for the year without making the business better. A person who is judged only by the figure may be tempted to change it.

The rule to remember is short. Profit maximisation is a useful first aim, but it ignores when profits arrive and how certain they are. A better aim must take both into account. Section 3.2 supplies it.

Before moving on. Profit maximisation means earning the largest profit in a period, usually one year. Its limits are timing, risk, and the fact that a yearly profit can be shaped by choices about spending. You should now be able to say why a larger profit this year is not always a better result for the owner.

3.2 WEALTH MAXIMISATION

Suppose you own a used bicycle and a friend asks what it is worth. You do not answer with the price you paid two years ago, and you do not answer with how much you enjoy riding it. You think about what a buyer would hand over today. A shop nearby might say Tk 6,000. That answer is the idea we need first.

The name for this idea is **market value**. Market value is the

The same profit has different usefulness at different times.

A choice that raises this year's profit and lowers the next year's.

Adding the two years shores a net loss, though the rule favours the cut.

Two plans may promise the same profit and still differ in how likely the profit is.

Profit is an accounting figure, and some choices change it without improving the business.

Profit maximisation is useful but incomplete.

A short summary of section 3.1 with a test sentence.

GO SLOWLY HERE.

Everyday items teach us what the words "worth" and "price" mean.

Market value. The price at which something could be

5.T2 Explain in plain words why an annuity due is worth more than an ordinary annuity with the same payment, interest rate and number of payments.

WHERE PEOPLE GO WRONG

Mistake: "Four payments of Tk 25,000 are worth Tk 100,000." That is their total, not their value today. Payments in the future are worth less than money now, so their present value is lower than the total.

Mistake: using the future value factor to value an offer today. The future value factor answers "what will the payments grow to?" The present value factor answers "what are the payments worth now?" Choose by the question asked.

Mistake: using a yearly interest rate with monthly payments. Either convert the rate to a monthly rate and count months, or use yearly payments. Mixed periods give wrong answers that look reasonable.

Mistake: treating a due as ordinary, or trusting $P \div i$ for any rate. The first loses one period of interest. The second holds only while the interest rate stays the same for ever.

IN HISTORY

Edmond Halley (1656–1742) is best known for predicting the return of the comet that now bears his name. In 1693 he also published, in the Philosophical Transactions of the Royal Society of London, a paper on the mortality of mankind. His data were the birth and death records of the city of Breslau, now Wrocław in Poland, which a pastor of the city, Caspar Neumann, had sent to the Royal Society.

From these records Halley made a table showing how many people of a group born in the same year would survive to each age. He then used the table to compute the price of life annuities, which pay a sum each year for as long as the buyer lives. In outline, the fair price is the present value of each yearly payment, weighted by the chance that the buyer is alive to receive it. The calculation joins two ideas of this chapter: equal payments over time, and discounting.

REVIEW QUESTIONS

5.1 Find the future value of Tk 12,000 deposited each year for 6 years at 9% a year, first as an ordinary annuity and then as an annuity due.

5.2 Find the present value of a yearly payment of Tk 45,000 for 5 years at 10% a year, paid at each year end.

5.3 Find the equal yearly deposit that reaches Tk 300,000 in 4 years at 7% a year, with deposits at each year end.

5.4 Find the level yearly payment that repays a loan of Tk 250,000 over 3 years at 12% a year.

5.5 Explain the difference between an ordinary annuity and an

refuse further loans. The law and honest dealing also set limits that no aim may break.

The aim has limits that you should know. A market price may be wrong for a time, and prices change as news arrives. A business whose common stock is not traded has no price to read, so its value must be estimated, and different people may estimate differently. Even so, the aim gives a more useful guide than profit alone, and it is the aim this book uses.

Before moving on. Market value is the price at which something could be sold today. Wealth maximisation aims to make the market value of the owners' stake as large as possible over time, and it takes account of timing and risk. Stakeholders are all the people affected by a business, and treating them well usually supports long-run value. You should now be able to say, in one sentence, why wealth maximisation is a better aim than profit maximisation.

3.3 AGENCY THEORY

In 1776 the Scottish thinker Adam Smith (1723–1790) published *The Wealth of Nations*. In it he doubted that the directors of large companies would guard the money of their shareholders with the same care that owners give to their own. Directors, he wrote, manage other people's money. Two hundred years later, in 1976, the economists Michael Jensen and William Meckling gave this old worry a precise form and a name. This section explains their idea in layers, beginning with an everyday case.

A farmer cannot go to the market, so a neighbour sells the rice harvest and receives a payment for every sack sold. The farmer wants a good price for the whole harvest. The neighbour earns more by selling many sacks quickly, even at a low price. The two people want different things, though each may be honest. Write this picture down, because every idea in this section is a version of it.

The word for the farmer in the story is **principal**. A principal is a person who hires another person to act for them. The owners of a business are its principals. The owners of a bookshop, for example, are the principals when they hire someone to run a branch.

The word for the neighbour is **agent**. An agent is a person who acts for a principal and takes decisions in the principal's name. A hired branch manager is an agent. The relation between the two is called an agency relation, and it exists wherever one person decides on behalf of another: a taxi owner and a driver, a school principal and a bursar, a business owner and a manager.

Two facts make this relation difficult. First, the agent decides, and the principal often cannot see every decision. Second, the result of the decision falls mainly on the principal. A manager who spends the owner's money freely gives up nothing personally, while the owner gives up the money.

When the aims of the two people differ, a clash may follow. An **agency conflict** is a clash between what a principal wants and what an agent chooses to do. The owners of a business want wealth maximisation. The manager wants a large bonus, a secure

Wealth maximisation has limits too.

A short summary of section 3.2 with a test sentence.

GO SLOWLY HERE.

A historical hook: in 1776 Adam Smith doubted that hired directors guard other people's money as owners do.

An everyday bridge: a neighbour paid for each sack sold.

Principal. A person who hires another person to act for them.

The person who hires is called the principal.

Agent. A person who acts for a principal and takes decisions in the principal's name.

The person hired is called the agent.

The agent decides, but the principal bears much of the result.

Agency conflict. A clash between what a principal wants and what an agent chooses to do.

The case file gives a scholarship fund that pays Tk 50,000 for ever.

The division reproduces the case figure, and a check shores why the fund never runs out.

Four slips cause most errors: the wrong factor, mixed periods, wrong timing and early rounding.

Recap: present value is the payment times a factor; for a due multiply by $1 + i$; for a perpetuity divide by the interest rate.

Sometimes the payment is the unknown, and the same factors answer by division.

Sinking fund. Money put aside in equal deposits to reach a target sum by a set date.

A loan repaid in equal instalments is an annuity whose present value is the amount borrowed.

Amortisation. Repaying a loan by equal payments that cover the interest and part of the amount borrowed.

Worked example 5 uses a small loan with illustrative figures that are not part of the case file.

Each year the interest share of the instalment falls and the

Case file, illustrative. Meghna and Sons, 15 August 2024. A scholarship fund pays Tk 50,000 a year for ever. The interest rate is 8% a year. The present value is Tk 625,000.

The calculation is $Tk\ 50,000 \div 0.08 = Tk\ 625,000$. The check: a fund of Tk 625,000 earns $8\% \times Tk\ 625,000 = Tk\ 50,000$ each year. If the fund pays out only that interest, the fund itself stays whole, and the payments can continue for ever. The rule has limits. It holds only while the interest rate stays at 8%, which no one can promise for ever. It also assumes that the first payment comes at the end of the first period; if the first payment is made today, add one payment to the result.

Common slips are few. Using the future value factor when a present value is wanted gives a sum that grows instead of shrinks. Mixing periods, such as monthly payments with a yearly interest rate, spoils every figure. Treating an annuity due as ordinary forgets the multiplication by $1 + i$. Rounding a factor to two or three places loses whole taka on large payments, so keep six places.

Before moving on. The present value of an ordinary annuity is the payment times the factor $[1 - (1 + i)^{-n}] \div i$. For an annuity due, multiply by $1 + i$. For a perpetuity, divide the payment by the interest rate. A factor table stores the factors ready made. *You should now be able to say* why a series of payments is worth less today than its total.

* * *

5.4 FINDING THE PAYMENT

So far the payment was given and the value was unknown. Often the position is reversed. A business knows the sum it needs, or the sum it has borrowed, and wishes to find the payment. The same two factors serve, but we divide instead of multiplying. The first case is a **sinking fund**. A sinking fund is money put aside in equal deposits to reach a target sum by a set date. The rule is: deposit = target sum $\div$ future value factor. The worked example that follows solves the van problem of the case file in full.

The second case is a loan. The lender receives equal payments, usually at period ends, so the lender holds an ordinary annuity whose present value is the amount borrowed. **Amortisation** means repaying a loan by equal payments that cover the interest and part of the amount borrowed. The rule is: instalment = amount borrowed $\div$ present value factor. Each instalment pays that year's interest first, and the rest repays part of the loan.

Given. A loan of Tk 100,000, repaid by 3 equal year-end instalments; interest rate 10% a year. *Find.* The yearly instalment. *Formula.* Instalment = amount borrowed $\div$ present value factor. The factor for 10% and 3 payments is 2.486852. *Steps.* Tk 100,000 $\div$ 2.486852 = Tk 40,211.48. *Answer.* The yearly instalment is Tk 40,211.48. *Check.* The table below shows the loan falling to zero after the third instalment.

Read each row from left to right. The opening balance earns 10% interest. The rest of the instalment. Tk 40,211.48 less the interest.

Conflicts of this kind have a price, and the price has a name.

An **agency cost** is the loss in value, plus the cost of prevention, that comes from an agency conflict. In the case file the loss in value is Tk 105,000 if the manager chooses Plan Y. The cost of prevention is the money and time spent to stop such a choice, for example the owners' time spent reading branch records each month. Jensen and Meckling showed that the loss and the cost of prevention belong together, because the owners can reduce one only by paying for the other.

IN SHORT

An agency cost is the loss from an agency conflict plus the cost of preventing it.

Owners cannot remove agency cost completely, but they can reduce it by bringing the interests of the agent closer to their own. Common ways are these. They can pay the manager by a measure of long-term value, not only one year. They can give the manager a stake in the business, so that a loss is a loss for the manager too. They can check records and require reports. They can also rely on pressure from outside: a manager who does poor work finds it harder to be hired again. Each way costs something, and the owners choose where the cost of prevention is smaller than the loss prevented.

Return to the restocking choice of 15 May 2024. A manager who is judged only on this year's profit sees a gain of Tk 80,000 and does not feel the fall of Tk 150,000 that comes later. The owners, who own the business in both years, see a net fall of Tk 70,000. The conflict here is again a difference in the time that each person looks at.

Agency theory also has limits. It does not say that managers are selfish, and many managers act in the owners' interest without any rule that forces them. It does not say that every cost of prevention is worth paying. It describes a pressure that exists, so that owners will think about pay rules before the pressure does harm.

Before moving on. A principal hires an agent, who takes decisions in the principal's name. An agency conflict is a clash between what the principal wants and what the agent chooses. An agency cost is the loss from the conflict plus the cost of preventing it. You should now be able to say who the principal and the agent are in a simple case, and why their aims may differ.

WORKED CASE: THE SECOND BRANCH

A worked case applies the ideas of the chapter to one situation. We follow five steps: *Situation. Question. Reasoning. Conclusion. Check.*

Interest. Word in use. In daily life, a wish to learn more about something, or the benefit a person gains from a result. In finance, the price paid for the use of borrowed money.

Agency conflict has a cost.

Agency cost. The loss in value, plus the cost of prevention, that comes from an agency conflict.

The loss and the cost of preventing it are together the agency cost.

Owners reduce the conflict by lining up the rewards of the agent with their own aim.

The restocking choice shows how a short-term measure tempts an agent.

The theory has limits as well.

A short summary of section 3.3 with a test sentence.

The worked case follows five steps: situation, question,

$1.10 = \text{Tk } 3,641$. Check by deposit: $\text{Tk } 1,000 \times (1.10^3 + 1.10^2 + 1.10) = \text{Tk } 1,331 + \text{Tk } 1,210 + \text{Tk } 1,100 = \text{Tk } 3,641$.

Case file, illustrative. Meghna and Sons, 15 August 2024. If the same Tk 60,000 is deposited at the start of each of the 5 years, at 8% a year, the future value is Tk 380,155.74.

The calculation is $\text{Tk } 351,996.06 \times 1.08 = \text{Tk } 380,155.74$. The check is the difference: $\text{Tk } 380,155.74 - \text{Tk } 351,996.06 = \text{Tk } 28,159.68$, which is 8% of Tk 351,996.06. In other words, depositing at the start of each year earns one extra year of interest on the final sum of the ordinary annuity.

PAUSE AND CHECK 5.P2 Find the future value of three deposits of Tk 1,000 at 10% a year when each deposit is made at the start of the year.

Before moving on. The future value of an ordinary annuity is the payment times the factor $[(1 + i)^n - 1] \div i$. For an annuity due, multiply that result by $1 + i$. *You should now be able to say why the annuity due is worth more at the end: each payment earns interest for one extra period.*

* * *

5.3 PRESENT VALUE OF AN ANNUITY

Return to the opening puzzle. The offer of three payments of Tk 1,000 has a value today, and we can now find it. The present value of an annuity is the sum today that is worth the same as all the payments together. Each payment is discounted back to today, and the discounted amounts are then totalled.³

Given. Three payments of Tk 1,000, at the end of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the three payments. *Formula.* Present value of a single sum = amount $\div (1 + \text{discount rate})$ to the power of the number of years until the amount is received. *Steps.* Payment 1: $\text{Tk } 1,000 \div 1.10 = \text{Tk } 909.09$. Payment 2: $\text{Tk } 1,000 \div 1.21 = \text{Tk } 826.45$. Payment 3: $\text{Tk } 1,000 \div 1.331 = \text{Tk } 751.31$. The total is $\text{Tk } 909.09 + \text{Tk } 826.45 + \text{Tk } 751.31 = \text{Tk } 2,486.85$. *Answer.* The present value of the three payments is Tk 2,486.85. *Check.* A present value must be less than the Tk 3,000 received in total, and Tk 2,486.85 is. A second check: the future value Tk 3,310 from worked example 1, divided by 1.331, also gives Tk 2,486.85.

So the puzzle is answered. At 10% a year the three payments are worth Tk 2,486.85 today, and the offer of Tk 3,000 today is better by Tk 513.15. The payments are worth less than their total because each one arrives after a delay.

The rule, in words first: the present value of an ordinary annuity equals the payment multiplied by the present value factor. The factor equals 1 less $(1 + \text{interest rate})$ to the power of minus the number of payments, all divided by the interest rate. In symbols: $\text{PVA} = P \times [1 - (1 + i)^{-n}] \div i$ PVA stands for the present value of the

³ Recall: present value and discount rate, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'. Here the discount rate is the interest rate that money could earn on a deposit.

The case file sets the annuity due beside the ordinary annuity.

Multiplying by 1.08 reproduces the case figure, and the difference equals one year of interest on the ordinary value.

Recap: future value of an ordinary annuity is the payment times a factor; for an annuity due, multiply by $1 + i$.

GO SLOWLY HERE.

The present value of an annuity is the total of every payment brought back to today.

Worked example 3 discounts each of the three payments in the puzzle at 10% and totals the results.

The puzzle is answered: at 10%, Tk 3,000 today is worth more than three payments of Tk 1,000.

The present value formula gives the factor for all payments in one step.

THINK IT THROUGH

3.T1 A manager is paid 4% of branch sales. Plan P gives sales of Tk 1,000,000 and profit before bonus of Tk 120,000. Plan Q gives sales of Tk 1,400,000 and profit before bonus of Tk 60,000. Find the bonus and the amount the owners keep under each plan, say which plan each side prefers, and find the loss in value to the owners if the manager chooses Plan Q.

3.T2 Suppose Mr Shafiq were paid 5% of the change in profit over the two years instead of 5% of sales. Using the restocking choice of 15 May 2024, find his gain or loss and say whether he would now cut restocking.

WHERE PEOPLE GO WRONG

The table lists four common mistakes. Read each row from left to right: the first column gives the mistake and the second gives the correction.

COMMON MISTAKES AND CORRECTIONS

MISTAKE	CORRECTION
The plan with the largest profit this year is always best	It may lower later profit or carry more risk; wealth maximisation weighs both
Wealth maximisation means ignoring workers and customers	Stakeholders affect long-run value, and the law and honest dealing still apply
An agency conflict means the manager is dishonest	The conflict comes from different rewards; an honest manager may still follow the pay rule
Agency cost exists only in large companies	It exists wherever one person decides for another, even in a small shop

IN HISTORY

In 1776 Adam Smith, a Scottish thinker (1723–1790), published *The Wealth of Nations* in London. In the part on companies owned by many shareholders, he doubted that the directors, who manage other people's money, would watch it with the care that owners give to their own. He expected carelessness and waste.

In 1976 two economists of the University of Rochester in the United States, Michael C. Jensen and William H. Meckling, published "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure" in the *Journal of Financial Economics* (volume 3, pages 305–360). They defined agency costs, related them to the separation of ownership from control, and studied the agency costs created by debt and by outside owners. They also grouped the costs under three heads: the cost of watching the agent, the cost the agent bears to give assurance, and the loss that remains.

Four common misunderstandings are stated and corrected in the table.

Smith in 1776 and Jensen and Meckling in 1976 form the history of agency.

Jensen and Meckling gave the worry a precise form exactly two hundred years later.

Recap: an annuity has equal payments at equal intervals; ordinary means period ends, due means period starts.

GO SLOWLY HERE.

The future value of an annuity is the total of every payment grown to the date of the last payment.

Worked example 1 treats each deposit as a single sum and totals the results.

The multipliers 1.21, 1.10 and 1.00 depend only on the interest rate and the number of payments, so they can be totalled once and reused.

The formula multiplies the payment by one factor that depends on the interest rate and the number of payments.

without limit. Section 5.3 shows that its present value is nevertheless a fixed, finite sum.

Before moving on. An annuity is a series of equal payments at equal intervals for a set number of periods. It is ordinary when the payments fall at period ends and due when they fall at period starts. A perpetuity never ends. *You should now be able to say* where each payment of an ordinary annuity and of an annuity due falls on a timeline.

* * *

5.2 FUTURE VALUE OF AN ANNUITY

The future value of an ordinary annuity is the sum that the payments reach on the date of the last payment. Each payment is first grown at compound interest to that date, and the grown amounts are then totalled. We begin with the smallest numbers, so that the method is the only new thing to learn.²

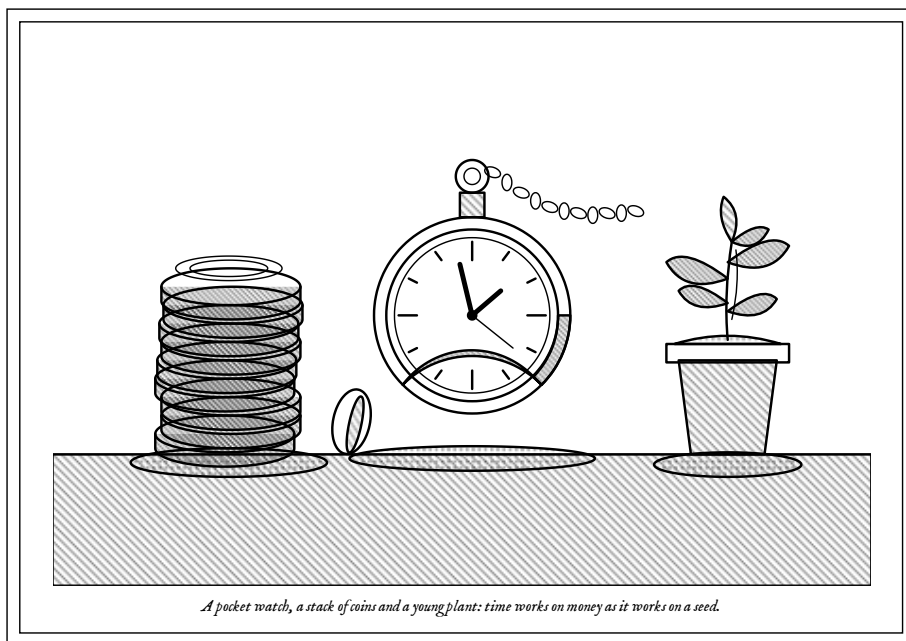
Given. Three deposits of Tk 1,000, one at the end of each year for three years; interest rate 10% a year. *Find.* The sum held at the end of year 3. *Formula.* Future value of a single sum = amount $\times$ (1 + interest rate) to the power of the number of years the amount earns interest. *Steps.* The first deposit earns interest for 2 years: Tk $1,000 \times 1.10^2 = \text{Tk } 1,210$. The second earns for 1 year: Tk $1,000 \times 1.10 = \text{Tk } 1,100$. The third earns for no years: Tk 1,000. The total is Tk $1,210 + \text{Tk } 1,100 + \text{Tk } 1,000 = \text{Tk } 3,310$. *Answer.* The deposits reach Tk 3,310 at the end of year 3. That is Tk 310 more than the Tk 3,000 deposited, and the extra Tk 310 is interest. *Check.* Interest on the first deposit is Tk 210 and on the second Tk 100, and Tk $210 + \text{Tk } 100 = \text{Tk } 310$. The figures agree.

Look at the multipliers in the steps: 1.21, 1.10 and 1.00. Their total is 3.31, and $3.31 \times \text{Tk } 1,000 = \text{Tk } 3,310$. This total does not depend on the size of the payment. It depends only on the interest rate and the number of payments. We may therefore calculate it once and reuse it for any payment. It is called the future value factor of an ordinary annuity. For long series, a formula gives it in one step.

The rule, in words first: the future value of an ordinary annuity equals the payment multiplied by the future value factor. The factor equals (1 + interest rate) to the power of the number of payments, less 1, all divided by the interest rate. In symbols: $\text{FVA} = P \times [(1 + i)^n - 1] \div i$ Here FVA stands for the future value of the annuity, P for the payment made each period, i for the interest rate per period written as a decimal, and n for the number of payments. The expression in square brackets is the factor. It comes from totalling the multipliers, each of which is the one before it multiplied by 1 + i. For our example, $(1.10^3 - 1) \div 0.10 = (1.331 - 1) \div 0.10 = 0.331 \div 0.10 = 3.31$, the same total as before.

PAUSE AND CHECK 5.P1 Using the formula, find the future value factor for 10% and 4 years. Then find the future value of four

² Recall: future value and compound interest, Chapter 4, 'The Time Value of Money: Single Sums', section 4.2 'Future Value of a Single Sum'.



CHAPTER IV

The Time Value of Money: Single Sums

Is a taka today worth the same as a taka next year?



Money has a price that depends on time. This chapter explains why a taka received sooner is worth more than a taka received later, and teaches two moves with one amount at a time: forward in time to find what it will grow to, and back in time to find what it is worth today.

YOU WILL NEED

Chapter 4, sections 4.2 and 4.3: the future value and the present value of a single sum. Every idea in this chapter is built from those two.

A choice between Tk 3,000 now and three payments of Tk 1,000 cannot be settled by adding the payments.

The same shape appears in saving: equal deposits grow, because each one earns interest for a different length of time.

Single sums can answer both questions, but the labour grows with every extra payment.

A friend offers you a choice. You may take Tk 3,000 today, or you may receive Tk 1,000 at the end of each of the next three years. Which is the better offer? Many people notice that both offers total Tk 3,000 and conclude that the two are equal. They are not equal. Money received today can be deposited and can earn interest; money received next year cannot do so until next year. The Tk 3,000 offer is therefore worth more. But how much more? To answer, you must give each Tk 1,000 payment its own value today.

Now reverse the picture. Suppose that you deposit Tk 1,000 at the end of each of the next three years in a bank that pays 10% a year. How much will you hold at the end of the third year? The answer is not Tk 3,000. The first deposit earns interest for two years, the second for one year, and the last for no time at all. Each deposit is a single sum with its own length of time, and Chapter 4 showed how to treat a single sum.¹

You could answer both questions with the tools of Chapter 4. You would treat each payment as a separate single sum and then total the results. For three payments this is tolerable. For a loan repaid over twenty years, or a pension paid over thirty, it would fill pages, and one slip would spoil every line below it. Yet series of equal payments are everywhere in business: rent, loan instalments, savings plans, pensions. This chapter gives such a series a name and a shortcut. It also asks you to see time in a new way, as a row of equal steps with a payment placed on some of them. Go slowly; each idea is short, and each rests on the one before.

* * *

5.1 WHAT AN ANNUITY IS

Begin with three ordinary households. One family pays Tk 8,000 rent on the last day of every month. A student deposits Tk 500 on the first day of every month. A father repays a loan with Tk 4,000 each month for two years. The purposes differ, but the payments share three features. Every payment has the same size. The same length of time separates each payment from the next. And the series stops after a known number of payments. Finance gives this shape a name.

That name is **annuity**. An annuity is a series of equal payments made at equal time intervals for a set number of periods. Three deposits of Tk 1,000, made one at the end of each of three successive years, form an annuity. So does the monthly loan repayment above. A period is the length of the interval between payments.

¹ Chapter 4, 'The Time Value of Money: Single Sums', section 4.2 'Future Value of a Single Sum' and section 4.3 'Present Value of a Single Sum'.

GO SLOWLY HERE.

Rent, loan instalments and savings plans share one shape: equal payments at equal intervals.

The word annuity names any series of equal payments made at equal intervals for a set number of periods.

Annuity. A series of equal payments made at equal time intervals for a set number of

The earning power of money has a price, and the price has a name. **Interest** is the price paid for using money for a period, received by the lender. If you lend Tk 100,000 to a bank for a year and the bank pays you Tk 8,000 for the use of it, the Tk 8,000 is interest. The borrower pays the interest and the lender receives it. A business that borrows through debt pays interest, and a business that deposits spare funds receives it.

The amount of interest depends on how much is lent, so we also need a way to compare loans of different sizes. The **interest rate** is the interest for one period, shown as a percentage of the amount lent or borrowed. Tk 8,000 earned on Tk 100,000 in one year gives $\text{Tk } 8,000 \div \text{Tk } 100,000 = 0.08$, which is 8% a year. When no period is named, this book means one year. In formulas we write the rate as a decimal, so 8% becomes 0.08, and we use the letter *i* for it.

The farmer who eats the seed gives something up. **Opportunity cost** is the value of the best choice given up when you pick another. The farmer's opportunity cost of eating the seed is the harvest that sowing would have given. In the same way, Tk 100,000 kept idle in a drawer for a year has an opportunity cost: the interest a bank would have paid, which is Tk 8,000 at 8%. The cost is real although no one sends a bill, because it is a gain that never arrived.

The words change with the viewpoint, but the idea stays the same. A lender sees interest as a reward for waiting. A borrower sees it as the price of using money before it is earned. An owner or manager sees it as the opportunity cost of the funds placed in the business. Chapter 3 explained why earlier wealth is preferred, because wealth maximisation looks at the market value of the owners' stake over time.² A manager who prefers wealth sooner, and an owner who asks what the funds could have earned elsewhere, both apply the time value of money.

The idea has limits. The size of the effect depends on the interest rate available: at a higher rate, waiting costs more. The idea also assumes that the later amount is certain. If the friend might not pay, the later promise is worth less for that reason as well. We set that matter aside until Chapter 9.

Before moving on. The time value of money is the idea that money received sooner is worth more than the same amount received later. Interest is the price of using money for a period, the interest rate is that price as a percentage per period, and opportunity cost is the value of the best choice given up. You should now be able to say why Tk 100,000 today is worth more than Tk 100,000 in a year.

² Chapter 3, 'The Goal of the Firm and Agency', section 3.2 'Wealth Maximisation': why a manager prefers earlier wealth.

Interest. The price paid for using money for a period, received by the lender.

Interest. Word in use. In daily life, a wish to learn about something, or the benefit a person gains from a result. In finance, a payment for the use of money.

Interest is the price of using money for a period.

Interest rate. The interest for one period, shown as a percentage of the amount lent or borrowed.

Rate. Word in use. In daily life, a speed or a count in a period, such as a rate of travel. In finance, a percentage of an amount for a period, such as 8% a year.

The interest rate puts the price as a percentage, so loans of different sizes can be compared.

Opportunity cost. The value of the best choice given up when you pick another.

The farmer who eats the seed gives up the harvest, and that loss is an opportunity cost.

Lender, borrower and owner see the same idea in different words.

The effect depends on the rate and on the later amount being certain.

A short summary of section 4.1 with a test sentence.

4.5 Explain the opportunity cost of idle cash in plain words, with a small example.

TK 200,000 AT 8% A YEAR (TK)

YEAR	SIMPLE INTEREST	COMPOUND INTEREST
0	200,000.00	200,000.00
1	216,000.00	216,000.00
2	232,000.00	233,280.00
3	248,000.00	251,942.40
4	264,000.00	272,097.79
5	280,000.00	293,865.62

In year 1 the two totals agree, because there is no earlier interest to earn on. After that, compounding pulls ahead. By year 5 the compound total is Tk 13,865.62 above the simple total, and the gap grows each year.

The figure draws the same two lines. To read it, look at the horizontal axis first: it shows the years from today. The vertical axis shows the amount in thousands of taka, and each point shows what the account holds at the end of that year. The dashed line is simple interest and the solid line is compound interest. A straight line means the same amount is added each year. A line that bends upward means each year adds more than the year before.

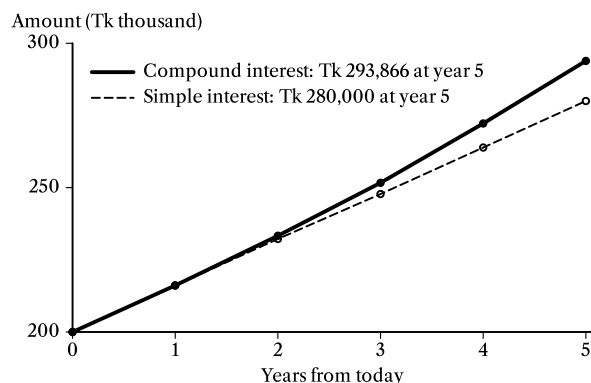


FIG. 4.1. Growth of Tk 200,000 at 8% a year over five years: simple interest against compound interest

Three cautions apply. The rate and the time must use the same period: a yearly rate needs the time counted in years. The rate enters the formula as a decimal, so 8% is 0.08. And the formula assumes interest is added once a year; section 4.4 changes that. Compounding also works against borrowers, because a debt grows in the same way.

Before moving on. A single sum is one payment at one point in time. Simple interest is worked out on the original amount only, and compound interest on the original amount plus interest already earned. Future value is what an amount today will grow to, and $FV = S \times (1 + i)$ to the power of n . You should now be able to say how much Tk 10,000 becomes in two years at 10% a year, and why

In year 1 the two agree; after that compounding pulls ahead.

How to read the figure.

Three cautions apply to the formula.

A short summary of section 4.2 with a test sentence.

A: Tk 70,000 $\times$ 0.793832 = Tk 55,568.24. B: Tk 85,000 $\times$ 0.680583 = Tk 57,849.56. *Steps at 12%.* The factors are $1 \div 1.12$ to the power of 3 = 0.711780 and $1 \div 1.12$ to the power of 5 = 0.567427. A: Tk 70,000 $\times$ 0.711780 = Tk 49,824.60. B: Tk 85,000 $\times$ 0.567427 = Tk 48,231.30. *Answer:* At 8% Offer B is better, by about Tk 2,281. At 12% Offer A is better, by about Tk 1,593. *Check.* A higher rate shrinks every present value, and it shrinks the later amount by more. So the nearer offer gains ground as the rate rises, and the ranking can change.

COMMON SLIPS

Four slips appear again and again. The table gives each slip and a way to avoid it. Read each row from left to right.

COMMON SLIPS AND HOW TO AVOID THEM

SLIP	HOW TO AVOID IT
Using 8 in the formula instead of 0.08	Write the rate as a decimal and say it aloud: 8% is 0.08
Using a yearly rate with half-years	Divide the rate and multiply the years by the periods in a year
Multiplying when you should divide, or the reverse	Forward in time multiplies; back in time divides; a present value is below the later amount
Forgetting the 1 in $(1 + i)$	The factor is $(1 + i)$ to the power of n , never i to the power of n

SUMMARY

Money received sooner is worth more than the same amount received later: this is the time value of money. Interest is the price of using money, and the interest rate states it as a percentage of the amount per period. Compound interest earns interest on interest, and future value is $FV = S \times (1 + i)$ to the power of n . Present value is found by discounting, $PV = FV \div (1 + i)$ to the power of n , at the discount rate. When interest is added within the year, divide the rate and multiply the periods, and compare offers by the effective annual rate.

THINK IT THROUGH

4.T1 A student chooses between Tk 90,000 today and Tk 130,000 in 5 years. Money can earn 7% a year, compounded yearly. Find the present value of Tk 130,000, decide which offer is better, and show a check by finding the future value of Tk 90,000.

4.T2 Tk 15,000 grows to Tk 21,277.79 at 6% a year. Use the future value table to find the number of years.

WHERE PEOPLE GO WRONG

The table lists five common mistakes. Read each row from left to right: the first column gives the mistake and the second gives the

Four slips appear often, and each has a simple defence.

Summary of the chapter.

Five common misunderstandings are stated

FUTURE VALUE FACTORS, $(1 + i)$ TO THE POWER OF N

YEARS	6%	8%	10%
1	1.060000	1.080000	1.100000
2	1.123600	1.166400	1.210000
3	1.191016	1.259712	1.331000
4	1.262477	1.360489	1.464100
5	1.338226	1.469328	1.610510
6	1.418519	1.586874	1.771561

PRESENT VALUE FACTORS, $1 \div (1 + i)$ TO THE POWER OF N

YEARS	6%	8%	10%
1	0.943396	0.925926	0.909091
2	0.889996	0.857339	0.826446
3	0.839619	0.793832	0.751315
4	0.792094	0.735030	0.683013
5	0.747258	0.680583	0.620921
6	0.704961	0.630170	0.564474

Case file, illustrative. Date: 1 July 2024. Meghna and Sons needs Tk 500,000 on 30 June 2029 for a premises deposit. The present value at 8% a year is Tk 340,291.60, using a present value factor of 0.680583.

The time from 1 July 2024 to 30 June 2029 is 5 years. The table gives 0.680583 for 5 years at 8%, and $\text{Tk } 500,000 \times 0.680583 = \text{Tk } 340,291.50$. A calculator that keeps every digit of the factor gives Tk 340,291.60, which is the figure in the case file; the gap of Tk 0.10 comes from rounding the factor. So a deposit of Tk 340,291.60 today at 8% will grow to Tk 500,000 on the date needed. Check: $\text{Tk } 340,291.60 \times 1.469328$ is about Tk 500,000.

The figure shows a timeline. Time runs from left to right, and the first mark is today. The upper arrow points forward and is labelled compounding: moving along it multiplies by $(1 + i)$ for each year. The lower arrow points back, is drawn with a dashed line and is labelled discounting: moving along it divides by $(1 + i)$ for each year.

The business case needs a future amount to be met by a deposit today.

Working of the case file: five years, factor, answer and check.

How to read the timeline figure.

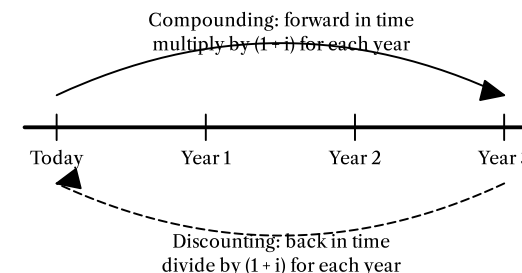


FIG. 4.2. Compounding moves an amount forward in time and discounting moves it back

Four cautions apply to present value.

A short summary of section 4.3 with a test sentence.

GO SLOWLY HERE.

A bank that adds interest twice a year pays more than one that adds it once.

Compounding period. The time between one addition of interest and the next.

The compounding period is the time between additions of interest.

Method: change the rate and the time to match the period.

Four cautions apply. A present value is below the future amount whenever the rate is above zero. A higher rate or a longer time makes the present value smaller: Tk 1 due in 5 years is worth 0.680583 at 8% but only 0.620921 at 10%. Rounding the factors can change the last digits of an answer. And the rate and the time must use the same period, as in section 4.2.

Before moving on. Present value is what an amount to be received later is worth today at a stated rate. Discounting finds it by dividing by $(1 + i)$ once for each year, and the rate used is the discount rate. You should now be able to say why Tk 12,100 due in two years at 10% is worth Tk 10,000 today.

4.4 COMPOUNDING WITHIN THE YEAR

A bank offers 12% a year on Tk 10,000 but adds interest every six months. Is that the same as adding 12% once at the end of the year? Work it through. After six months the bank adds half a year's interest, 6% of Tk 10,000, which is Tk 600, so the account holds Tk 10,600. After twelve months it adds 6% of Tk 10,600, which is Tk 636, so the account holds Tk 11,236. Adding once a year would give Tk 11,200. The bank that adds twice pays Tk 36 more.

The reason lies in the **compounding period**, which is the time between one addition of interest and the next. In the example it is six months. Because the bank adds interest twice a year, the second addition earns interest on the first. The shorter the compounding period, the sooner interest starts to earn interest.

To handle this by hand, change both the rate and the time to match the period. The rate for one period is the yearly rate divided by the number of periods in a year. The number of periods is the years times the number of periods in a year. In symbols: $FV = S \times (1 + i \div m)$ to the power of $(m \times t)$, where m is the number of periods in a year and t is the number of years. In the example, $i \div m = 0.12 \div 2 = 0.06$ and $m \times t = 2$, so $FV = \text{Tk } 10,000 \times 1.06$ to the power of 2, which is Tk 11,236.

THREE SHAPES OF CASH FLOW

SHAPE	NUMBER OF PAYMENTS	SIZES	TOOL FOR VALUING IT
Single sum	One	Not applicable	One discounting step
Annuity	Several	All equal	The annuity factor
Mixed stream	Several	May differ	Discount each, then total

Notice that the shapes are nested. An annuity is a mixed stream whose sizes happen to be equal, and a single sum is a mixed stream with one payment. The method for a mixed stream therefore works for every shape, although it is slower. The annuity factor is a shortcut for the equal case.

The shape appears from several viewpoints. The owner of a shop sees a mixed stream in yearly profit that grows as the business matures. The financial manager sees it in the cash that a new machine will bring. A lender sees it in a loan whose repayments are uneven. Each of them must turn the stream into one number before deciding anything.

Before moving on. A mixed stream is a series of payments or receipts that differ in size from one period to the next. A single sum and an annuity are special cases of it. Because the sizes differ, we value each payment alone and then total the results.

* * *

6.2 PRESENT AND FUTURE VALUE OF A MIXED STREAM

The method has a name. **Discounted cash flow** is a way of valuing a series of future cash flows by turning each one into a present value. The present values share one date, today, so they can be added. We begin with the smallest numbers, so that the method is the only new thing to learn.²

Given. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the stream. *Formula.* Present value of a single sum = amount ÷ (1 + discount rate) to the power of the number of years until it is received. *Steps.* Year 1: Tk 1,000 ÷ 1.10 = Tk 909.091. Year 2: Tk 2,000 ÷ 1.21 = Tk 1,652.893. Year 3: Tk 3,000 ÷ 1.331 = Tk 2,253.944. The total is Tk 909.091 + Tk 1,652.893 + Tk 2,253.944 = Tk 4,815.928, or Tk 4,815.93. *Answer.* The present value of the stream is Tk 4,815.93. *Check.* The present value must be less than the Tk 6,000 received in total, and it is.

PAUSE AND CHECK 6.P1 Value the opposite stream, Tk 3,000, Tk 2,000 and Tk 1,000 at the ends of years 1, 2 and 3, at 10%. Is it

² Recall: present value and discounting, Chapter 4, 'The Time Value of Money: Single Sums', section 4.3 'Present Value of a Single Sum'.

A mixed stream is the general shape; the other two are special cases of it.

The owner, the manager and the lender each meet mixed streams in different places.

Recap: a mixed stream has payments of different sizes, and single sums and annuities are special cases.

Discounted cash flow turns each future amount into a present value and totals the results.

Discounted cash flow. A way of valuing a series of future cash flows by turning each one into a present value.

Worked example 1 discounts three uneven receipts at 10% and totals the present values.

Two jobs, one certain and one uncertain: which would you take?

Neither choice is foolish, because the question is about how much uncertainty you will accept.

This chapter gives three tools: a way to state a gain, a way to describe uncertainty, and a way to compare choices.

The chapter relies on one earlier idea, required return, from Chapter 8.

Money put to work earns a gain or suffers a loss, and we need a name for it.

Return. the total gain or loss from an investment over a period, income plus any change in value

Return. Word in use. In daily life, to give back or to come back. In finance, the gain or loss from an investment.

Return is everything gained or lost on an investment during a period.

Return as an amount cannot be compared between investments of different size.

Rate of return. the return shown as a percentage of the amount first invested

Rate of return divides the return by the amount first invested.

Imagine that you are offered two jobs selling books at a fair. The first pays Tk 12,000 every month, whatever happens. The second pays Tk 2,000 in a poor month, Tk 14,000 in an ordinary month and Tk 26,000 in a busy month, and nobody can tell you in advance which kind of month will come. Which job would you take?

Many people choose the first job, although the second pays more over a long run of months. Others choose the second. Both groups answer a question that the amount of money cannot settle alone: how much uncertainty will you accept for the chance of a larger gain? A business meets the same question each time it buys stock, borrows money or starts a project.

Three smaller questions follow from the puzzle. How do we state what an investment pays, in one number that can be compared? How do we describe uncertainty in numbers rather than in feelings? How do we compare two choices when one pays more and also varies more? By the end of the chapter you will answer each question with a figure that you can calculate by hand.

You will need the idea of required return from Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value'. Required return is the least return that makes a project worth accepting. Here we learn to measure the returns that are compared with it.

* * *

9.1 WHAT RETURN IS

Suppose you lend Tk 1,000 to a cousin, who repays Tk 1,100 after one year. You have gained Tk 100. Suppose instead that you buy a goat for Tk 1,000 and sell it a year later for Tk 900. You have lost Tk 100. In both cases you placed funds in something and later counted the result.

Finance gives the name **return** to this result. Return means everything you gain or lose from an investment in a period: the payments you receive, plus any rise or fall in its price. An *investment* is money placed in something in the hope of gain. If a bond pays Tk 60 in interest and its price rises by Tk 40, the return is Tk 100.

A return stated in taka has a weakness. A return of Tk 100 is large on an investment of Tk 500 and small on an investment of Tk 50,000. To compare fairly, we scale the return by the size of the investment.

The **rate of return** is the return shown as a percentage of the amount first invested. In words: rate of return equals return divided by amount first invested, multiplied by 100. In symbols, with R for the return and P for the amount first invested, the rate is $(R \div P) \times 100\%$. The Tk 100 gained on the loan of Tk 1,000 is a rate of return of 10%.

Check. The future value equals the present value grown for five years, and $1.12^5 = 1.762342$. For stream A, Tk 212,147.18 $\times$ 1.762342 is about Tk 373,876. For stream B, Tk 227,589.53 $\times$ 1.762342 is about Tk 401,091. Both agree with the tables to within the rounding of the multiplier.

Many streams contain a stretch of equal receipts with one or two odd amounts. Then the stream can be split into an annuity part and an odd part. The annuity part is valued with the annuity factor of Chapter 5, and the odd part as a single sum. The two present values are added. Chapter 5 gave the factor for 10% and 3 payments as 2.486852.

Given. Receipts of Tk 20,000, Tk 20,000 and Tk 30,000 at the ends of years 1, 2 and 3; discount rate 10% a year. *Find.* The present value of the stream. *Formula.* Present value = annuity payment $\times$ annuity factor + odd amount $\times$ discount multiplier of its year. *Steps.* The level part is Tk 20,000 in each of 3 years: Tk 20,000 $\times$ 2.486852 = Tk 49,737.04. The odd part is the extra Tk 10,000 in year 3: Tk 10,000 $\div$ 1.331 = Tk 7,513.15. The total is Tk 49,737.04 + Tk 7,513.15 = Tk 57,250.19. *Answer.* The present value of the stream is Tk 57,250.19. *Check.* Discount each receipt alone: Tk 18,181.82 + Tk 16,528.93 + Tk 22,539.44 = Tk 57,250.19. The two methods agree.

Three things fix the value of a stream: the size of each receipt, the date of each receipt, and the discount rate. Nothing else matters to the arithmetic. The name of the project, the person who pays and the total of the receipts do not enter the formula. This is why two plans that look very different can have the same present value, and two plans that look alike can differ. When you meet a new stream, list the receipts with their years first, as in the tables of this chapter, and only then begin to discount.

A table is the safest way to value a stream by hand. Write the years down the left side. Next to each year write the receipt, then the multiplier, then the product. The total of the last column is the present value. If a year has no receipt, write zero and keep the row, so that no year is skipped. Check the table by reading the multipliers down the column: they must fall steadily, because each later year has a smaller multiplier.

Before moving on. Discounted cash flow values a stream by turning each receipt into a present value and adding. The future value grows each receipt to the last date and adds. A level part may use the annuity factor. *Common slip:* using the wrong year for a receipt, so that it is discounted too much or too little.

* * *

6.3 COMPARING STREAMS WITH A COMMON RATE

A buyer who must choose between two streams needs one number for each. Totals do not serve, because they ignore timing. The fair way is to value both at the same date, usually today, and at the same discount rate. The stream with the higher present value is

The check: a present value grown for five years gives the future value, to within a few poisha.

Streams that contain a level part can use the annuity factor for that part.

Worked example 3 splits a stream of Tk 20,000, Tk 20,000 and Tk 30,000 into a level part and an odd part.

A stream's value depends only on the sizes of the receipts, their dates and the discount rate.

A tidy table prevents most errors: one row for each year, one column for each quantity.

Recap: discount each receipt, or split the stream into an annuity part and an odd part, and total the present values.

Two streams can be compared fairly only at the same date and the same discount rate.

The viewpoint of the decision-maker also counts. An owner who must repay a loan next year may prefer the stream with early receipts even at a low rate, because the cash is needed on that date. A saver who will not need the money for ten years may care little about timing. The discount rate expresses these needs in one number, so choosing the rate is itself a decision, to be made with care and stated openly.

Before moving on. Compare streams by present value at one common rate. Test the ranking at several rates, because it can reverse when totals differ. With equal totals, the stream with the larger early receipts has the higher value.

* * *

WORKED EXAMPLE: TWO STREAMS AT THREE RATES

Here we compare a rising stream R of Tk 1,000, Tk 2,000 and Tk 3,000 with a falling stream F of Tk 3,000, Tk 2,000 and Tk 1,000, all at year ends. Both total Tk 6,000. The multipliers below are the discount multipliers for years 1, 2 and 3.

Given. Stream R: Tk 1,000; 2,000; 3,000. Stream F: Tk 3,000; 2,000; 1,000. Discount rates 9%, 12% and 15% a year. *Find.* The present value of each stream at each rate, and which is worth more. *Formula.* Present value = total of (receipt $\times$ discount multiplier of its year), where the multiplier is $1 \div (1 + \text{rate})$ to the power of the year. *Steps.* At 12%: $R = \text{Tk } 892.857 + \text{Tk } 1,594.388 + \text{Tk } 2,135.341 = \text{Tk } 4,622.59$. $F = \text{Tk } 2,678.571 + \text{Tk } 1,594.388 + \text{Tk } 711.780 = \text{Tk } 4,984.74$. The steps at 9% and 15% use the multipliers in the table. *Answer.* F is worth more than R at all three rates. The results are in the second table. *Check.* At a rate of 0% both streams are worth Tk 6,000, so the gap must vanish as the rate falls. The gap in the table shrinks as the rate falls, which agrees.

Read the first table by choosing a rate in the left column and moving right to the multiplier for each year.

DISCOUNT MULTIPLIERS

RATE	YEAR 1	YEAR 2	YEAR 3
9%	0.917431	0.841680	0.772183
12%	0.892857	0.797194	0.711780
15%	0.869565	0.756144	0.657516

Read the second table by rate. For each rate it gives the present value of R, of F, and the difference between them.

PRESENT VALUES OF THE TWO STREAMS (TK)

RATE	PRESENT VALUE OF R	PRESENT VALUE OF F	GAP (F - R)
9%	4,917.34	5,207.84	290.50
12%	4,622.59	4,984.74	362.15

A viewpoint matters: an owner who needs cash early values early receipts more than the arithmetic alone shows.

Recap: compare streams at the same date and rate, test the rate, and expect the earlier-heavy stream to win at equal totals.

The worked example compares a rising and a falling stream, with equal totals, at 9%, 12% and 15%.

The steps follow the standard order from Given to Check, with the work at 12% shown in full.

The first table lists the discount multipliers for the three rates and three years.

The second table shows that the gap between the falling and rising streams widens as the rate rises.

Answer and check: accept Press.

Answer. Accept Press. Both projects pass the net present value test, but Press adds Tk 197,055 against Tk 86,464, and the internal rate of return and the profitability index agree. The only tool that prefers Fit-out is payback, and its advantage is a recovery 0.46 years earlier, in exchange for Tk 110,591 less value. *Check.* $197,055 - 86,464 = 110,591$. Press passes a stricter test as well: its internal rate of return is 19.9%, well above 12%. A common slip is to compare payback figures first and stop; another is to compare projects at different discount rates.

* * *

SUMMARY

The payback period is the time to recover the initial investment; it is simple but ignores later cash and timing. Net present value discounts the inflows at the required return and subtracts the initial investment; accept when it is zero or more. The internal rate of return is the discount rate that makes the net present value zero. The profitability index is the present value of inflows divided by the initial investment. When rankings conflict for mutually exclusive projects, net present value decides.

THINK IT THROUGH

8.T1 Two mutually exclusive projects each have an initial investment of Tk 500,000 and a required return of 10%. Project X has inflows of 300,000, 200,000, 100,000 and 100,000 in years 1 to 4. Project Y has inflows of 100,000, 150,000, 200,000 and 400,000. Find each payback period and net present value, rank the projects by each tool, and decide.

8.T2 In plain words, explain why a project's net present value is exactly zero when the discount rate equals its internal rate of return.

WHERE PEOPLE GO WRONG

Many students believe the project with the shorter payback is the better one. It is only the faster one. Project A in the pause question recovers its money in 2 years but never earns more than its cost.

Another mistake is to compare net present values worked out at different discount rates. Use one required return for all projects of similar risk. Otherwise the comparison measures the choice of rates, not the projects.

A third mistake is to treat the internal rate of return as a guaranteed yearly gain. It is the rate at which the net present value reaches zero. It also does not state the size of the gain in taka.

A fourth mistake is to use accounting profit instead of the cash flows built in Chapter 7. Depreciation, for example, must be added again, as that chapter showed.

A short payback is not the same as a good project.

Discount rates must stay the same when comparing.

The internal rate of return is not always the whole story.

Cash flows, not accounting profit, go into every tool.

REVIEW QUESTIONS

- 6.1 Find the present value of receipts of Tk 20,000, Tk 35,000 and Tk 50,000 at the ends of years 1, 2 and 3, at 10% a year.
- 6.2 Find the future value at the end of year 3 of the stream in question 6.1, at 10% a year.
- 6.3 A stream pays Tk 15,000 at the end of each of years 1 to 4 and Tk 40,000 at the end of year 5. Split it into an annuity part and an odd part, and find its present value at 10% a year. The annuity factor for 10% and 5 payments is 3.790787.
- 6.4 Compare the rising stream of question 6.1 with the falling stream Tk 50,000, Tk 35,000 and Tk 20,000, at 9% and at 15% a year. Which is worth more at each rate?
- 6.5 Explain how a mixed stream differs from an annuity, and say why the annuity factor cannot be used for a mixed stream as a whole.

Rate. Word in use. In daily life, speed or price per unit. Here, the rate in internal rate of return is a yearly return on the money invested, not a charge set by a lender.

Internal rate of return. The discount rate at which a project's net present value is exactly zero.

The internal rate of return is the discount rate that makes the net present value zero.

Trial and interpolation find the rate by hand.

Profitability index. The present value of a project's inflows divided by its initial investment.

The profitability index shows the present value earned for each taka invested.

Both tools have limits.

The **internal rate of return** is the discount rate at which a project's net present value is exactly zero. Often shortened to IRR, it is the yearly return the project itself earns. Figure 8.2 shows how it is found. The horizontal axis shows the discount rate; the vertical axis shows the net present value of Press in thousands of taka. As the discount rate rises, the value falls. The point where the line crosses zero, near 19.9%, is the internal rate of return.

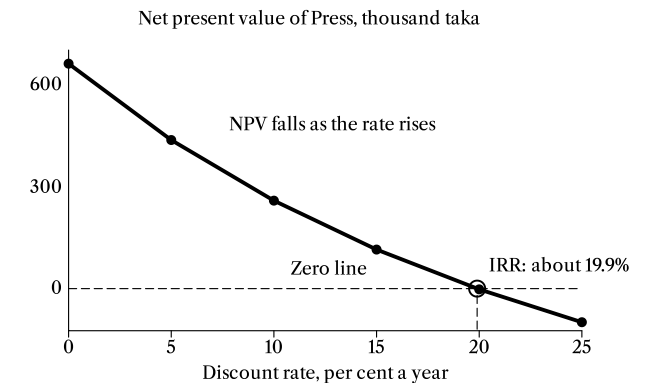


FIG. 8.2. Net present value of Project Press at different discount rates, crossing zero at the internal rate of return

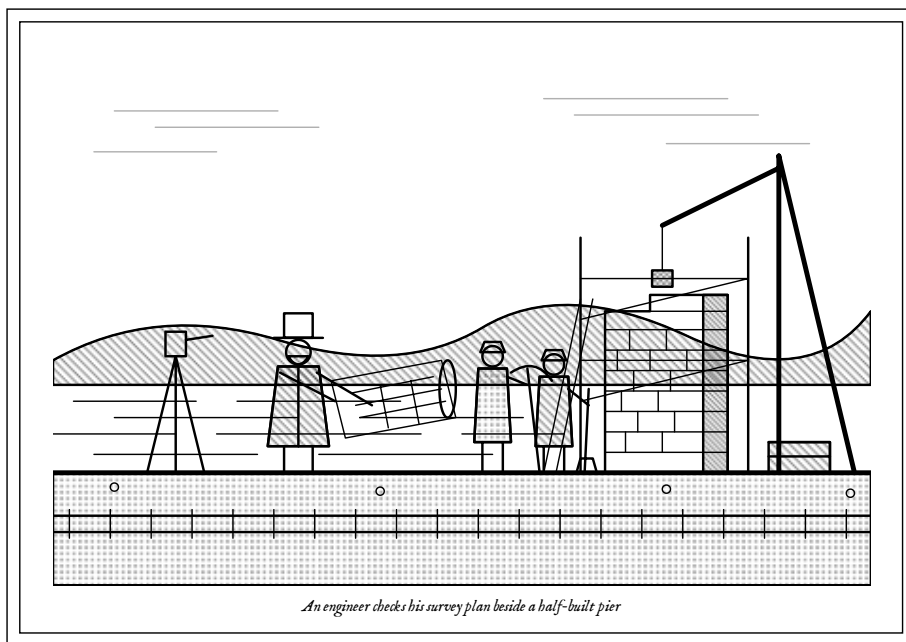
To find it by hand, calculate the net present value at two rates, one giving a positive value and one a negative value, then interpolate. For Press, NPV at 15% is +114,687 and at 20% is -2,253. The rate is $15 + 5 \times 114,687 \div (114,687 + 2,253) = 15 + 4.90 = 19.9\%$.¹ The decision rule is to accept when the internal rate of return is at least the required return: 19.9% exceeds 12%. For Fit-out, NPV at 15% is +27,975 and at 20% is -56,925, giving about 16.6%, which also exceeds 12%.

The **profitability index** is the present value of a project's inflows divided by its initial investment. Press: $1,097,055 \div 900,000 = 1.22$. Fit-out: $986,464 \div 900,000 = 1.10$. A value above 1 means each taka invested returns more than a taka in present value. Accept when the index is at least 1, which is the same test as a net present value of zero or more.

Both tools have limits. The internal rate of return can mislead when projects differ much in size or timing. The profitability index ignores the size of the gain: a small project can have a high index and still add little. Here the two projects have equal initial investments, so neither problem arises.

* * *

¹ A spreadsheet finds the rate with a built-in function. It works by the same repeated trial, only faster.



CHAPTER VII

Capital Budgeting: Decisions and Cash Flows

Which purchase will still be paying for itself in five years?



Some decisions tie up a business's money for many years and are hard to undo. This chapter explains what such decisions are called, the steps a business follows to make them, and how to measure the cash that one project spends and receives. We follow one machine from its first payment to its last.

Recap: payback measures the speed of recovery only.

A promise of more taka later is worth less than it seems.

Net present value. The present value of a project's cash inflows minus its initial investment.

Net present value is the worth today of the inflows, less what we pay.

Return. Word in use. In daily life, the act of giving something back. In finance, the gain from an investment, shown as a percentage of the money invested.

Required return. The lowest return a project must earn to be worth accepting.

The required return is the discount rate and the test.

The decision rule follows from the meaning of zero.

A clean example shows the method.

PAUSE AND CHECK 8.P1 Project A has an initial investment of Tk 60,000 and inflows of Tk 30,000 in each of years 1 and 2, then nothing. Project B has the same initial investment and inflows of Tk 20,000 in each of years 1 to 6. Find each payback period. What does payback miss?

Before moving on. The payback period is the time to recover the initial investment. Add the inflows into a running total, find the recovery year, and add the fraction of that year. It is simple but blind to later cash and to timing.

* * *

8.2 NET PRESENT VALUE

Return to the machine of Tk 10,000 with inflows of Tk 5,000 for three years. The inflows total Tk 15,000, so the machine seems to gain Tk 5,000. But money received later is worth less than money received today, because today's money can be used or invested at once. To compare fairly we convert every inflow into its worth today, using the method of Chapter 6, section 6.2, in which each receipt of a mixed stream is discounted separately.

Net present value is the present value of a project's cash inflows minus its initial investment. Net present value is often shortened to NPV. For a project with a negative result, the inflows are worth less today than the initial investment. The discount rate used to find each present value is chosen next.

The **required return** is the lowest return a project must earn to be worth accepting. Other books call it the required rate of return or the hurdle rate. In this chapter we use it as the discount rate. It depends on the risk of the project and on the cost of the funds; Chapter 10 and Chapter 17 explain how it is found. For Meghna and Sons it is 12%.

The rule is: accept the project when its net present value is zero or positive, and reject it when it is negative. A zero net present value means the project earns exactly the required return. A positive value means it earns more, and the amount is the extra worth added, in taka today.

Given. Initial investment Tk 10,000; inflows of Tk 5,000 at the end of years 1, 2 and 3; discount rate 10%. *Find.* The net present value. *Formula.* The present value of an inflow equals the inflow divided by $(1 + r)$ to the power of t , where r is the discount rate and t the year. Net present value equals the sum of these present values minus the initial investment. *Steps.* Year 1: $5,000 \div 1.10 = 4,545$. Year 2: $5,000 \div 1.21 = 4,132$. Year 3: $5,000 \div 1.331 = 3,757$. Sum = 12,434. NPV = $12,434 - 10,000 = \text{Tk } 2,434$. *Answer.* Net present value is Tk 2,434, so the machine should be accepted. *Check.* The inflows are equal, so Chapter 5 gives a shortcut: $5,000 \times 2.4869 = 12,434.5$.

Second, the amount is large compared with the size of the business. Third, the decision is hard to reverse, because a used machine sells for far less than a new one. At 1 January 2024, a year before the machine was considered, the assets of Meghna and Sons were Tk 2,400,000. A machine with an initial cost of Tk 900,000 is therefore more than a third of everything the shop owned. All three marks are present.

Notice that size is measured against the business, not in taka alone. A sum of Tk 900,000 is large for a bookshop but small for a national bank. The marks also work together. A cheap item that lasts for years is not a capital budgeting decision, because a mistake costs little. A large payment that the seller returns in full is easier to manage, because the mistake can be undone. The decisions that need the most care are those with all three marks.

The money paid out has a name too. **Capital expenditure** means spending on assets that will serve the business for many years. A machine, a delivery van and a new shop floor are capital expenditure. A month of electricity is not, because it is used up at once. Capital budgeting is the planning; capital expenditure is the payment.

Why spend time on weighing a proposal before accepting it? There are four reasons. The first is size: one mistake can use a large part of the business's funds. The second is length: the machine works for years, so a wrong decision is paid for in every one of those years. The third is that the asset is hard to sell, so the owner cannot escape a poor choice cheaply. The fourth is that funds spent on the machine cannot also be spent on the next proposal, so every decision closes other doors.

Careful work has limits. The future is uncertain, and forecasts will sometimes be wrong. What careful work does is to put every assumption on paper, so that others can question it and the business can check it later. A forecast that is written down can be improved; a hope that is never written down cannot.

Different people see the same decision differently. The owner asks whether the funds will earn more than they cost. The financial manager must prepare the forecasts and defend them. An employee may gain work when a machine arrives, or lose it if the decision goes wrong. A customer gains a new service if the decision is good, and may face higher prices if it is not.

Before moving on. Capital budgeting is the planning of long-term investments, and capital expenditure is the money paid. A decision belongs here when it is long, large for the business and hard to reverse. Careful appraisal matters because errors are costly and last for years.

* * *

7.2 THE STEPS OF THE PROCESS

A builder who must cross a river does not begin by pouring concrete. She first asks where a bridge might go, then measures the ground, then chooses a design, then builds, and later inspects the

Size is relative, and the marks work together.

Capital expenditure. Spending on assets that will serve the business for many years.

The money paid out on such assets has its own name.

Four reasons make careful appraisal worth the effort.

Careful work cannot remove error, but it exposes assumptions.

Each viewpoint sees the decision differently.

Recap: capital budgeting is planning for large, long and hard-to-reverse choices.

A large task becomes manageable when it is split into steps.

A merchant must choose one cargo, and the two manifests point in different directions.

You will need: Chapter 7, section 7.3, and Chapter 6, section 6.2.

Speed, size and timing each pull the choice a different way.

This chapter gives four tools for the judgement.

The first question most owners ask is how soon the money returns.

Payback period. The time a project takes to recover its initial investment from its cash flows.

Payback period is the time needed to recover the initial investment.

Equal inflows allow a one-step formula.

Unequal inflows need a running total.

Case: two projects with the same initial investment compete.

A merchant in a busy port has money for one cargo only. Two captains bring him lists of goods, called manifests, and each captain asks for the same sum. The first captain promises that the merchant will have his money back within a year, but the cargo then earns little. The second captain needs longer, but his cargo earns far more in the end. Which captain should the merchant choose?

The merchant can argue for either captain. Speed is attractive, because money returned early can be used again, and because the future may disappoint. Size is attractive, because the larger total is worth more. Timing complicates both: a coin received in a year's time is worth less than a coin received today, so the totals cannot be compared as they stand.

In Chapter 7 you learned to measure the cash flows of a project. This chapter teaches how to judge them. We study four tools in turn, and we apply each one to the same pair of projects at Meghna and Sons, the family bookshop in Dhaka, so that you can compare the answers. Where the tools disagree, the last sections explain which one to trust and why.

* * *

8.1 PAYBACK PERIOD

An owner who has paid Tk 10,000 for a machine soon asks a plain question: when will I have my money back? Suppose the machine brings in Tk 5,000 at the end of each year for three years. After year 1 the owner has recovered Tk 5,000, after year 2 the full Tk 10,000. The answer is two years, and that answer has a name.

The **payback period** is the time a project takes to recover its initial investment from its cash flows. For the machine above, the payback period is 2 years. A shorter payback period means the money is tied up for less time, which is why many owners like the tool. It also means the business is exposed to the unknown future for less time.

Given. Initial investment Tk 10,000; inflows of Tk 5,000 at the end of each of 3 years. *Find.* The payback period. *Formula.* With equal inflows, payback period (years) equals initial investment divided by the yearly inflow. *Steps.* $10,000 \div 5,000 = 2$. *Answer.* The payback period is 2 years. *Check.* Two years of Tk 5,000 equal Tk 10,000, so the money is recovered exactly at the end of year 2.

Real projects rarely give equal inflows, so we need a second method. Add the inflows year by year to make a running total, called the cumulative cash flow, starting from the negative initial investment. Find the year in which the total changes from negative to positive. Then use this rule: payback period equals the number of full years before recovery, plus the amount still unrecovered at the start of the recovery year, divided by the inflow of that year. This assumes the inflows arrive evenly through the year.

Case file, illustrative. 2 February 2025: Meghna and Sons compares two mutually exclusive projects. Project Press: initial investment Tk 900,000; inflows Tk 276,000 in years 1 to 4 and Tk 456,000 in year 5. Project Fit-out: initial investment Tk 900,000;

Independent projects are projects where accepting one has no effect on whether another is accepted. For example, the print-on-demand machine and a new set of shelves for the front room are independent: the shop may accept both, either or neither.

Mutually exclusive projects are projects where accepting one means rejecting the others. The machine and the branch fit-out are mutually exclusive, because the case file says they cannot be run together. Two clinics choosing between two scanners, when only one will be bought, face the same relation.

Before moving on. The process has five steps: find proposals, estimate cash flows, evaluate and choose, implement, and review. Projects are independent when one does not affect another, and mutually exclusive when choosing one rejects the rest.

* * *

7.3 CASH FLOW FROM AN INVESTMENT

Return to the rickshaw owner. He pays Tk 90,000 for the rickshaw today. Last month he paid Tk 2,000 for a mechanic's report about rickshaws, and that money cannot be recovered. He also pays Tk 500 a month for the garage where his three rickshaws stand, and he pays the same whether or not he buys a fourth. Which of these amounts should he count against the new rickshaw? The answer decides whether his forecast is honest.

RELEVANT CASH FLOW

A **relevant cash flow** is a cash flow that changes only because a project is accepted. If the garage rent stays at Tk 500 a month with or without the new rickshaw, it is not relevant. Three tests help. The item must be cash, meaning money that really moves, not an accounting entry. It must lie in the future, because money already spent cannot be recovered. And it must change if the project goes ahead; an item that would be the same either way is not relevant.

The second test needs a name. A sunk cost is money already spent that cannot be recovered, whatever the owner decides. Sunk costs do not change with the decision, so they are not relevant. Many students find this hard to accept, because the money feels lost if the project is rejected. But the money is lost in both cases, so it cannot help to choose between them.

Case file, illustrative. 5 January 2025: a feasibility survey for the print-on-demand machine, costing Tk 15,000, was paid in December 2024. It is a sunk cost and is not a relevant cash flow.

PAUSE AND CHECK 7.P1 A friend says the Tk 15,000 survey makes the machine less attractive, because the shop has already paid for it. Is the friend right? Answer in one or two sentences.

INITIAL INVESTMENT

Before a machine earns anything, someone must pay for it. The **initial investment** is the cash spent at the start to get a project going. It includes the price of the asset and everything needed to

Independent projects. Projects where accepting one has no effect on whether another is accepted.

Independent projects do not affect each other.

Mutually exclusive projects. Projects where accepting one means rejecting the others.

Mutually exclusive projects compete: one only.

Recap: five steps, and two kinds of relation between projects.

Measuring cash flow means deciding which money to count.

Relevant cash flow. Cash flow that changes only because a project is accepted.

Only cash that changes because of the project counts.

A cost already paid is not a reason for or against a project.

Case: the survey is sunk, so it is left out.

The first cash flow is what must be paid to start.

Initial investment. Cash

7.6 A machine with a book value of Tk 50,000 sells for Tk 70,000 at the end of its life. Tax is 25%, and Tk 20,000 of working capital returns. Find the terminal cash flow.

TERMINAL CASH FLOW

When the project ends, some cash may return. The **terminal cash flow** is cash received at the end of a project life, such as the resale value of equipment. It has two parts: the sale of the asset, and the working capital that comes back as stock is sold and customers pay what they owe. These amounts are not earned by operating, so they are kept apart from the operating cash flow.

Case file, illustrative. End of year 5 of the print-on-demand machine: resale Tk 120,000, equal to book value, so no tax on the sale; working capital of Tk 60,000 returned. Terminal cash flow Tk 180,000.

Here book value means the cost of the asset less the depreciation charged so far. If an asset sells for more than its book value, the gain is usually taxed. For example, a machine with a book value of Tk 120,000 that sells for Tk 140,000 has a gain of Tk 20,000, and tax at 25% is Tk 5,000, so the business keeps Tk 135,000. Rules for taxing such gains differ by country, so a student must always read the local rule.

Before moving on. A relevant cash flow changes only because the project is accepted; sunk costs do not count. The initial investment is paid at the start, the operating cash flow each year, and the terminal cash flow at the end. Operating cash flow equals EBIT, less tax, plus depreciation.

* * *

WORKED EXAMPLE: CASH FLOWS OF THE PRINT-ON-DEMAND MACHINE

The tailor's figures were small so that the method stood out. Now we use the same method on the machine for Meghna and Sons, in the standard order: Given, Find, Formula, Steps, Answer and Check.

Case file, illustrative. *Given.* 5 January 2025. Machine Tk 780,000; installation Tk 60,000; working capital Tk 60,000. Each year for five years: added revenue Tk 700,000; cash costs Tk 380,000; depreciation Tk 144,000, straight line over 5 years; tax at 25%, assumed for illustration only. End of year 5: resale Tk 120,000, equal to book value; working capital Tk 60,000 returned. The Tk 15,000 survey is a sunk cost.

Find. The net cash flow of each year from year 0 to year 5, counting relevant cash flows only. *Formula.* In words, net cash flow in a year equals the initial investment (year 0 only, shown as a negative number), plus the operating cash flow, plus the terminal cash flow (year 5 only). In symbols, $NCF = -I + OCF + TCF$, where NCF is net cash flow, I is the initial investment and TCF is terminal cash flow.

Steps. Step 1, the initial investment: $780,000 + 60,000 + 60,000 =$ Tk 900,000. The survey of Tk 15,000 is left out, because it is sunk. Step 2, a check on depreciation: the asset cost is $780,000 + 60,000 =$ Tk 840,000, so $(840,000 - 120,000) \div 5 = 720,000 \div 5 =$ Tk

Terminal cash flow. Cash received at the end of a project life, such as the resale value of equipment.

At the end of its life a project may return cash.

Case: resale and working capital make up the terminal cash flow.

Book value decides whether a sale creates a taxable gain.

Recap: four kinds of cash flow, and which ones count.

The same method now meets realistic figures.

Given: the facts of the case, as recorded.

Find and Formula: the question and the rule.

Steps 1 and 2: the initial investment and depreciation.

Table 7.1 collects the results: read by column and by row.

Table 7.1 gathers the results. Read each column as one year, and each row as one kind of cash flow. A negative number is cash paid out; a positive number is cash received.

NET CASH FLOW OF THE MACHINE, IN TAKA

ITEM	YEAR 0	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Initial investment	-900,000	0	0	0	0	0
Operating cash flow	0	276,000	276,000	276,000	276,000	276,000
Terminal cash flow	0	0	0	0	0	180,000
Net cash flow	-900,000	276,000	276,000	276,000	276,000	456,000

Answer: the shop pays Tk 900,000 now and receives Tk 276,000 a year, with Tk 456,000 in year 5.

Answer. The machine needs Tk 900,000 at the start, brings Tk 276,000 in each of years 1 to 4, and brings Tk 456,000 in year 5. *Check.* Run the second route for operating cash flow: $700,000 - 380,000 - 44,000 =$ Tk 276,000, the same as before. Adding the net cash flows gives $4 \times 276,000 + 456,000 - 900,000 =$ Tk 660,000. That sum ignores the time value of money, so it does not yet show whether the machine is worth buying; Chapter 8 completes that step. For the present value of level cash flows, which it will use, see Chapter 5, 'Annuities', section 5.3 'Present Value of an Annuity'.

Watch for three common slips. The first is to forget the sign: the initial investment is cash paid out, so it is negative. The second is to forget depreciation in the second move, so that EBIT is reported as the cash flow. The third is to mix periods, such as placing the working capital at the end of year 1 instead of at year 0. Writing the units, taka, on every line catches many errors.

* * *

SUMMARY

Capital budgeting is the process of planning, evaluating and choosing long-term investments. Such decisions are large, long and hard to reverse, so they follow five steps: find proposals, estimate cash flows, evaluate and choose, implement, and review. Projects may be independent or mutually exclusive. Only relevant cash flows count. The initial investment is paid at the start, the operating cash flow equals EBIT less tax plus depreciation, and the terminal cash flow arrives at the end.

THINK IT THROUGH

Two situations to test the chapter's ideas.

7.T1 A shop is considering a delivery van. Price Tk 560,000; registration and fitting Tk 40,000; working capital Tk 50,000. The van works for 4 years and then sells for Tk 100,000, equal to its book value. Each year it adds revenue of Tk 500,000 and cash costs of Tk 260,000. Depreciation is straight line on the asset cost less the resale value. Tax is 25%. Build the table of net cash flows for years 0 to 4.

PAUSE AND CHECK 9.P1 A list of outcomes gives probabilities of 0.2, 0.5 and 0.4. Is it a probability distribution? Say why.

Where do probabilities come from? A business may use its past records or the judgement of people who know the trade. Either way they are estimates, not promises. A careful manager states them openly so that other people can question them.

Three outcomes with three probabilities are hard to compare with another set. We want one number that sums up the whole distribution: the average outcome. The plain average of 10, 25 and 40 minutes treats each outcome as equally likely, which is wrong, because 25 minutes happens twice as often as each of the others. We must weight each outcome by its probability.

Expected value is the average outcome, weighting each outcome by its probability. In words: multiply each outcome by its probability, then add the products. In symbols, with x for an outcome and p for its probability, expected value = $(p_1 \times x_1) + (p_2 \times x_2) + (p_3 \times x_3)$. Many students find this idea confusing, because *expected* sounds like a prediction. It is a weighted average and promises nothing about any one day.

EXPECTED TIME OF TWO ROUTES

TRAFFIC	PROBABILITY	ROUTE A MINUTES	P × MINUTES	ROUTE B MINUTES	P × MINUTES
Clear	0.25	20	5.0	10	2.5
Usual	0.50	25	12.5	25	12.5
Heavy	0.25	30	7.5	40	10.0
Total	1.00		25.0		25.0

Given. The distributions of the two routes in the table. *Find.* The expected time of each route. *Formula.* Expected value equals the sum of probability × outcome. *Steps.* Route A: $0.25 \times 20 = 5.0$; $0.50 \times 25 = 12.5$; $0.25 \times 30 = 7.5$; the sum is 25.0 minutes. Route B: $0.25 \times 10 = 2.5$; $0.50 \times 25 = 12.5$; $0.25 \times 40 = 10.0$; the sum is 25.0 minutes. *Answer.* Both routes have an expected time of 25 minutes. *Check.* An expected value must lie between the smallest and the largest outcome: 25 lies between 20 and 30, and between 10 and 40.

Two limits matter. First, the expected value is the average over many repeats, not a forecast for one day. A coin that pays Tk 0 or Tk 10 with equal probability has an expected value of Tk 5, which can never be paid on one toss. Second, the answer is only as good as the probabilities. Common slips are probabilities that do not add up to one, and a plain average used in place of the weighted one.

Before moving on. A probability is a number from 0 to 1. A probability distribution lists the outcomes with their probabilities, which add up to one. The expected value weights each outcome by its probability. *You should now be able to say* what an expected value is and why both routes have the same one.

* * *

Probabilities come from records or from judgement, so they are estimates.

A whole distribution needs to be summed up in one number.

Expected value. the average outcome, weighting each outcome by its probability

Expected value multiplies each outcome by its probability and adds the products.

The table sets out the weighted calculation for both routes, one row for each kind of day.

The worked Steps repeat the table in the standard order.

Expected value is a long-run average, and it need not be an outcome that can happen.

The section's ideas are probability, the distribution and the expected value.

Financial leverage raises financial risk, and it helps only while EBIT covers the charges.

The section's ideas are fixed financing charges, the two kinds of stock and financial leverage.

Meghna and Sons Limited provides a full data set for measuring the swing.

Earnings per share. profit left for common stock owners divided by the number of common shares

Earnings per share is the profit for the common stock owners, shared out per common share.

Five subtractions and one division take EBIT to earnings per share.

A rise in EBIT shows the swing: earnings per share moves by much more.

Financial leverage is the cause of the financial risk that Chapter 9 defined: the extra risk to owners that comes from borrowing.² Two limits follow. First, the lever helps only while EBIT is high enough to cover the charges. If EBIT falls below the interest, the owners make a loss on the unit. Second, borrowing is not wrong in itself; the question is how much swing the owners can bear.

Before moving on. A fixed financing charge is interest or preferred dividend owed at any level of EBIT. Common stock receives what remains, and preferred stock receives a fixed dividend first. Financial leverage is the extra swing in profit for each share that these charges cause.

* * *

12.2 DEGREE OF FINANCIAL LEVERAGE

Case file, illustrative. Meghna and Sons Limited, 1 July 2025. The business was registered to own the shop and the printing unit, and the 8,000-pack data of Chapter 11 is repeated. Sales are 8,000 packs at Tk 90 each. Variable cost is Tk 50 a pack. Fixed operating cost is Tk 240,000. EBIT is therefore Tk 80,000. The printing unit is financed by a bank loan of Tk 200,000 at 10%, giving interest of Tk 20,000, and by Tk 100,000 of preferred stock at 12%, giving a dividend of Tk 12,000. There are 5,000 common shares. The tax rate is 25%.

The owners of common stock care about what is left for them. **Earnings per share** is the profit left for common stock owners divided by the number of common shares. If Tk 33,000 is left and there are 5,000 shares, each share has a claim on Tk 6.60. To find the profit left, we work down from EBIT, taking away each claim in the order of the table above.

Given. The case file above. *Find.* Earnings per share. *Formula.* In words: take EBIT, subtract interest to reach profit before tax, subtract tax, subtract the preferred dividend, and divide by the number of common shares. In symbols, with I for interest, T for the tax rate, D_p for the preferred dividend and N for the number of common shares: earnings per share = $[(EBIT - I) \times (1 - T) - D_p] \div N$. *Steps.* Profit before tax: $80,000 - 20,000 = \text{Tk } 60,000$. Tax at 25%: $0.25 \times 60,000 = \text{Tk } 15,000$. Profit after tax: $60,000 - 15,000 = \text{Tk } 45,000$. Less preferred dividend: $45,000 - 12,000 = \text{Tk } 33,000$. Divide by shares: $33,000 \div 5,000 = \text{Tk } 6.60$. *Answer.* Earnings per share is Tk 6.60. *Check.* The profit left, Tk 33,000, is below profit after tax, Tk 45,000, as it must be, because the preferred owners were paid first.

Now raise EBIT. At 9,000 packs, Chapter 11 gives EBIT of Tk 120,000. Profit before tax is $120,000 - 20,000 = \text{Tk } 100,000$; tax is Tk 25,000; profit after tax is Tk 75,000; less the preferred dividend of Tk 12,000 leaves Tk 63,000. Earnings per share is $63,000 \div 5,000 = \text{Tk } 12.60$. EBIT rose by 50%, from Tk 80,000 to Tk 120,000, but earnings per share rose from Tk 6.60 to Tk 12.60, which is a rise of $6.00 \div 6.60 = 90.9\%$.

² Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', defined financial risk.

9.4 MEASURING THE SPREAD OF OUTCOMES

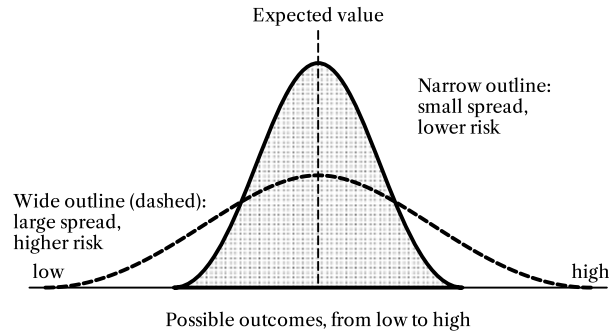


FIG. 9.1. Two sets of outcomes with the same expected value but different spread

Standard deviation is stated in the units of the outcomes, so it fits a comparison only when the expected values are equal. A standard deviation of Tk 8,000 is small beside an expected value of Tk 500,000 and large beside one of Tk 10,000. We need risk for each unit of return.

The **coefficient of variation** is the standard deviation divided by the expected value, showing risk for each unit of return. In symbols, coefficient of variation = standard deviation ÷ expected value. For route A it is $3.54 \div 25 = 0.14$, and for route B it is $10.61 \div 25 = 0.42$. A larger figure means more risk for each unit of the expected result. The coefficient cannot be used when the expected value is zero or negative.

Before moving on. The standard deviation measures spread around the expected value in four steps: deviate, square, weight and add, then take the root. The coefficient of variation divides it by the expected value so that choices of different size can be compared. Common slips are forgetting the root and forgetting to weight by probability. *You should now be able to say* what the standard deviation measures and why we divide it by the expected value.

* * *

WORKED CASE: TWO PRODUCT LINES

The case file gives the facts. We will find the expected value, the standard deviation and the coefficient of variation for each line, and then judge.

Case file, illustrative. Meghna and Sons, 2 March 2025. Two product lines are compared on Tk 100,000 invested in each. Trade is slow with probability 0.25, normal with probability 0.50 and busy with probability 0.25. Study guides return 9%, 12% and 15% in those states. Art supplies return 2%, 14% and 26%.

To read the figure, look at the three groups of bars: slow, normal

Standard deviation has a limit when the expected values of two choices differ.

Coefficient of variation. the standard deviation divided by the expected value, showing risk for each unit of return

The coefficient of variation divides the standard deviation by the expected value.

The section ends with a recap and a test sentence.

Meghna and Sons compares two product lines, using every tool of the chapter.

The case file gives the trade states, their probabilities and the returns of each line.

The figure shows the returns of

Two printing units with the same operating profit react very differently when sales rise.

The answer is a lever: a fixed payment that does not change while profit does.

This chapter measures the lever in three steps.

The chapter extends the data set of Chapter 11 and the idea of financial risk.

A tea stall shows the effect of borrowing in the smallest numbers.

In a poor year the same loan turns the gain into a loss.

Charge. Word in use. In daily life, a price asked for a service, or a claim made against a person. In finance, a cost that a business must pay.

Fixed financing charge. interest or preferred dividend

Picture two printing units. Both sell the same number of packs, both have the same costs, and both make an operating profit of Tk 80,000. The first was paid for entirely with the owners' own money. The second was paid for partly with a bank loan. Sales rise by 12.5%, and in each unit the operating profit rises by 50%. The owners of the first unit see their profit rise by half. The owners of the second see theirs nearly double. Why?

The difference lies in what the second unit must pay to the bank. The bank wants its interest in a good year and in a poor year alike. When operating profit rises, the owners keep all of the rise, because the interest stays the same. A small hand pressing a long lever can lift a heavy weight, and a fixed payment works in the same way. But a lever works downward as well as upward, and the owners of the second unit would feel a fall equally sharply.

We take three steps. First, we see why fixed payments to lenders and to a second kind of owner make the profit for each share swing more widely. Second, we measure that swing with one number. Third, we join it to the swing from operating costs that Chapter 11 measured, and read the combined number as a sign of risk.

You will need Chapter 11, 'Operating Leverage and Break-even Analysis', sections 11.2 and 11.3, which set out the printing unit data and the degree of operating leverage. You will also need Chapter 9, 'Return and Risk', section 9.2 'What Risk Is', where financial risk was defined.

* * *

12.1 BORROWING AND FIXED FINANCING CHARGES

Suppose you plan a tea stall that costs Tk 10,000. Option one: you pay the whole sum from your savings. Option two: you pay Tk 5,000 and borrow Tk 5,000, and the loan costs Tk 500 in interest each year. In a good year the stall makes Tk 2,000 before any interest. Under option one you keep Tk 2,000, which is 20% of your Tk 10,000. Under option two you keep Tk 2,000 - Tk 500 = Tk 1,500, which is 30% of your own Tk 5,000. Borrowing has lifted your gain, measured on your own money.

Now suppose a poor year, when the stall makes only Tk 400 before interest. Under option one you keep Tk 400, which is 4% of your Tk 10,000. Under option two you owe Tk 500 and receive only Tk 400, so you are Tk 100 short: a loss of 2% on your own Tk 5,000. The loan helped in the good year and hurt in the poor year, because the Tk 500 did not change. The rest of this section gives that fixed payment its name.

A **fixed financing charge** is interest or preferred dividend that must be paid whatever the level of EBIT. *Financing* means paying for a business with funds from outside the day-to-day trade. The tea stall's Tk 500 is a fixed financing charge. Do not confuse it with the fixed operating cost of Chapter 11, such as rent. Both stay the same when sales change. The operating cost is paid to run the

Read each row across. Q has the highest expected value. P has the lowest risk for each unit of return, at 0.25. R has the highest, at 0.50, yet its expected value lies below Q's, so R is the weakest choice of the three on both counts.

* * *

SUMMARY

Return is the gain or loss on an investment, and the rate of return scales it by the amount invested. Risk is the chance that results differ from those expected. A probability distribution lists outcomes with probabilities adding to one. The expected value is its weighted average, the standard deviation measures the spread around it, and the coefficient of variation shows the risk for each unit of return.

Asset Q has the highest return, and asset R has the most risk for each unit of return.

The chapter's tools are return, risk, probability, expected value, standard deviation and the coefficient of variation.

THINK IT THROUGH

9.T1 Project X has an expected value of Tk 50,000 and a standard deviation of Tk 10,000. Project Y has an expected value of Tk 80,000 and a standard deviation of Tk 24,000. Which carries more risk for each unit of return? Show the figures.

WHERE PEOPLE GO WRONG

Error: "Risk is the chance of losing money." Risk is the chance that results differ from those expected. A result better than expected is also a difference, and the standard deviation counts it. A certain loss carries no risk at all.

Error: "The expected value is what will happen." It is a weighted average over many repeats. In one trial the actual result may differ widely, and may never equal the expected value.

Error: "The line with the smaller standard deviation is the safer one." A smaller standard deviation is safer only when the expected values are equal. When they differ, compare the coefficients of variation. Also check that the probabilities add up to one, and do not forget the square root.

A common error is to think that risk means a loss.

A second error is to treat the expected value as a forecast.

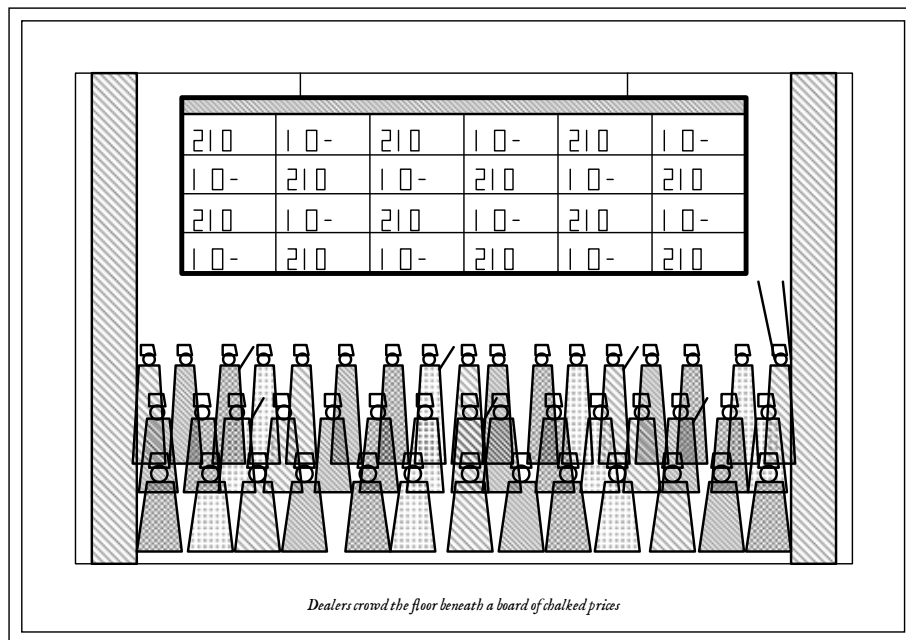
A third error is to compare standard deviations when the expected values differ.

IN HISTORY

Edward Lloyd kept a coffee house in Tower Street, in the City of London near the River Thames. The London Gazette first mentioned it in 1688. Lloyd did not insure ships himself. He provided premises and reliable news of shipping, and the ship owners, merchants and captains who met there arranged insurance among themselves.¹ Each wealthy person took a share of the risk of a ship and its cargo, and signed his name beneath the policy with the amount he would cover. For this reason such people were called underwriters. No single storm could ruin one of them, because each had placed only a part of the cargo at risk.

Merchants who shared the risk of ships began the great insurance market of London.

¹ The date 1688 is the first recorded mention of the coffee house. Its exact opening date is not known.



CHAPTER X

Risk, Required Return and Market Efficiency

Why must an uncertain gain promise more than a certain one?



This chapter explains why a risky investment must promise more than a safe one, and how much more. You will separate risk that can be removed from risk that cannot, measure how strongly an investment moves with the whole market, and turn that measure into a required return. The chapter ends with the question of how markets use information.

The more of a business's costs are fixed, the higher its leverage, the larger its swings in EBIT, and the higher its business risk.

Fixed costs raise risk because they cannot be reduced when sales fall.

The notebook seller's numbers are used to find break-even and leverage at two volumes in one problem.

The steps compute the contribution margin, then the break-even point, then EBIT and leverage at each volume.

The answer and check show leverage is higher close to break-even and agree with a direct calculation.

Fixed and variable costs lead to a break-even point and a degree of operating leverage, which together describe business risk.

A printer, separate from ...

cuts it by 50%. The fixed cost of Tk 240,000 must be paid in a poor year and in a good one alike.

The rule follows. The larger the share of fixed cost in total cost, the higher the degree of operating leverage, the larger the swings in EBIT, and the higher the business risk. Owners may accept this in return for larger gains in good years. Lenders worry that EBIT may fall too low to pay what is owed. Employees may fear lost jobs when sales fall. Operating leverage is only one source of business risk, because demand and input prices also change.

Before moving on. Higher fixed cost means higher leverage and a larger swing in EBIT in both directions. That swing is why operating leverage raises business risk.

* * *

WORKED EXAMPLE: BREAK-EVEN AND LEVERAGE TOGETHER

Given. Selling price Tk 20, variable cost Tk 12, fixed cost Tk 8,000 a year. *Find.* The break-even point, and the EBIT and degree of operating leverage at 1,200 and at 2,000 notebooks, then test the result with a rise of 10% in sales from 2,000. *Formula.* $Q = F \div CM$, $EBIT = Q \times CM - F$, and $DOL = (Q \times CM) \div EBIT$.

Steps. $CM = 20 - 12 = Tk\ 8$. Break-even $Q = 8,000 \div 8 = 1,000$ notebooks. At 1,200 notebooks, contribution = $1,200 \times 8 = Tk\ 9,600$, $EBIT = 9,600 - 8,000 = Tk\ 1,600$, and $DOL = 9,600 \div 1,600 = 6.00$. At 2,000 notebooks, contribution = $2,000 \times 8 = Tk\ 16,000$, $EBIT = 16,000 - 8,000 = Tk\ 8,000$, and $DOL = 16,000 \div 8,000 = 2.00$.

Answer. Break-even is 1,000 notebooks. The degree of operating leverage is 6.00 at 1,200 notebooks and 2.00 at 2,000 notebooks. *Check.* A rise of 10% in sales from 2,000 gives 2,200 notebooks. EBIT is then $2,200 \times 8 - 8,000 = 17,600 - 8,000 = Tk\ 9,600$. The rise is 1,600 on 8,000, which is 20%, and $10\% \times 2.00 = 20\%$. The volume closer to break-even has the higher leverage.

* * *

SUMMARY

A fixed operating cost does not change with units sold, and a variable operating cost does. The contribution margin is price minus variable cost per unit. The operating break-even point equals fixed cost divided by contribution margin. EBIT equals units sold times contribution margin, minus fixed cost. The degree of operating leverage equals contribution divided by EBIT; it is highest near break-even and shows how strongly EBIT responds to sales. Greater leverage raises business risk.

THINK IT THROUGH

A printer sells at Tk 90 per pack. Today the variable operating cost

return at which the investor accepts the danger. A return below it does not pay enough for the risk; a return above it leaves a gain.

Given. Risk-free rate 8%; the lender demands 12% from the tea seller. *Find.* The risk premium. *Formula.* Risk premium equals required return less risk-free rate. *Steps.* $12\% - 8\% = 4\%$. *Answer.* The risk premium is 4 percentage points. *Check.* Adding the premium back to the risk-free rate, $8\% + 4\%$, gives 12%, which is the required return we began with.

The table sets three loans in order of rising danger. Read each row from left to right: the borrower, the premium the lender demands, and the required return, which is the risk-free rate of 8% plus that premium. The figures are illustrative.

REQUIRED RETURN RISING WITH RISK (ILLUSTRATIVE FIGURES)

BORROWER	RISK PREMIUM	REQUIRED RETURN
Government, three-month bill	0%	8%
Long-established bookshop	2%	10%
New tea seller with a cart	4%	12%

The viewpoints differ. The lender asks what extra return would make the loan worth the danger. The borrower sees the premium as a cost of funds. The financial manager sees both, because the required return is the benchmark for judging a project, as Chapter 8 taught,² and the manager must therefore know what premium to add.

Two limits matter. First, the premium is what investors *demand*, not what they receive. An investment that promises 12% may pay 12%, more, or nothing; the premium is the reason investors accept the danger, not a guarantee. Second, the premium is not the same for every investor or in every period. It rises when people grow fearful and falls when they grow confident. A common slip is to call the premium itself the required return, and to forget to add the risk-free rate.

Before moving on. The risk-free rate pays for waiting alone. The risk premium is the extra that investors demand for risk, and the required return is the sum of the two. *You should now be able to say* why a loan to a tea seller must offer more than a government bill.

* * *

10.2 DIVERSIFICATION AND BETA

Imagine a trader who sells umbrellas. In a wet week the profit is Tk 20,000, and in a dry week it is nothing. Each kind of week is equally likely, so the expected value is Tk 10,000 a week, with a wide spread around it. A long dry spell could ruin the trader. Now suppose the trader puts half the money into a sunshade stall. In a wet week the umbrellas earn Tk 10,000 and the sunshades nothing in a dry week the reverse. Either way the profit is Tk 20,000. Chapter 8, 'Capital Budgeting: Evaluation Techniques', section 8.2 'Net Present Value', uses the required return as the benchmark for a project.

A short worked example shows the two steps in the standard order.

The table lists three loans in order of rising danger, with the premium rising too.

Each person who sees the same loan sees a different side of the premium.

The premium is a demand and not a promise, and it changes with the mood of the market.

The section's ideas are the risk-free rate, the risk premium and the required return.

GO SLOWLY HERE.

Two stalls that suffer in opposite weather make a steadier living than one.

Break-even analysis assumes that price, variable cost and fixed cost stay the same across the volumes studied.

The break-even point is found by dividing the fixed cost by the contribution margin, and EBIT rises with every unit above it.

At the printing unit, a rise in sales of one eighth lifts EBIT by one half; we can now explain why.

Operating leverage. the effect of fixed operating costs in making EBIT change by more than sales change

Operating leverage means fixed costs make EBIT move by a larger percentage than sales.

Degree of operating leverage. the percentage change in EBIT caused by a one percent change in sales
The degree of operating leverage puts a number on the effect: the percentage change in EBIT per one percent change in sales.

The worked Steps find the degree of operating leverage at 8,000 packs and at 9,000 packs.

The case file records both values, and the check agrees with the percentages in the puzzle.

The break-even rule has limits. It assumes that price and variable cost per unit do not change as volume grows, and that the fixed cost stays the same across the range studied. A manager who plans to sell far more than today may meet a new fixed cost, such as a second machine, and then the rule must be applied again with the new figures.

Before moving on. The operating break-even point equals the fixed operating cost divided by the contribution margin. For the printing unit it is 6,000 packs. Below it EBIT is negative, and above it EBIT rises by the contribution margin with each extra pack.

* * *

11.3 DEGREE OF OPERATING LEVERAGE

Return to the puzzle that opened the chapter. At 8,000 packs EBIT is Tk 80,000, and at 9,000 packs it is Tk 120,000. Why does a rise of 12.5% in sales give a rise of 50% in EBIT? Each extra pack adds Tk 40 of contribution, and no extra fixed cost. The extra 1,000 packs therefore add Tk 40,000, and all of it reaches EBIT. Measured against a base EBIT of only Tk 80,000, that is a rise of 50%.

The word comes from the lever, where a small push at one end moves a heavy load at the other. **Operating leverage** is the effect of fixed operating costs in making EBIT change by more than sales change. A business with no fixed operating cost has none, because its EBIT would change by the same percentage as its sales. The larger the fixed cost, the stronger the effect.

We want one number that measures the strength of this effect. The **degree of operating leverage** is the percentage change in EBIT caused by a one percent change in sales. A value of 4.00 means that each 1% change in sales changes EBIT by 4%. It is found from contribution and EBIT at one starting volume. In words: degree of operating leverage equals contribution divided by EBIT. In symbols, $DOL = (Q \times CM) \div EBIT$. The formula arises because a 1% change in sales changes contribution by 1% of $Q \times CM$, all of which reaches EBIT, and dividing by EBIT states that change as a percentage of EBIT.

THE PRINTING UNIT AT TWO VOLUMES

Given. Price Tk 90, variable cost Tk 50, contribution margin Tk 40, fixed cost Tk 240,000. *Find.* The degree of operating leverage at 8,000 packs and at 9,000 packs. *Formula.* $DOL = (Q \times CM) \div EBIT$, where Q is the starting volume. *Steps.* At 8,000 packs, contribution = $8,000 \times 40 = \text{Tk } 320,000$, EBIT = $320,000 - 240,000 = \text{Tk } 80,000$, and $DOL = 320,000 \div 80,000 = 4.00$. At 9,000 packs, contribution = $9,000 \times 40 = \text{Tk } 360,000$, EBIT = $360,000 - 240,000 = \text{Tk } 120,000$, and $DOL = 360,000 \div 120,000 = 3.00$.

Case file, illustrative. Meghna and Sons, 6 April 2025, printing unit. Degree of operating leverage at 8,000 packs = $320,000 / 80,000 = 4.00$. At 9,000 packs = $360,000 / 120,000 = 3.00$. A 12.5%

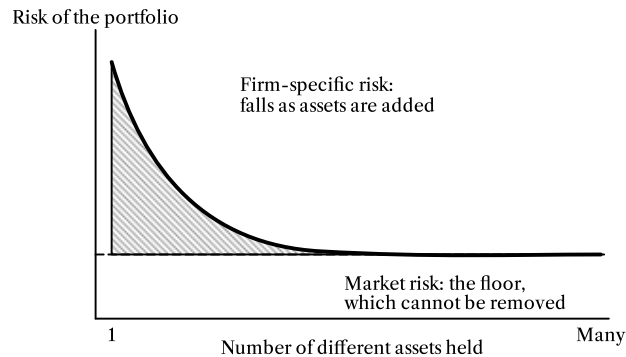


FIG. 10.1. Risk of a portfolio falling as different assets are added, then levelling at a floor

Here is the idea that carries the rest of the chapter. Because an investor can remove firm-specific risk by diversifying, the market does not pay extra for bearing it. Only market risk earns a premium. To find a fair required return for one investment, we must therefore measure how much market risk it carries. There is one exception to remember: a family that owns a single bookshop cannot diversify its wealth, so for that family firm-specific risk is real. The theory fits best an investor who holds many different assets.

Picture ten shops in one bazaar. When the number of visitors rises by 10%, a shop selling festival goods may see its sales rise by 15%, and a shop selling rice may see a rise of only 6%. Each shop follows the crowd, but by a different amount. The crowd plays the part of the market, and the amount of following is what we want to measure.

The **beta** of an investment is a number showing how strongly it moves with the whole market. By agreement the whole market has a beta of exactly 1. An investment with a beta of 1.5 tends to move one and a half times as far as the market: when the return of the market rises by 10%, the return of this investment tends to rise by about 15%, and when the market falls by 10%, it tends to fall by about 15%. An investment with a beta of 0.6 moves about six-tenths as far.

The table lists four betas. Read each row from left to right: the beta, what it means, and the usual movement of the investment when the return of the market changes by 10%.

HOW TO READ A BETA

BETA	MEANING	USUAL MOVEMENT FOR A 10% MARKET CHANGE
0	Does not move with the market	0%
0.6	Moves less than the market	6%

Investors are paid for market risk and not for risk that anyone can remove.

Several shops in one bazaar follow the crowd by different amounts.

GO SLOWLY HERE.

Beta, a number showing how strongly an investment moves with the whole market

Beta compares the movement of an investment with the movement of the whole market.

The table gives four betas and what each means.

* * *

11.2 THE OPERATING BREAK-EVEN POINT

The seller wants to know how many notebooks must be sold before the loss ends.

Operating break-even point, the sales volume at which EBIT is exactly zero

The operating break-even point is the volume at which EBIT is exactly zero.

Return to the seller who lost Tk 4,000 on 500 notebooks. Each notebook gives Tk 8 of contribution, and the fixed cost of Tk 8,000 must be covered first. How many notebooks are needed? Each one pays Tk 8 towards Tk 8,000, so we need $8,000 \div 8 = 1,000$ notebooks. At that volume the seller neither gains nor loses. Below it the seller has a loss, and above it a profit.

The **operating break-even point** means the sales volume at which EBIT is exactly zero. For the seller it is 1,000 notebooks. It is measured in units, such as notebooks or packs, and not in taka. The word *operating* reminds us that we look at EBIT, before interest and tax.

THE BREAK-EVEN RULE

EBIT equals contribution minus fixed cost. EBIT is zero when contribution equals fixed cost, so break-even volume equals fixed operating cost divided by contribution margin.

CLEAN NUMBERS FIRST

Given. Selling price Tk 20 per notebook, variable operating cost Tk 12 per notebook, fixed operating cost Tk 8,000 a year. *Find.* The operating break-even point.

Formula. In words: break-even volume equals fixed operating cost divided by contribution margin. In symbols, $Q = F \div CM$. Here Q is the number of units at break-even, F is the fixed operating cost, and CM is the contribution margin per unit, which is the price minus the variable cost per unit.

Steps. First, $CM = 20 - 12 = \text{Tk } 8$ per notebook. Second, $Q = 8,000 \div 8 = 1,000$ notebooks. *Answer.* The seller reaches break-even at 1,000 notebooks a year. *Check.* At 1,000 notebooks, revenue is $20 \times 1,000 = \text{Tk } 20,000$. Total operating cost is $8,000 + 12 \times 1,000 = \text{Tk } 20,000$. EBIT is $20,000 - 20,000 = 0$, as it should be.

PAUSE AND CHECK 11.P1 A stall has a contribution margin of Tk 25 per unit and a fixed operating cost of Tk 5,000 a year. How many units must it sell to reach the operating break-even point?

EBIT AT DIFFERENT VOLUMES

The break-even point is one volume out of many, so it helps to see EBIT on both sides of it. The table below uses the notebook seller. Each row multiplies the units sold by the contribution margin of Tk 8, then subtracts the fixed cost of Tk 8,000. Read across a row to follow that calculation.

EBIT OF THE NOTEBOOK SELLER AT THREE VOLUMES

NOTEBOOKS SOLD	CONTRIBUTION (TK)	FIXED COST (TK)	EBIT (TK)
800	6,400	8,000	-1,600

To read the next figure, find an investment by its beta on the horizontal axis. Go up to the line, then across to the vertical axis. The height of the dot is the required return for that beta. An investment whose expected return lies above the line promises more than it must; one below the line promises less.

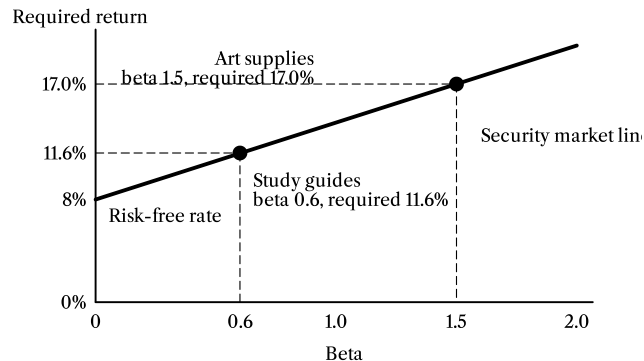


FIG. 10.2. The security market line, with the required returns of two product lines

The model gives a rule for decisions. Accept an investment when its expected return is equal to or above its required return, which means it lies on or above the line. Reject it when the expected return is below the required return. This is the same test as Chapter 8 used with net present value: a project is worth accepting when it earns at least the benchmark.

The model has limits. It assumes that investors hold diversified portfolios and that one market return and one beta explain required returns. Betas change over time, and the market return and risk-free rate are estimates, so different estimates give different required returns. Even so, it is the standard first tool, because it turns a feeling about risk into a number that can be argued about.

Before moving on. The capital asset pricing model sets required return as the risk-free rate plus beta times the market risk premium. The security market line draws it as a straight line. *You should now be able to say* how a higher beta raises the required return.

* * *

10.4 EFFICIENT MARKETS

Suppose a business announces a large new order. When should the price of its common stock rise: before the news, when the news is announced, or a week later? If buyers and sellers are quick and well informed, the price should move at the moment of the announcement, and not after. A reader who waits a week to buy finds that the gain has gone.

An **efficient market** is a market in which prices already reflect

The figure places two investments on the line by their betas.

The decision rule compares the expected return with the line.

The model rests on simplifying assumptions, so it guides and does not prove.

The section's ideas are the pricing formula and the line that draws it.

If a business announces good news, when should its share price change?

Efficient market. a market in which prices already

Sales rise by one eighth, yet profit rises by one half; the reason is the subject of this chapter.

A tea seller shows the same effect: once the rent is paid, each extra cup adds mostly profit.

The chapter answers three questions: where the loss ends, how strongly profit responds to sales, and what this means for risk.

A business pays some costs whatever it sells and other costs only as it sells.

Consider two numbers from the case file of the printing unit of Meghna and Sons, the family-run bookshop in Dhaka that appears throughout this book. In one scenario the unit sells 8,000 packs in a year and earns an EBIT of Tk 80,000. EBIT is the profit of a business before interest and tax are taken away.¹ In the next scenario the unit sells 9,000 packs. Sales have grown by one eighth, which is 12.5%. A reader might expect EBIT to grow by about the same share. It does not. EBIT rises from Tk 80,000 to Tk 120,000, a rise of 50%.

Think of a tea seller who rents a stall. The rent is the same on a quiet day and on a busy day. The tea leaves and milk, however, are used up one cup at a time. On a quiet day the seller may barely cover the rent. On a busy day, once the rent is covered, almost every extra cup adds to the seller's profit. A rise in cups sold of one tenth can therefore raise profit by far more than one tenth. The printing unit behaves in the same way, and so does every business that has costs which do not move with sales.

This chapter answers three questions. First, how many units must a business sell before it stops making a loss? Second, by what percentage does profit change when sales change by one percent? Third, why does the answer matter to anyone who owns, lends to or works for the business? We shall build each answer one idea at a time, starting with the way costs behave.

YOU WILL NEED

Chapter 9, 'Return and Risk', section 9.2 'What Risk Is': business risk. Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', section 7.3 'Cash Flow from an Investment': EBIT.

* * *

11.1 FIXED AND VARIABLE COSTS

Start with a small business that you could run yourself. A seller rents a table in a market for Tk 8,000 a year and sells notebooks. Each notebook is bought from a wholesaler for Tk 12 and sold for Tk 20. The seller soon notices that two kinds of money leave the business. One kind is the same whether the table sells ten notebooks or a thousand. The other kind grows with every notebook sold. Businesses of every size, from this table to a large factory, contain both kinds, and separating them is the first step of this chapter.

¹ Recall: EBIT, Chapter 7, 'Capital Budgeting: Decisions and Cash Flows', section 7.3 'Cash Flow from an Investment'. From here on the book uses the short form EBIT.

11.6%. The expected return is higher, so accept. *Check.* A beta between 0 and 1 must give a required return between 8% and 14%, and 11.6% does.

Given. Art-supplies beta 1.5; expected return 14%; same rates. *Find.* The required return, and whether to accept. *Steps.* Beta times premium: $1.5 \times 6\% = 9.0\%$. Required return: $8\% + 9.0\% = 17.0\%$. Compare: expected return 14% is below 17.0%. *Answer.* The required return is 17.0%. The expected return is lower, so reject. *Check.* A beta above 1 must give a required return above the market return of 14%, and 17.0% does.

Chapter 9 found that art supplies had the higher expected return, 14% against 12%. Here the higher return does not pay for the higher market risk. The two lines also agree with the standard deviation of Chapter 9, which found art supplies the riskier, although beta and standard deviation measure different things. The owner sees a line that fails the test, and the financial manager sees the reason: its return is too low for its beta.

The table combines the two results. Read each row from left to right: the beta, the required return from the line, the expected return, and the line's position.

BOTH LINES AGAINST THE SECURITY MARKET LINE

LINE	BETA	REQUIRED RETURN	EXPECTED RETURN	POSITION AND DECISION
Study guides	0.6	11.6%	12%	0.4 points above the line: accept
Art supplies	1.5	17.0%	14%	3.0 points below the line: reject

Watch for three slips. Some students multiply beta by the market return and not by the premium. Some forget to add the risk-free rate at the end. Some compare the expected return with the market return and not with the required return.

* * *

SUMMARY

The required return equals the risk-free rate plus a risk premium. Diversification removes firm-specific risk but not market risk, so only market risk earns a premium. Beta shows how strongly an investment moves with the market. The capital asset pricing model sets required return as the risk-free rate plus beta times the market risk premium, and the security market line draws it. An efficient market is one whose prices already reflect the information available.

THINK IT THROUGH

10.T1 The risk-free rate is 8% and the market return is 14%. A stationery line has a beta of 1.0 and an expected return of 13%. Find the required return, say whether the line lies above or below the

The second line is worked the same way and reaches the opposite decision.

Ranking by expected return alone would have chosen the wrong line.

The table places both lines against the security market line.

Common slips arise at each step of the formula.

The chapter's rule is that only market risk earns a premium, and beta sets its size.

A second error is to think that a high beta marks a bad investment.

A third error is to think that an efficient market is always right.

A fourth error is to treat the required return as a forecast.

On one day in 1987 the market fell by about a fifth, and no single business was to blame.

The crash tested how markets take in information.

Sharpe, 1964, and Fama, 1970, are the landmark works for this chapter.

Error: "A high beta means a bad investment." A high beta means that the investment moves strongly with the market, in both directions. It raises the required return; it does not make the investment good or bad. The test is whether the expected return meets the required return.

Error: "An efficient market never makes mistakes." Efficiency means prices reflect the information available, which may be incomplete. Prices can still fall sharply, and investors can still lose.

Error: "The required return is what the investment will earn." It is the benchmark: the least return that pays for the market risk. The expected return is the estimate of what the investment may earn.

IN HISTORY

On Monday 19 October 1987, the Dow Jones Industrial Average, an index that follows the share prices of 30 large United States businesses, fell by 508 points, a little over 22% in one day.⁴ Markets in many other countries fell as well. Nearly every business lost value together, whatever news it had announced, which is what market risk means: an investor who held many different businesses could not escape the fall.

The crash also tested the idea of an efficient market. No single piece of public news seemed large enough to explain so great a fall in a day, and economists still debate the causes; a presidential commission later pointed to trading by computer programs among them. Yet the market absorbed the shock within a day, and the index regained its loss in about two years.

SOURCES

William Sharpe, "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk" (*Journal of Finance*, 1964), set out the model that links required return to beta. Eugene Fama, "Efficient Capital Markets: A Review of Theory and Empirical Work" (*Journal of Finance*, 1970), gave the three forms of efficiency their names and reviewed the evidence. Harry Markowitz, 1952, whose work is described in Chapter 9, supplied the idea of measuring risk by spread.

REVIEW QUESTIONS

10.1 The risk-free rate is 7%, the market return is 13% and an investment has a beta of 1.2. Find the required return.

10.2 Explain why diversification lowers some risk but not all.

10.3 Draw a security market line with a risk-free rate of 6% and a market return of 10%, with beta from 0 to 2. Read from it the required return at a beta of 0.5 and at a beta of 1.5.

10.4 Name the three forms of market efficiency and give one example for each of information that would not help an investor if

⁴ Sources round the figures differently; the index closed at about 1,739, a fall of about 22.6%.

higher number means a wider swing in earnings per share for the same change in sales.

The two levers are not equally easy to change. The trade fixes much of a business's operating leverage: a printing unit needs machines whatever the owners prefer. The financing is a choice. A business with a high degree of operating leverage may therefore choose to borrow little, to keep the total from growing too large. A business with steady sales and little operating leverage can bear more borrowing.

Before moving on. Total leverage joins business risk and financial risk. Because the financing is a choice, owners can lower the total by borrowing less.

* * *

WORKED PROBLEM: DEBT REPLACED BY EQUITY

An illustrative business, not the case file, has EBIT of Tk 80,000, a contribution margin of Tk 240,000 and no preferred stock; the tax rate is 25%. Under Plan A it borrows Tk 200,000 at 10% (interest Tk 20,000) and has 5,000 common shares. Under Plan B it replaces the loan by selling 2,000 more common shares at Tk 100 each, giving 7,000 shares and no interest. We compare the plans.

Given. The data above. *Find.* Earnings per share, the degree of financial leverage and the degree of total leverage for each plan, then the effect of a 50% change in EBIT. *Steps for Plan A.* Earnings per share = $(80,000 - 20,000) \div 5,000 = 45,000 \div 5,000 = \text{Tk } 9.00$. Degree of financial leverage = $80,000 \div 60,000 = 1.33$. The degree of operating leverage is $240,000 \div 80,000 = 3.00$, so the degree of total leverage is $3.00 \times 1.33 = 4.00$. *Steps for Plan B.* Earnings per share = $80,000 \times 0.75 \div 7,000 = 60,000 \div 7,000 = \text{Tk } 8.57$. The degree of financial leverage is $80,000 \div 80,000 = 1.00$, and the degree of total leverage is $3.00 \times 1.00 = 3.00$.

The table lists earnings per share at three levels of EBIT. Read each row from left to right: the level of EBIT, then the earnings per share and the change for each plan.

EARNINGS PER SHARE UNDER THE TWO PLANS

EBIT	PLAN A	PLAN A CHANGE	PLAN B	PLAN B CHANGE
Tk 40,000	Tk 3.00	-66.7%	Tk 4.29	-50.0%
Tk 80,000	Tk 9.00	0%	Tk 8.57	0%
Tk 120,000	Tk 15.00	+66.7%	Tk 12.86	+50.0%

Answer. Plan A pays more at the base level, Tk 9.00 against Tk 8.57, and swings by 66.7% for a 50% change in EBIT, against 50.0% for Plan B. *Check.* Each swing equals the degree of financial leverage times 50%: $1.33 \times 50\% = 66.7\%$ and $1.00 \times 50\% = 50\%$. Neither plan is right in every case. Plan A suits owners who accept more swing for a higher base figure, and Plan B suits owners who value steadiness.

A business can choose its financing more freely than its trade.

The section's idea is that total leverage is a measure of total risk.

A reshaped problem tests the same method on two financing plans.

Both plans are worked from EBIT down to earnings per share.

The table shows what each plan does when EBIT changes by half.

Plan A gives more in good years and less in poor years, so it carries more risk.

The steps take each item in turn and then combine them.

The answer is 20 days and Tk 16,000, and the check explains the Tk 4,000 above cost.

Example 2 repeats the method with the case file figures.

The steps carry two decimal places and round only at the end.

The cycle is 50 days and the net investment is Tk 1,010,959, which passes a reasonableness check.

Four slips cause most wrong answers in this kind of problem.

Example 3 changes the situation: the suppliers ask to be paid sooner.

The steps change only the payables and compare the new need with the old.

The cycle grows by 15 days and the need by Tk 258,904, as the check confirms.

Example 4 combines the cash needed with the cost of float.

inventory days $\div 365$) + (sales $\times$ receivable days $\div 365$) - (cost of sales $\times$ payable days $\div 365$).

Steps. Cycle: $30 + 10 - 20 = 20$ days. Inventory: $219,000 \times 30 \div 365 = \text{Tk } 18,000$. Receivables: $365,000 \times 10 \div 365 = \text{Tk } 10,000$. Payables: $219,000 \times 20 \div 365 = \text{Tk } 12,000$. Net investment: $18,000 + 10,000 - 12,000 = \text{Tk } 16,000$.

Answer. The cycle is 20 days, and the trader needs Tk 16,000 of its own money in it. *Check.* Cost of sales is Tk 600 a day, and 20 days at that rate is Tk 12,000. The answer is Tk 4,000 higher because receivables are valued at the selling price: $10 \text{ days} \times (\text{Tk } 1,000 - \text{Tk } 600) = \text{Tk } 4,000$.

Example 2: realistic numbers. *Given.* Sales Tk 9,000,000 and cost of sales Tk 6,300,000. Inventory is held 75 days, customers pay in 20 days and suppliers are paid in 45 days. *Find.* The same two answers. *Formula.* The same as in Example 1.

Steps. Cycle: $75 + 20 - 45 = 50$ days. Inventory: $6,300,000 \times 75 \div 365 = 472,500,000 \div 365 = \text{Tk } 1,294,520.55$. Receivables: $9,000,000 \times 20 \div 365 = 180,000,000 \div 365 = \text{Tk } 493,150.68$. Payables: $6,300,000 \times 45 \div 365 = 283,500,000 \div 365 = \text{Tk } 776,712.33$. Net investment: $1,294,520.55 + 493,150.68 - 776,712.33 = \text{Tk } 1,010,958.90$, which rounds to Tk 1,010,959.²

Answer. The cycle is 50 days, and the shop needs Tk 1,010,959 of its own money in it. *Check.* Fifty days of cost of sales is about Tk 863,014. The answer is higher because receivables are counted at the selling price, which includes the shop's profit.

Common slips are these. Using sales where cost of sales belongs, for inventory and payables. Forgetting to divide by 365. Adding payables instead of subtracting them. Mixing periods, such as months for one item and days for another. Write the units beside every number, and check the sign of each line.

Example 3: a reshaped case. This is an exercise built on the case file, not part of it. *Given.* Suppose the suppliers of Meghna and Sons shorten the payment time from 45 days to 30 days. All other figures stay as in Example 2. *Find.* The new cycle and the extra money needed. *Formula.* The same as before.

Steps. New cycle: $75 + 20 - 30 = 65$ days. New payables: $6,300,000 \times 30 \div 365 = 189,000,000 \div 365 = \text{Tk } 517,808.22$. New net investment: $1,294,520.55 + 493,150.68 - 517,808.22 = \text{Tk } 1,269,863.01$, which rounds to Tk 1,269,863. Extra money: $1,269,863 - 1,010,959 = \text{Tk } 258,904$.

Answer. The cycle lengthens by 15 days to 65 days, and the shop must find Tk 258,904 more. *Check.* One day of cost of sales is about Tk 17,260. Fifteen days $\times$ Tk 17,260 is about Tk 258,900, which agrees.

Example 4: a combined problem. *Given.* The case file figures. Suppose every sale of the year is paid by cheque, and each cheque has the 6 days of float found in section 15.3. The interest rate is 10% a year. *Find.* The money held in float on average, its yearly cost, and the total need. *Formula.* Money in float = sales $\times$ float

² Each amount is rounded to the nearest taka in the case file. Adding the rounded parts gives Tk 1,010,960. The unrounded total rounds to Tk 1,010,959, and that is the figure we use.

12.3 A business has a degree of operating leverage of 2.50 and a degree of financial leverage of 1.60. Find its degree of total leverage.

12.4 The degree of total leverage is 4.00 and sales rise by 10%. By about what percentage does earnings per share rise? If earnings per share was Tk 5.00, what is it now?

12.5 Two plans give the same EBIT of Tk 100,000 and ignore tax. Plan X pays interest of Tk 40,000 and has 4,000 shares. Plan Y pays no interest and has 8,000 shares. Compare earnings per share at EBIT of Tk 100,000 and Tk 60,000, and say which plan is riskier.

be able to say why the supplier's 45 days reduce the shop's need for its own money.

* * *

15.3 FLOAT

Start with one cheque. A customer of Meghna and Sons pays a large bill by cheque. The customer has paid, but can the shop spend the money that same day? It cannot. The case file records what happened.

Case file, illustrative. Meghna and Sons Limited. On 3 February 2025 a customer posted a cheque for Tk 200,000. Mail took 2 days, processing took 1 day and clearing took 3 days. Total float was 6 days. Money unused at 10% a year costs $200,000 \times 0.10 \times 6/365 = \text{Tk } 329$.

While the cheque travelled, the customer had paid but the shop could not use the money. That delay has a name. **Float** is money that has been paid by one person but cannot yet be used by the receiver. It has three parts. Mail float is the days the cheque spends travelling to the receiver. Processing float is the days the receiver takes to record the cheque and pay it into a bank. Clearing float is the days the banks take to move the money from the payer's account to the receiver's. For the shop, $2 + 1 + 3 = 6$ days.

The next figure is a second timeline. Read it from left to right. Each unit is one day. Day 0 is the day of posting, and day 6 is the day the money can be used. Each bar is one part of float, and the line below it shows the total.

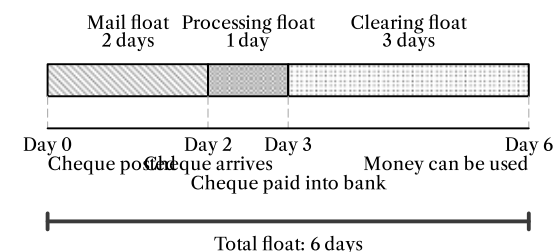


FIG. 15.2. Timeline of one cheque, from posting to usable money

Money that cannot be used has a cost, because it could have reduced a loan or earned a return. In words: cost of float = amount $\times$ yearly interest rate $\times$ days of float $\div$ 365. Here, $\text{Tk } 200,000 \times 0.10 \times 6 \div 365 = \text{Tk } 329$. The sum is small for one cheque. A business that receives hundreds of cheques loses much more. Float can also help the payer, who keeps the use of the money until the cheque clears.

A cheque takes days to become usable money, and the delay has a cost.

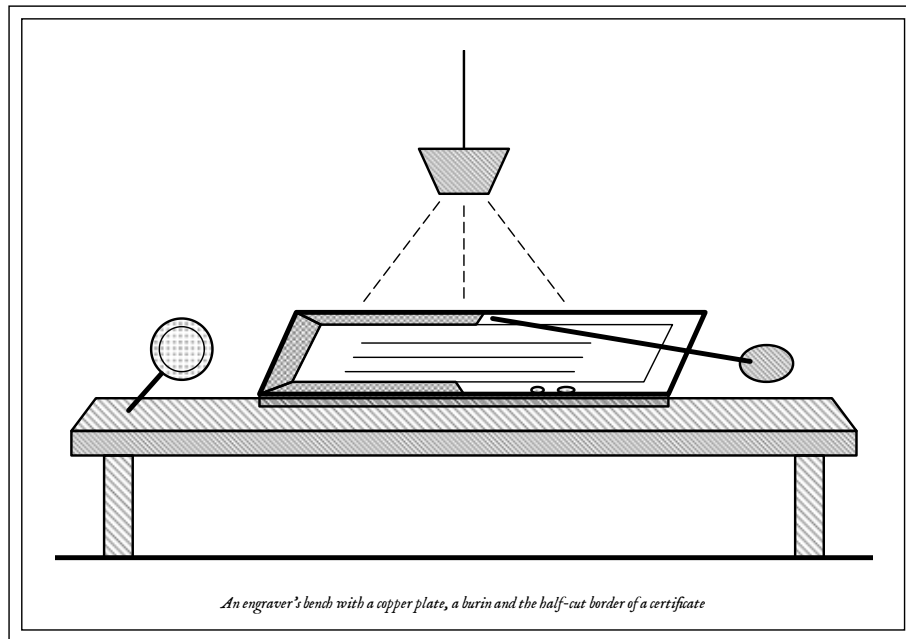
The case file records the three delays one cheque met before the shop could use its money.

Float is money paid by one person that the receiver cannot yet use.

Float. money that has been paid by one person but cannot yet be used by the receiver

The figure places the three parts of float on one timeline, from posting to usable money.

The cost of float is the amount, times the interest rate, times the days lost, divided by 365.



CHAPTER XIII

Valuing Bonds

What is a promise of future payments worth today?



A certificate can promise payments every year and a larger payment at the end. This chapter shows how to find what such a certificate is worth, why its worth rises and falls as market conditions change, and how to measure the return an investor earns by buying it at a given price.

The case file gives the trading days and the yearly amounts for Meghna and Sons Limited.

Cost of sales is what the goods sold in the year cost the business to buy.

A current asset is only a hope of cash, so a large working capital does not make payment certain.

Recall: liquidity, Chapter 2, section 2.3 'How Finance Shapes Business Success'.

Owner, supplier and financial manager each look at working capital differently.

Recap: working capital is short-term money; net working capital is what is left after short-term debts.

Follow one parcel of goods from arrival to cash, counting the days.

GO SLOWLY HERE.

The operating cycle is the time from buying goods to receiving the customer's cash.

Operating cycle. the days from buying stock to collecting cash from the customer

The shop did not pay its supplier on day 0, so the supplier's money was in the shop's hands.

would warn her that she could not pay her short-term debts from her short-term assets.

Case file, illustrative. Meghna and Sons Limited, 1 September 2025 (figures for the 12 months to 31 August 2025). Sales were Tk 9,000,000. Cost of sales was Tk 6,300,000. Inventory was held for 75 days. Customers (school accounts) paid in 20 days. Suppliers were paid in 45 days.

Cost of sales is the price the business paid for the goods it sold in the year. We use it here only as a measure. The shop's inventory and its accounts payable are valued at cost, so we compare them with cost of sales and not with sales.

Working capital has limits. A current asset is a hope of cash, not cash. Rice may not sell. A customer may pay late, or not at all. So a high net working capital does not guarantee that bills can be paid on Monday, if most of it is unsold goods. This is a question of liquidity, which you met in Chapter 2.

The owner sees working capital as money that cannot be taken out of the business. The supplier sees it as a sign that the business can pay its bills. The financial manager asks two questions: is the amount enough to keep trading, and is it larger than it needs to be?

Before moving on. Working capital is the money a business holds in short-term assets. Current assets turn into cash within about a year, and current liabilities fall due within about a year. Net working capital is the difference. Inventory, accounts receivable and accounts payable are the three items we follow in the next section. *You should now be able to say* how the stall's net working capital is found, and why it is Tk 1,200.

* * *

15.2 THE CASH CONVERSION CYCLE

Follow one parcel of books through Meghna and Sons. On day 0 the books arrive from the supplier. They stand on the shelves for some time. Then a school buys them on credit, and some days later the school pays. Each stage takes days, and we can count them. In the case file, the books stand for 75 days, and the school takes 20 days to pay.

The time from the day the goods arrive to the day the customer's cash is collected has a name. The **operating cycle** is the days from buying stock to collecting cash from the customer. It has two parts. The inventory days are the days the goods wait for a buyer. The receivable days are the days a customer takes to pay. In words: operating cycle = inventory days + receivable days. For the shop, 75 + 20 = 95 days.

PAUSE AND CHECK 15.P1 A business holds its goods for 40 days before they are sold, and its customers pay in 15 days. How long is its operating cycle?

But the shop did not pay the supplier on day 0. It paid on day 45. For those 45 days the supplier's money, not the shop's own, was in the shop's hands. The operating cycle leaves this out. The shop must find

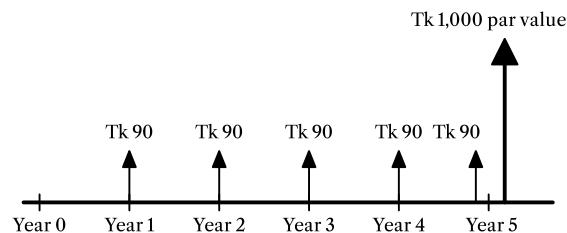
A bond has three facts printed or agreed at the start. The first is the **par value**. It means the amount printed on a bond that is repaid at maturity, a date explained below. Par value is also called face value. If the investor lends Tk 1,000 and the bond is repaid with Tk 1,000, then Tk 1,000 is the par value.

The second fact is the **coupon interest**. It is the fixed interest a bond pays each period, set as a percentage of par value. The percentage is called the coupon rate. A bond with a par value of Tk 1,000 and a coupon rate of 9% pays $9\% \times 1,000 = \text{Tk } 90$ each period. In the past, bond certificates carried printed coupons that the investor cut off and presented for payment, and the name has remained.

The third fact is **maturity**, the date on which a bond is repaid. A bond with five years to maturity makes its last coupon payment and repays its par value at the end of year 5. In this chapter, coupon interest is paid at the end of each year, unless we say otherwise. Section 13.3 treats interest paid twice a year.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. A corporate bond: par value Tk 1,000, coupon 9% a year (Tk 90), 5 years to maturity.

Figure 13.1 shows the payments that this bond promises. Read the line from left to right: it is a timeline, and each mark on it is the end of a year, from year 0 (today) to year 5. An arrow pointing up is money received by the investor. The five small arrows are the coupon payments of Tk 90. The one large arrow at year 5 is the repayment of par value, Tk 1,000. The investor receives the last coupon and the par value on the same date.



Arrows pointing up are money received by the investor
Small arrows: coupon interest. Large arrow: par value at maturity

FIG. 13.1 A five-year bond: five coupon payments and a larger payment at the

Par value. the amount printed on a bond that is repaid at maturity

Par. Word in use. In daily life, par means an equal level, as in "on a par with". In bonds, par is the amount printed on the bond.

Face. Word in use. In daily life, the front of the head or the front of an object. In bonds, face value is another name for par value.

The par value is the amount printed on the bond and repaid at the end.

Coupon interest. the fixed interest a bond pays each period, set as a percentage of par value

Coupon interest is the fixed interest paid each period, found by applying the coupon rate to par value.

Maturity. the date on which a bond is repaid

Maturity is the date on which the bond is repaid.

The case file records the terms of the corporate bond used throughout the chapter.

The timeline shows five small coupon payments and one larger payment of par value at maturity.

A good year of sales and a record profit, yet an empty cash drawer: how can all three be true?

You will need: Chapter 2, 'The Financial Manager's Work', section 2.3 'How Finance Shapes Business Success'; Chapter 1, 'Money, Finance and the Firm', section 1.3 'Where Funds Come From and Where They Go'.

The answer lies in time: the shop paid before it was paid.

This chapter names the gap, measures it in days, and estimates the money needed to cross it.

A rice stall, with small numbers, shows the idea before any new word appears.
GO SLOWLY HERE.

The table lists what she holds and what she owes on Wednesday evening.

Imagine a bookshop that has had its best year. Sales rose in every month, and the profit is the largest in the history of the business. On the last evening of the year, the owner opens the cash drawer. It is almost empty. A supplier's bill falls due on Monday, and the staff must be paid on Thursday. The owner has a profit but no money. How can both be true?

The answer is not found in the sales. It is found in time. The shop paid for its books before it sold them. It then sold many of them to schools on credit, which means the schools promised to pay later. Between paying and being paid, the shop's money lay in two places where it could not be spent. Some lay on the shelves, in the form of books. The rest lay in the promises of customers. The profit was real, but it was not yet cash.

This gap between paying out and collecting is the subject of this chapter. It matters to every financial manager, because many businesses have failed for lack of cash while their sales were strong. We shall name the parts of the gap and measure it in days. We shall learn why even a cheque needs days to become money that can be used. Then we shall estimate how much money a business must hold to cross the gap. We begin with a market stall, because the idea is the same at any size.

* * *

15.1 WHAT WORKING CAPITAL IS

Before any new word, take the smallest picture we can find. A woman keeps a rice stall in a market. On Monday morning she buys ten sacks of rice from a wholesaler at Tk 1,000 a sack, Tk 10,000 in all. The wholesaler trusts her and lets her pay on Friday. By Wednesday evening she has sold six sacks at Tk 1,200 each. A tea stall and a canteen took Tk 1,500 of that rice on credit, which means they will pay next week. The rest was paid in cash, and the cash is in her tin. Four sacks remain on the stall.

Let us list what she holds and what she owes at that moment. Each row is one item and its value in taka. The first three rows are what she holds, and the fourth row adds them. The fifth row is what she owes. The last row is the difference.

WEDNESDAY EVENING AT THE RICE STALL

ITEM	TK
Cash in the tin	5,700
Four sacks of rice, valued at cost	4,000
Owed to her by the tea stall and the canteen	1,500
Total she holds	11,200
Owed to the wholesaler	10,000
Total held minus total owed	1,200

the pattern of a single sum. We already know how to find the present value of each pattern. The value of the bond is the two present values added together, using the required return as the discount rate.

Here is the rule. In words: the value of a bond equals coupon interest times the annuity factor, plus par value times the single-sum factor. In symbols, $V = C \times a + P \times f$. Here V is the value of the bond, C is the coupon interest per period, and P is the par value. The annuity factor a equals $[1 - 1 \div (1 + r)^n] \div r$, and the single-sum factor f equals $1 \div (1 + r)^n$. In both, r is the required return per period, and n is the number of periods. The factors can be read from tables or found with a calculator.

THE SMALLEST CASE

Begin with one year, so that only one division is needed. A bond has a par value of Tk 1,000, a coupon rate of 10% and one year to maturity. The required return is 10%. The investor receives the coupon of Tk 100 and the par value of Tk 1,000 at the end of the year, a total of Tk 1,100. The value is $1,100 \div 1.10 = \text{Tk } 1,000$. Here the value equals the par value, and Section 13.4 explains why.

PAUSE AND CHECK 13.P2 A bond has a par value of Tk 1,000, a coupon rate of 5% and one year to maturity. The required return is 10%. What is its value?

CLEAN NUMBERS

Given. Par value Tk 1,000, coupon rate 10%, so $C = \text{Tk } 100$, three years to maturity, required return 10%. *Find.* The value of the bond. *Formula.* $V = C \times a + P \times f$, with $r = 10\%$ and $n = 3$.

Steps. First the single-sum factor: $(1.10)^3 = 1.10 \times 1.10 \times 1.10 = 1.331$, so $f = 1 \div 1.331 = 0.751315$. Second the annuity factor: $a = (1 - 0.751315) \div 0.10 = 2.486852$. Third the present value of the coupons: $100 \times 2.486852 = \text{Tk } 248.685$. Fourth the present value of the par value: $1,000 \times 0.751315 = \text{Tk } 751.315$. Fifth the sum: $V = 248.685 + 751.315 = \text{Tk } 1,000.000$. *Answer.* The bond is worth Tk 1,000.00. *Check.* The coupon rate equals the required return, so the value equals par, as in the one-year case.

REALISTIC NUMBERS

The case bond has a coupon rate of 9%, but investors in bonds of this risk require 12%. *Given.* Par value Tk 1,000, $C = \text{Tk } 90$, five years, required return 12%. *Find.* The value. *Formula.* $V = C \times a + P \times f$. *Steps.* $(1.12)^5 = 1.762342$, so $f = 1 \div 1.762342 = 0.567427$. Then $a = (1 - 0.567427) \div 0.12 = 3.604776$. The present value of the coupons is $90 \times 3.604776 = \text{Tk } 324.430$. The present value of the par value is $1,000 \times 0.567427 = \text{Tk } 567.427$. The sum is $324.430 + 567.427 = \text{Tk } 891.857$.

Case file, illustrative. Meghna and Sons Limited, 3 August 2025. Same corporate bond, required return 12%: value Tk 891.86.

Answer. The bond is worth Tk 891.86. *Check.* The bond pays Tk 90 a year while the market requires 12%, so an investor will pay less than par. The result is below Tk 1,000, as expected, and it is more than the par value's present value alone, Tk 567.43, because the

GO SLOWLY HERE.

The formula adds the present value of the coupons to the present value of the par value.

A one-year bond is solved first, because it needs only one division.

The first full example uses round numbers so that the method is the only thing to learn.

The steps find the two factors, multiply each by its payment, and add the results.

The case bond is valued at a required return of 12%, which differs from its coupon rate.

The case file records the value at 12%, which is below par.

The value is below the par value of Tk 1,000, and a check by estimate confirms that this is sensible.

Shares that could be sold gave each investor a way out.

The new feature was that the shares could be transferred. An investor who did not want to wait for the voyages to finish could sell the shares to another person, and buyers began to trade them in Amsterdam within days of the launch (Petram, 2014). Every sale needed a price, and a price needs a way to value a part of a business that cannot be seen. Buyers asked the questions of this chapter: what will the share pay, and what return do I require? Many historians treat the company as the first widely traded joint-stock company, meaning a business whose capital is divided into shares held by many owners. The historian Fernand Braudel cautioned that Amsterdam was not strictly the first stock market, so the claim is best made with care.

SOURCES

John Burr Williams, in *The Theory of Investment Value* (1938), argued that a stock is worth the present value of the dividends it will pay. This is the idea of the whole chapter. Myron Gordon and Eli Shapiro (1956) set out the growth formula that is often called the Gordon model, and that this book calls the dividend growth model. William Sharpe (1964) built the capital asset pricing model, which supplies the required return used in section 14.3. Lodewijk Petram (2014) tells the history of trade in the Dutch East India Company's shares.

REVIEW QUESTIONS

- 14.1 Preferred stock pays Tk 9 a year for ever. The required return is 12%. What is the value of one share?
- 14.2 Common stock has a last dividend of Tk 3.50, a growth rate of 5% and a required return of 11%. Find its value today.
- 14.3 Explain why the growth rate must stay below the required return in the dividend growth model.
- 14.4 Give three differences between a bond and a stock.
- 14.5 A common stock pays a level dividend of Tk 6 for ever. The required return rises from 12% to 15%. Find the new value and the fall in value.
- 14.6 Two stocks each have a last dividend of Tk 4.00 and a required return of 12%. Stock A grows at 4% a year and stock B at 7%. Find the value of each and explain the gap.
- 14.7 An investor requires 10%. She is offered a bond paying Tk 80 a year for 3 years plus Tk 1,000 at the end of year 3, and a common stock with a last dividend of Tk 2.00 and growth of 5% a year. Value one bond and ten shares, and give the total.

VALUE OF THE CASE BOND AT THREE REQUIRED RETURNS

REQUIRED RETURN	VALUE (TK)	KIND OF BOND
12%	891.86	Discount
9%	1,000.00	Par
7%	1,082.00	Premium

The names come from the price. A **discount bond** is a bond that sells for less than its par value. At 12%, the case bond sells for Tk 891.86, a discount of Tk 108.14. A **premium bond** is a bond that sells for more than its par value. At 7% it sells for Tk 1,082.00, a premium of Tk 82.00. A bond that sells at exactly par, as at 9%, is a par bond.

The rule is this. If the required return is above the coupon rate, the bond sells at a discount. If it equals the coupon rate, the bond sells at par. If it is below the coupon rate, the bond sells at a premium. The reason is that the coupon is fixed. When the market requires more than the coupon pays, the price must fall until the buyer's return reaches the required level. When the market requires less, buyers bid the price up.

Figure 13.2 draws the idea. The horizontal axis shows the required return, and the vertical axis shows the value of the bond. Each point on the curve is the value at one required return. The dashed horizontal line marks par value, Tk 1,000. The curve slopes downward: a higher required return always gives a lower value. It crosses the par line at 9%, the coupon rate. To the left lie premium bonds, and to the right lie discount bonds.

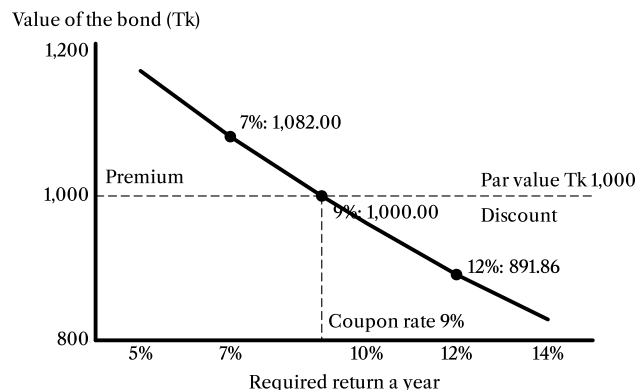


FIG. 13.2. Value of the case bond falling as the required return rises, with par value as a horizontal line

One limit should be noted. A discount or premium does not last. As maturity approaches, the value moves toward par, because the par value is repaid on that date.

Discount bond. a bond that sells for less than its par value

Premium bond. a bond that sells for more than its par value

A discount bond sells below par, and a premium bond sells above par.

The rule links the required return to the coupon rate: higher means discount, equal means par, lower means premium.

Figure 13.2 shows the value falling as the required return rises, crossing the par line at the coupon rate.

A bond's value moves toward par as maturity approaches, because the par value is then repaid.

BOND AND COMMON STOCK COMPARED

FEATURE	BOND	COMMON STOCK
Holder is	A lender	An owner
Payment	Interest fixed by contract	Dividend decided by the directors
Voting right	None	Usually one vote a share
Claim if the business ends	Paid before owners	Residual claim, paid last
Reward	Interest and repayment	Dividends and capital gain
Risk to the holder	Lower	Higher

Viewpoint matters. A cautious investor values certain payments and chooses the bond. An investor who accepts risk for a chance of greater reward may choose the stock. The business has its own view. Interest on a bond must be paid on time, whatever the business earns, but a dividend can be reduced when profit falls. Preferred stock sits between the two: its dividend is fixed like interest, yet it is not a legal debt.

Before moving on. A bond holder lends and is paid first, in fixed amounts. A stockholder owns and is paid last, in amounts that can vary. Both are valued as the present value of the cash they bring.

* * *

WORKED CASE: VALUING THREE STOCKS

Case file, illustrative. Meghna and Sons Limited, 7 September 2025. Preferred stock paying a fixed Tk 8 a year, required return 10%: value Tk 80.00. Common stock with a level dividend of Tk 5 and required return 10%: value Tk 50.00. Common stock of a publishing business: last dividend Tk 4.00, growth 6% a year for ever, required return 12%: next dividend Tk 4.24; value $4.24 / (0.12 - 0.06) = \text{Tk } 70.67$.

We now check each entry, beginning with the two that use the level formula. *Given.* For the preferred stock, $D = \text{Tk } 8$ and $r = 10\% = 0.10$. For the common stock, $D = \text{Tk } 5$ and $r = 0.10$. *Find.* The value of each. *Formula.* $P = D \div r$. *Steps.* Preferred: $P = 8 \div 0.10 = \text{Tk } 80.00$. Common: $P = 5 \div 0.10 = \text{Tk } 50.00$. *Answer.* The preferred stock is worth Tk 80.00 and the common stock Tk 50.00. *Check.* Tk 80.00 at 10% gives Tk 8.00 a year, and Tk 50.00 at 10% gives Tk 5.00 a year. Both match their dividends. The common result also answers the opening puzzle: a piece paying Tk 5 each year is worth Tk 50.00 to an investor who requires 10%, so Tk 80 was too much.

Given. Last dividend $D_0 = \text{Tk } 4.00$; growth rate $g = 6\% = 0.06$; required return $r = 12\% = 0.12$. *Find.* The value of one share today, P_0 . *Formula.* Value today equals the next dividend divided by the required return minus the growth rate, or $P_0 = D_1 \div (r - g)$, with $D_1 = D_0 \times (1 + g)$. *Steps.* First, $D_1 = 4.00 \times 1.06 = \text{Tk } 4.24$. Second, $r - g = 0.12 - 0.06 = 0.06$. Third, $P_0 = 4.24 \div 0.06 = \text{Tk } 70.67$, rounded to

From the investor's viewpoint, the choice trades certainty for possible reward.

Recap of the section comparing bonds and stock.

The case file records three valuations made on one date.

The first two entries use the level formula; each step is shown.

Note the growing stock: data, formula, steps, answer and check.

Steps, part two. The price of Tk 940 is below Tk 949.24, so the rate must exceed 5%. Try 6%: $f = 0.704961$ and $a = 4.917324$, so the value is $40 \times 4.917324 + 1,000 \times 0.704961 = 196.693 + 704.961 = \text{Tk } 901.65$. The price lies between the two values. The fraction is $(949.24 - 940) \div (949.24 - 901.65) = 9.24 \div 47.59 = 0.194$, so the half-yearly rate is about $5\% + 0.194 \times 1\% = 5.19\%$. Bond markets state the yearly figure by doubling the half-yearly rate, which gives about 10.4%.

Answer. The value at 10% is Tk 949.24, and the yield to maturity at a price of Tk 940 is about 10.4% a year. *Check.* The bond is a discount bond, so its value is below par, and the coupon rate of 8% is below the required return of 10%, as the rule requires. A price of Tk 940 is below Tk 949.24, so the yield must exceed 10%, and 10.4% does.

* * *

SUMMARY

A bond is a loan sold to investors, with a par value, coupon interest and a maturity. Its intrinsic value is the present value of its coupons plus the present value of its par value, at the required return: $V = C \times a + P \times f$. For half-yearly interest, halve the coupon and the rate and double the periods. The value falls as the required return rises. A bond sells at a discount when the required return exceeds the coupon rate, at par when equal, and at a premium when below. The yield to maturity is the return from buying at the price and holding to maturity, found by trial.

THINK IT THROUGH

The case bond has a par value of Tk 1,000, a coupon rate of 9% and five years to maturity. It now sells for Tk 940.

13.T1 Does the market require more or less than the coupon rate? Show the check by calculating the value at 10% and at 11%.

13.T2 Is this a discount bond, a par bond or a premium bond? What would the price become if the required return fell to 9%?

WHERE PEOPLE GO WRONG

Some students treat the coupon rate as the investor's return. The coupon rate is fixed on par value. An investor who pays Tk 920 earns more than 9%, and one who pays Tk 1,082 earns less. The return depends on the price.

Some students expect a higher required return to give a higher value. The reverse is true. A higher required return means a larger discount on the fixed payments, so the present value falls.

Some students discount the coupons at the coupon rate. This always gives a value equal to par, whatever the market requires, and so answers nothing. The discount rate is the required return.

The second part tries 5% and 6% per half-year, then interpolates and doubles the result.

The answers are Tk 949.24 and about 10.4%, and each is tested against the other.

A bond is valued by discounting its coupons and par value at the required return, and its price and yield move in opposite directions.

A bond price of Tk 940 is compared with the case bond to learn what the market requires.

The coupon rate is not the return an investor earns, because the return depends on the price paid.

A higher required return lowers the value of a bond, and does not raise it.

The coupons are discounted at the required return, never at the coupon rate.

$\text{Tk } 6 \div 0.12 = \text{Tk } 50.00$. *Answer.* One share of this preferred stock is worth Tk 50.00. *Check.* Tk 50.00 placed in a deposit that pays 12% gives Tk 6 each year, which equals the dividend. An investor who requires 12% should therefore pay no more than Tk 50.00.

The rule has limits. A preferred dividend is fixed by the terms of the issue, so the rule suits it well. A common dividend is decided by the directors year by year, so a level dividend is only an assumption. A business that earns more profit may raise its dividend, and a business that earns less may cut it. The rule also takes the required return as steady, although investors change their minds. The next section removes the first limit by letting the dividend grow.

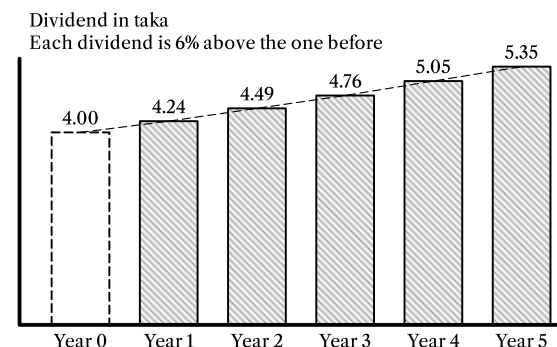
Before moving on. Stock is worth the present value of its dividends. When the dividend is level, the value is the dividend divided by the required return. This rule fits preferred stock well and common stock only when the dividend is level.

* * *

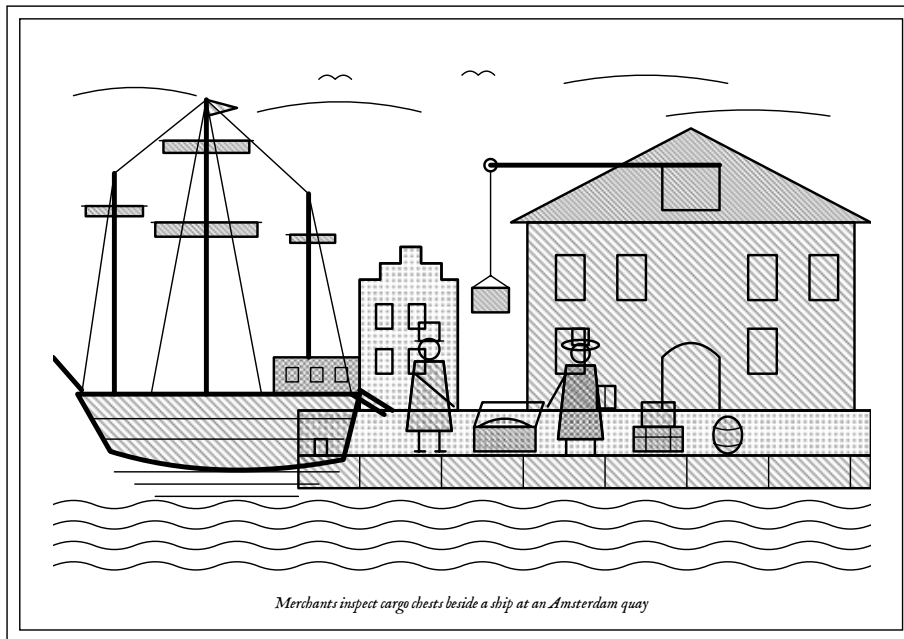
14.3 VALUING STOCK WITH GROWING DIVIDENDS

A publishing business paid a dividend of Tk 4.00 on each share this year, and its directors expect to raise the dividend by 6% every year for ever. Each payment is larger than the one before, and there is no final payment. Must we add up an endless list of present values? The answer is no, because a short formula does the work. First we must see how the dividends behave.

The dividend paid most recently is the last dividend, written D_0 . The next dividend, D_1 , is the last dividend increased by the growth rate g : $D_1 = D_0 \times (1 + g)$. With $D_0 = \text{Tk } 4.00$ and $g = 6\%$, $D_1 = 4.00 \times 1.06 = \text{Tk } 4.24$. The dividend after that is $4.24 \times 1.06 = \text{Tk } 4.49$, and so on. The figure below shows the first five payments. To read it, look at the horizontal axis, which shows the year, and at the height of each bar, which shows the dividend paid in that year in taka. The first bar is the dividend already paid. The rising bars are the dividends to come.



Dashed bar: last dividend, already paid. Hatched bars: dividends to come



CHAPTER XIV

Valuing Common and Preferred Stock

What is a piece of a business worth when you cannot see the profit it will make?



This chapter explains what the owners of a business hold and how to give that holding a price. You will value preferred stock and common stock with level dividends, then common stock whose dividends grow at a steady rate. You will end by comparing stock with bonds.

YOU WILL NEED

Chapter 12, section 12.1 'Borrowing and Fixed Financing Charges'; Chapter 5, section 5.1 'What an Annuity Is'; and Chapter 13, section 13.2 'The Idea of Value'.

A neighbour offers a piece of her business for Tk 80. Is that fair?

The fair price depends on the payments and on the return you require.

Three questions remain, and the chapter answers each in turn.

Owning stock makes you an owner, and ownership brings a voice and a claim.

Stock. Word in use. In a shop, the goods kept ready for sale. In finance, ownership of a business, held in units called shares.

Share. Word in use. In daily life, a part of something divided among people, such as a share of a meal. In finance, one unit of ownership in a business.

The first right is a vote on how the business is guided.

Voting right. the right of a shareholder to vote on matters such as choosing directors

Suppose a neighbour offers to sell you a small piece of her business. The piece pays its owner Tk 5 at the end of every year, and she expects the payments to continue for ever. She asks Tk 80 for it. Is that a fair price? You cannot see the future payments, and you cannot ask the business to return your money on demand. All you would hold is a paper that says you own a part. How would you decide what the paper is worth?

Most people begin by asking what they could earn elsewhere. Suppose a safe deposit pays 10% a year. Then Tk 50 in that deposit gives you Tk 5 each year. A price of Tk 80 now looks too high, because Tk 80 in the deposit would give Tk 8. A price of Tk 40 looks like a bargain. This reasoning carries most of the chapter: the worth of a piece of a business is linked to the payments it makes and to the return that you require from your money.

The neighbour's offer still hides three questions. What exactly do you own when you hold the paper? What if the yearly payment is not level but grows? And how does the paper differ from a bond, which also pays regular amounts? We take the questions in that order, beginning with the rights that come with ownership.

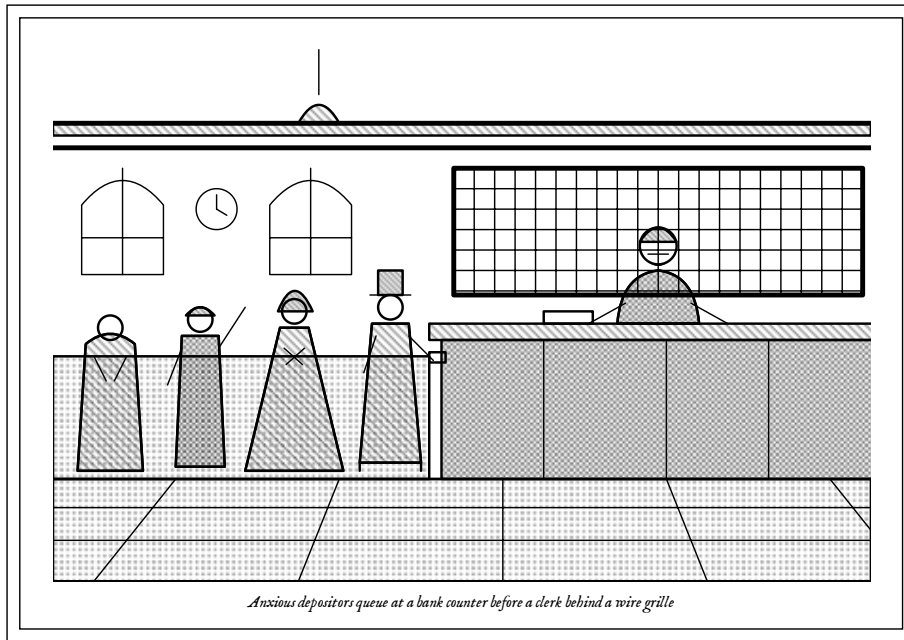
* * *

14.1 RIGHTS OF SHAREHOLDERS

Chapter 12 introduced common stock and preferred stock as the two kinds of stock that a business can issue.¹ Here we ask a new question: what does an owner actually hold? Picture a market stall owned by three cousins. Each cousin may speak and vote when the family meets to decide how to run the stall, and each is entitled to a part of whatever remains after the bills are paid. A voice and a claim: these two ideas are the centre of ownership, and each has a name. A person who owns shares is a shareholder.

The first right is the voice. A **voting right** means the right of a shareholder to vote on matters such as choosing directors. Directors are the people who guide a business on behalf of its owners. A holder of common stock usually receives one vote for each share held, so a holder of 100 shares casts 100 votes at the yearly meeting. The details differ between countries and between businesses, because the law of each country and the rules of each business set them.

¹ Chapter 12, 'Financial and Total Leverage', section 12.1 'Borrowing and Fixed Financing Charges': common stock and preferred stock were first defined there.



CHAPTER XVI

Short-Term Financing and Matching

Why can a sound bank fail on the day its customers ask for their money?



This chapter sorts financing by the time for which a business holds the money. You will price trade credit, the most common short-term source, and learn when a supplier's discount is worth taking. You will then apply the matching principle to decide which kind of money suits which need.

Business risk. The risk that operating profit varies because of sales, costs and trade conditions. It belongs to the trade, whatever the way the business is funded. A bookshop whose sales fall in the school holidays meets business risk. (*Chapter IX*)

Capital asset pricing model. A formula setting required return as risk-free rate plus beta times the market risk premium. With a risk-free rate of 8%, a market return of 14% and a beta of 0.6, the required return is $8\% + 0.6 \times 6\% = 11.6\%$. (*Chapter X*)

Capital budgeting. The process of planning, evaluating and choosing long-term investments, such as a machine that will work for five years. It covers finding proposals, estimating their cash flows, choosing among them and later reviewing the results. (*Chapter VII*)

Capital expenditure. Spending on assets that will serve the business for many years. A machine, a delivery van and a new shop floor are examples; a month of electricity is not, because it is used up at once. (*Chapter VII*)

Capital gain. The profit made when an asset is sold for more than its cost. For example, stock bought for Tk 50 and sold for Tk 56 gives a capital gain of Tk 6. Together with dividends, it forms the owner's return from holding the stock. (*Chapter XIV*)

Capital market. The market where long-term loans and shares are issued and traded. Securities are first sold there by the business that issues them, and later traded between investors. A stock exchange is part of it. (*Chapter XVII*)

Cash conversion cycle. The operating cycle minus the days a business takes to pay its suppliers. It equals inventory days plus receivable days minus payable days. For Meghna and Sons Limited, $75 + 20 - 45 = 50$ days. (*Chapter XV*)

Cash discount. A reduction in price for paying a supplier early. On an invoice of Tk 500,000 with a 2% cash discount, the discount is Tk 10,000. Refusing it is a costly loan from the supplier, so the buyer should compare its yearly cost with the interest rate of a bank. (*Chapter XVI*)

Cash flow. Money moving into or out of a business during a period such as a day, a month or a year. A customer paying Tk 3,000 on Monday is a cash flow in; paying a supplier Tk 2,000 is a cash flow out. (*Chapter I*)

Coefficient of variation. The standard deviation divided by the expected value, showing risk for each unit of return. It allows choices with different expected values to be compared. A standard deviation of 8.49% on an expected value of 14% gives a coefficient of 0.61. (*Chapter IX*)

Collateral. An asset a borrower pledges so the lender can take it if the loan is not repaid. It lowers the lender's risk, and so often lowers the interest rate. A press machine bought with the loan can be the collateral. (*Chapter XVII*)

Common stock. Ownership shares that receive what remains after all other claims, with no fixed dividend. Owners of common stock take the most risk and keep any gain. Some countries call these

FINANCING SPANS AND THEIR USUAL SOURCES

SPAN	LENGTH	USUAL SOURCES
Short-term	Up to 1 year	Trade credit, bank overdraft, short bank loans
Intermediate-term	1 to 5 years	Loans for equipment or vehicles, repaid in instalments
Long-term	More than 5 years	Bonds, common stock, profit kept in the business, long loans

The lines between the spans are not walls. A bank may renew an overdraft each year, so short-term money can stay with a business for many years. But the bank is free to refuse, and that freedom is the danger. The label describes the agreement, not the hope of the borrower. The long-term instruments, such as loans and bonds, are taught in Chapter 17.²

Viewpoint changes the picture. A lender who gives short-term money keeps the right to ask for it back soon, so the lender takes a smaller risk and usually asks for a lower promise of reward. The owner of the business sees the reverse: short money is easy to obtain but must be found again and again. Long-term money gives the owner peace of mind, but the owner must pay for it every month and cannot return it early without agreement. Each span is therefore a trade between ease, safety and cost.

Before moving on. Financing is sorted by time: up to one year is short, one to five years is intermediate, and more than five years is long. Each span has its usual sources. A short-term label is only as safe as the lender's promise to renew.

* * *

16.2 TRADE CREDIT

Imagine that a supplier sends you books worth Tk 500,000 and offers two choices. You may pay Tk 490,000 within 10 days, or Tk 500,000 within 40 days. The saving is only 2%. Is it worth hurrying to pay? Many people say no, because 2% seems small. We shall find that the answer depends on a short calculation, and that the answer is often surprising.

Start with the idea of delay. **Trade credit** means a supplier allowing a business to pay for goods some days after delivery. A bookshop that receives books on the 1st and pays on the 30th has used the supplier's money for 29 days without paying interest. In the shop's records the unpaid invoice is an account payable. Trade credit is the most common source of short-term financing, because almost every business buys on credit. To the supplier it is a way to win customers. To the buyer it is a loan that appears with each purchase.

Suppliers state their conditions in a standard code. **Credit terms** means the discount offered and the days allowed for payment,

² Chapter 17, 'Long-Term Financing and the Cost of Capital', section 17.1 'Term Loans and Debt Instruments': long-term debt instruments are taught there.

The label describes the agreement, not how long the business hopes to use the money.

Owners and lenders see the three spans from different sides.

Recap of the section on spans.

A supplier's early-payment discount looks small, yet refusing it can be very costly.

Trade credit lets a business pay a supplier after the goods arrive.

Trade credit, a supplier allowing a business to pay for goods some days after delivery

Credit terms are a short code that states the discount and the days allowed

IN HISTORY

Leasing is old, but the lease business is modern.

Why leasing grew: scarce credit and favourable tax rules.

Some historians trace leasing to the ancient Near East, around 2000 BC, when owners let farm tools to farmers under written agreements. In 1877 the Bell Telephone Company began to let telephones to customers instead of selling them. The United States Leasing Corporation, formed in San Francisco in 1952, is usually named as the first business to treat equipment leasing as its main work. Leasing then spread to Europe in the late 1950s and early 1960s, helped in the United States by tax rules that favoured it.

Why did leasing grow so fast? One reason, in the United States, was a shortage of medium-term credit, which left businesses wanting machines they could not pay for at once. Another was the tax treatment of lease payments and depreciation, which differs from country to country and can favour the lessor, the lessee or both. Banks and manufacturers later formed their own leasing businesses. A lease is therefore both a way of financing and a way of sharing tax savings, which is why the comparison in this chapter always counts tax.

REVIEW QUESTIONS

18.1 A rickshaw owner lets a puller use a rickshaw each day for an agreed payment under a contract. Name the lessor and the lessee, and say what a lease is.

18.2 Explain the difference between an operating lease and a financial lease, with one example of each.

18.3 A lease payment is Tk 80,000 a year and the tax rate is 25%. Find the after-tax cash outflow and say why it is lower than the payment.

18.4 State the four steps of the lease-or-buy decision in order.

18.5 A clinic can lease a scanner for Tk 100,000 at each year end for 3 years, or buy it with a loan. The after-tax outflows of buying are Tk 70,000, Tk 72,000 and Tk 74,000. The loan interest rate is 8% and the tax rate is 25%. Find the discount rate and the present value of each road, then decide.

16.3 OTHER SHORT-TERM SOURCES

When suppliers cannot supply enough, the business turns to a bank. A **bank overdraft** means a bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used. Suppose the limit is Tk 100,000 and the business uses Tk 50,000 for 30 days at 12%. The interest is $50,000 \times 0.12 \times 30 \div 365 = \text{Tk } 493.15$. Interest runs only on the amount used and only for the days used, which suits needs that rise and fall. A short loan is different: the bank lends a fixed amount for a fixed period, often a few months, and interest runs on the whole amount until repayment. Banks may refuse to renew either one, and rates and rules differ by bank and by country.

Before moving on. An overdraft charges interest only on the amount used. A short loan charges interest on the whole amount for the whole period. Both are short-term, and both depend on the bank's willingness to continue.

* * *

16.4 THE MATCHING PRINCIPLE

Return to the opening puzzle. The friend's one-month loan was wrong for a five-year motorcycle, and the ten-year loan was wrong for a six-week stock of toys. A rule can prevent both errors. First, though, we must sort a business's needs by time, because some needs never end and others come and go.

A bookshop shows the difference. It always needs some stock, some money in the till and some money owed by customers, whatever the month. **Permanent working capital** means the minimum working capital a business needs all year. Before the school term opens, the shop must also buy extra books. **Seasonal working capital** means extra working capital needed only at the busiest times of the year. The size of both comes from the cash conversion cycle, which sets how much money is tied up in operations.³

The figure below shows both kinds of need. To read it, look at the horizontal axis, which shows the twelve months of the year, and at the height of the shaded area, which shows the money the business needs at that time, in taka. The lower band is the permanent need, which stays level all year. The upper band is the seasonal need, which appears for three months and then disappears.

Banks supply short-term money as overdrafts and as short loans.

Bank overdraft. a bank allowing an account to go below zero up to an agreed limit, with interest charged on the amount used

Recap of the section on banks.

Return to the motorcycle puzzle: the money must fit the need.

Permanent needs stay all year, while seasonal needs appear at the busiest times.

Permanent working capital. the minimum working capital a business needs all year

Seasonal working capital. extra working capital needed only at the busiest times of the year

The figure shows a level permanent need and a seasonal peak above it.

interest in the year, D the depreciation, C the outflow in a year and r the discount rate.

Steps. Step 1, leasing: $210,000 \times (1 - 0.25) = \text{Tk } 157,500$ in each of the 5 years. Step 2, buying: the interest in year 1 is $10\% \times 780,000 = \text{Tk } 78,000$. The instalment of Tk 205,762.04 repays Tk 127,762.04 of the loan, leaving Tk 652,237.96 owed, so year 2 interest is $10\% \times 652,237.96 = \text{Tk } 65,223.80$. The same method gives the later years. The table that follows is read across each row: the year, the interest, the tax saved (25% of interest plus Tk 156,000 of depreciation) and the after-tax outflow, which is the instalment minus the tax saved.

AFTER-TAX OUTFLOW OF BUYING, IN TAKA TO THE NEAREST TAKA

YEAR	INTEREST	TAX SAVED	AFTER-TAX OUTFLOW
1	78,000	58,500	147,262
2	65,224	55,306	150,456
3	51,170	51,792	153,970
4	35,711	47,928	157,834
5	18,706	43,676	162,086

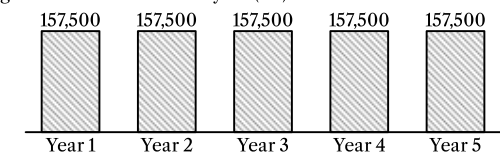
Step 3 converts each outflow to a present value at 7.5%.

The figure shows the two outflow streams side by side.

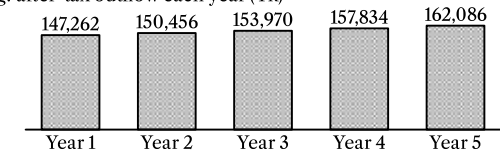
Step 3, present values at 7.5%. Leasing is an ordinary annuity, so the 5-year annuity factor of 4.045885 applies: $157,500 \times 4.045885 = \text{Tk } 637,227$. For buying, each year's outflow is multiplied by its present value factor: $147,262 \times 0.930233 = 136,988$; $150,456 \times 0.865333 = 130,195$; $153,970 \times 0.804961 = 123,940$; $157,834 \times 0.748801 = 118,186$; $162,086 \times 0.696559 = 112,902$. The present values add to Tk 622,210; the yearly figures here are rounded, so their sum can differ from this total by a taka.

Figure 1 draws the two streams. To read it, look at one row at a time: each bar is one year's after-tax outflow, and a taller bar means more cash paid out that year. The top row, with line hatching, is leasing; the bottom row, with cross hatching, is buying. Bars above one another belong to the same year.

Leasing: after-tax outflow each year (Tk)



Buying: after-tax outflow each year (Tk)



Same scale in both rows; lines hatching = leasing, cross hatching = buying

FIG. 18.1 Yearly after-tax outflows of leasing and of buying, aligned by year

Answer: Leasing has a present value of Tk 637,227 and buying has

³ Chapter 15, 'Working Capital and the Cash Cycle', section 15.2 'The Cash Conversion Cycle': the cycle sets the size of the need.

WORKED CASE: A DISCOUNT AND A SEASONAL PEAK

Case file, illustrative. Meghna and Sons Limited, 1 October 2025. A supplier invoice of Tk 500,000 on terms 2/10, net 40. Taking the discount saves Tk 10,000 and costs Tk 490,000 on day 10. Cost of giving up the discount = $2/98 \times 365/30 = 24.83\%$ a year (approximation). A bank overdraft costs 14%: borrowing Tk 490,000 for 30 days costs Tk 5,638, which is less than the Tk 10,000 discount, so take the discount and borrow. Seasonal need: permanent current assets Tk 900,000 financed long term; a seasonal peak of Tk 400,000 for 3 months before the January term. Short-term borrowing at 14% costs Tk 14,000; carrying the Tk 400,000 on a long-term loan at 11% for the whole year costs Tk 44,000.

Given. Invoice Tk 500,000; terms 2/10, net 40; overdraft interest rate 14% a year. *Find.* The yearly cost of giving up the discount, and whether to take the discount. *Formula.* $\text{Cost} = d \div (1 - d) \times 365 \div (N - D)$. *Steps.* First, the discount is $0.02 \times 500,000 = \text{Tk } 10,000$, so the payment on day 10 is Tk 490,000. Second, the extra days are $N - D = 40 - 10 = 30$. Third, $\text{cost} = 0.02 \div 0.98 \times 365 \div 30 = 0.020408 \times 12.1667 = 0.2483$. *Answer.* The cost of giving up the discount is 24.83% a year, which is above the overdraft rate of 14%, so the discount should be taken. *Check.* $\text{Tk } 10,000 \div \text{Tk } 490,000 = 2.04\%$ for 30 days. Twelve such periods give about 24.5%, close to 24.83%, because a year holds slightly more than twelve periods of 30 days.

Given. Borrowing Tk 490,000 for 30 days at 14%. *Find.* The interest cost, to compare with the Tk 10,000 discount. *Formula.* $\text{Interest} = \text{amount} \times \text{yearly rate} \times \text{days} \div 365$. *Steps.* $490,000 \times 0.14 = 68,600$. Then $68,600 \times 30 \div 365 = \text{Tk } 5,638$. *Answer.* The overdraft costs Tk 5,638, which is less than the Tk 10,000 discount, so the business gains $\text{Tk } 10,000 - \text{Tk } 5,638 = \text{Tk } 4,362$ by borrowing to take the discount. *Check.* $5,638 \div 490,000 = 1.15\%$, below the 2.04% that the supplier's terms charge for the same 30 days.

Given. Permanent current assets Tk 900,000 financed long term. Seasonal peak Tk 400,000 for 3 months. Short-term rate 14%; long-term rate 11%. *Find.* The interest cost of each way of financing the peak. *Formula.* $\text{Interest} = \text{amount} \times \text{yearly rate} \times \text{months} \div 12$. *Steps.* Short-term: $400,000 \times 0.14 \times 3 \div 12 = \text{Tk } 14,000$. Long-term for the whole year: $400,000 \times 0.11 \times 12 \div 12 = \text{Tk } 44,000$. *Answer.* Short-term borrowing costs Tk 14,000 and the long loan costs Tk 44,000, so matching saves Tk 30,000 for the year. *Check.* The short rate is higher, 14% against 11%, but it is paid for only 3 months instead of 12, so the total is smaller.

Many students make the same slips here, so check for them. First, using 40 days instead of 30 in the cost formula; the days that matter are the extra days, $N - D$. Second, forgetting that the bottom of the first fraction is $100\% - 2\% = 98\%$, not 100%. Third, mixing months and years: 3 months is $3 \div 12$ of a year. Fourth, comparing the cost of giving up the discount with the interest rate without checking that both are yearly rates.

* * *

The case file records a supplier discount and a seasonal need on two dates.

First question: is the discount worth taking?

Second question: what does the overdraft cost for the 30 days?

Third question: which source is cheapest for the seasonal peak?

Four slips are common in these calculations.

Sale and leaseback. Selling an asset to a lessor and at once leasing it back.

Selling an asset and leasing it back turns it into cash.

Accounting rules change the records, not the contract.

Recap of section 18.2.

The question: which road costs the business less?

Recall: net present value, Chapter 8, section 8.2 "Net Present Value".

After-tax cash outflow. Cash paid out in a year after taking away the tax saved by that payment.

Tax saved lowers the true cost of each payment.

For leasing, the whole payment is a deduction.

Buying with a loan has two tax savings; repayment of the loan is not one.

Recall: after-tax cost of debt.

operating lease also gives the lessee a service, such as repairs, so a comparison of cost alone would leave out part of what is bought.

A third arrangement is a **sale and leaseback**, which means selling an asset to a lessor and at once leasing it back. A baker who owns a delivery van may sell it to a lessor for cash and then lease the same van back. The business receives cash today and keeps using the van, but it now makes lease payments for as long as it needs the van.

Accounting rules affect how a lease appears in a business's records, but they do not change what the contract is.² The risk question stays useful, because it still tells us who bears the burden of owning.

Before moving on. An operating lease is short, can be cancelled and leaves most risk with the lessor. A financial lease is long, cannot be cancelled and leaves most risk with the lessee. A sale and leaseback turns an owned asset into cash while the business keeps using it.

18.3 THE LEASE OR BUY DECISION

Suppose a business has decided that it needs a machine. The investment decision is made. One question remains: should the business lease the machine, or buy it with money borrowed from a bank? The answer cannot come from adding the payments, because a taka paid in year five is worth less today than a taka paid in year one. The answer must come from present values, using the method of net present value.

Tax changes each payment, so we need one more idea. The **after-tax cash outflow** is the cash paid out in a year after taking away the tax saved by that payment. Suppose a lease payment is Tk 100 and the tax rate is 25%. The payment lowers taxable profit by Tk 100, so tax falls by Tk 25, and the after-tax cash outflow is Tk 75. **PAUSE AND CHECK 18.P1** A lease payment is Tk 60,000 and the tax rate is 25%. What is the after-tax cash outflow?

For leasing, the rule is direct. The lessee deducts the whole lease payment from taxable profit. In words: the after-tax cash outflow of leasing equals the lease payment multiplied by one minus the tax rate.

For buying with a loan, the rule needs more care. Each yearly instalment repays part of the loan and pays interest. Only the interest is deducted from taxable profit. The business may also deduct depreciation each year, although depreciation is not a cash payment. Many students find this step confusing, because the instalment is cash but depreciation is not. The explanation is that depreciation sends no cash out, yet it lowers the tax bill, so it still leaves the business with more cash. The after-tax cash outflow of buying therefore equals the instalment minus the tax rate multiplied by the sum of interest and depreciation.

Which discount rate should we use? Both streams are fixed by ² Under International Financial Reporting Standard 16, which applies to reporting periods beginning on or after 1 January 2019, a lessee records most leases as an asset and a liability. Standards differ by place.

REVIEW QUESTIONS

- 16.1 An invoice of Tk 10,000 carries terms 2/10, net 30. Find the cost of giving up the discount as a yearly rate.
- 16.2 Find the cost of giving up the discount on terms 3/10, net 45.
- 16.3 An invoice of Tk 200,000 carries terms 2/10, net 40. The bank charges 13% a year. Should the business borrow from the bank to take the discount? Compare the interest with the discount.
- 16.4 Compare the terms 3/10, net 30 and 1/10, net 60. Which is more costly to give up, and by how much?
- 16.5 Define short, intermediate and long spans of financing, and give one source for each.
- 16.6 A toy shop has a December peak in stock. Apply the matching principle: say how it should fund its permanent need and its December need.
- 16.7 Explain the difference between permanent and seasonal working capital, using a seller of school books as your example.
- 16.8 Explain one difference between a bank overdraft and a short loan, with a reason why a business might prefer the overdraft.

YOU WILL NEED

Chapter 17, "Long-Term Financing and the Cost of Capital", section 17.5 "The Cost of Capital", for the after-tax cost of debt; Chapter 8, "Capital Budgeting: Evaluation Techniques", section 8.2 "Net Present Value", for the method of discounting; Chapter 5, "Annuities", section 5.4 "Finding the Payment", for the payments on a loan.

A traveller hires a cab and owns nothing, yet the work is done.

Hiring and owning differ in when the money is paid.

A business meets the same puzzle with larger sums.

A visitor to London in the 1870s who wished to cross the city did not buy a carriage, a horse and a stable. He hired a cab for an hour, paid the driver and walked away owning nothing. The cab did his work as well as any carriage would have done. His neighbour, who crossed the city twice every day, took the opposite view and kept a carriage of his own.

Both men were sensible. Their choices differed because their needs differed, and because the two roads carry different costs. Hiring is paid in small amounts as time passes. Owning is paid mostly at the start, and the owner must also meet repairs, idle days and the day when the carriage is worth less than before.

A modern business meets the same puzzle with larger sums. Meghna and Sons Limited needs a second machine. The seller wants Tk 780,000 now. Another business, one that owns machines in order to let others use them, offers the same machine for a payment at the end of each year. Which road costs less? To answer, we must compare payments made in different years, with tax changing each payment. You already own the tools for that task. This last chapter brings them together.

* * *

18.1 WHAT A LEASE IS

Begin with something familiar. A student who rents a room pays the owner each month. The student may use the room but may not sell it, and must leave when the agreement ends. A lease follows the same pattern, applied to the assets of a business.

A **lease** is a contract in which the owner of an asset lets another party use it in return for regular payments. A baker, for example, may lease an oven for five years instead of buying it. The baker uses the oven every day, but the oven stays the property of its owner.

Every lease has two parties. The **lessor** is the owner who lets an asset out under a lease. The **lessee** is the party who pays to use the asset. In the cab story the cab owner is the lessor and the traveller is the lessee. Many students swap these two words, because they differ by one letter. The ending *-or* belongs to the one who does the letting; the ending *-ee* belongs to the one who receives the use.

What would a business lease? Consider the most obvious situation

A student who rents a room already knows the idea.

Rent. Word in use. In daily life, the money paid to live in a room or use a shop. In this chapter, the regular payment under a lease is called a lease payment.

Lease. A contract in which the owner of an asset lets another party use it for regular payments.

A lease lets one party use what another party owns.

Lessor. The owner who lets an asset out under a lease.

Lessee. The party who pays to use an asset under a lease.

Every lease has two parties: the one who owns and the one who uses.

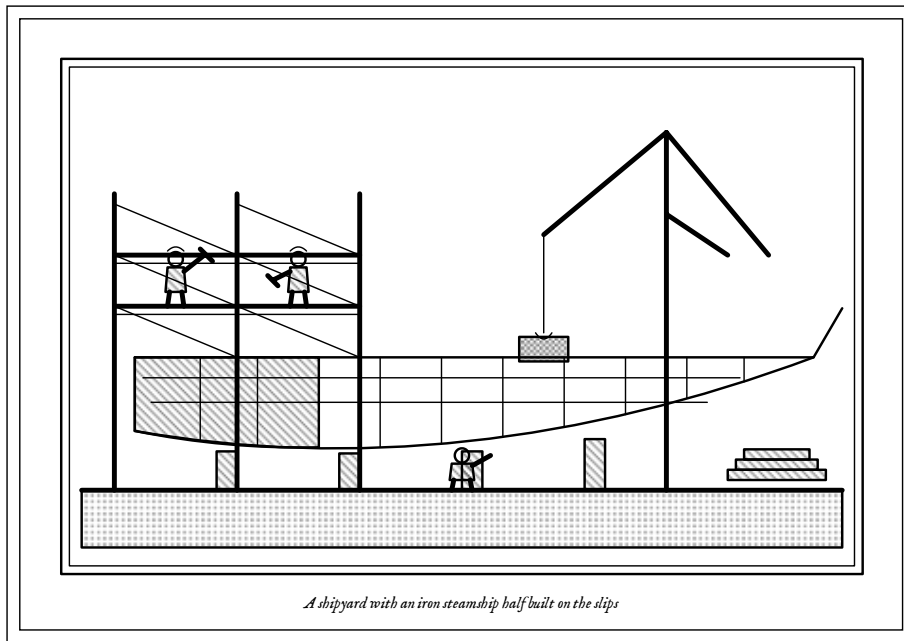
Every business lease is

The sale was led by an investment bank, and the money went to the Foundation, not to the business.

A group of 205 banks, led by Goldman Sachs, an American investment bank, handled the sale. The money went to the Foundation, which wanted to hold less of its wealth in one business, and not to Ford. The offering shows what an initial public offering does. It moves shares into many hands and creates a market where they can be traded, which is the work of a capital market.

REVIEW QUESTIONS

- 17.1 Name three conditions a lender may attach to a term loan, and say why a lender asks for each.
- 17.2 Compare a rights issue with a public issue. Say who may buy and what each does to the control of the existing owners.
- 17.3 List four types of lender or market venue for long-term funds in Bangladesh, and say what each supplies.
- 17.4 A loan costs 11% a year and the tax rate is 25%. Find the after-tax cost of the loan.
- 17.5 A business is funded 25% by debt costing 7.5% after tax, 5% by preferred stock costing 12%, and 70% by common equity costing 14%. Find the weighted average cost of capital.



CHAPTER XVII

Long-Term Financing and the Cost of Capital

Who lends money for five years, and what do they ask in return?



This chapter surveys the long-term funds a business can raise: loans, owners' money and shares sold in the capital market. It names the institutions that supply them in Bangladesh and shows how to compute what the funds cost.

The lender may also limit what the borrower does while the loan lasts. A **covenant** is a promise in a loan agreement that limits what the borrower may do. In the case file, owners' equity must stay above Tk 2,000,000. Other covenants may limit new borrowing or the profit paid out to owners. The reason is the same each time: money paid out, or lost, reduces the cushion that protects the lender. If a covenant is broken, the lender may ask for early repayment.

The instalment of Tk 158,278.49 comes from the payment method you learned in Chapter 5, 'Annuities'. Section 17.6 works it through in full.

A business that wants money from many lenders issues bonds. Chapter 13, section 13.1 'What a Bond Is', describes them. Bonds differ in what backs them. A **debenture** is a bond backed only by the general standing of the borrower, with no particular asset pledged. A holder of a debenture trusts the business's record and its profit. A mortgage is different. It is a loan secured by a pledge of land or buildings, which is a particular kind of collateral. A debenture holder therefore carries more risk than a lender with collateral, and usually asks a higher interest rate.

PAUSE AND CHECK 17.P1 A bank lends Tk 300,000 and takes a pledge over the borrower's shop building. Is this a debenture? Give a reason.

These protections have limits. A machine may sell for less than the debt. A covenant may stop a business from acting quickly. And debt must be repaid whether or not the business makes a profit, which is why too much of it is dangerous. The lender wants safety. The borrower wants freedom and a low cost. A loan agreement is the balance between the two.

Before moving on. A term loan runs for more than one year and is repaid in instalments. Collateral is an asset the lender may take. A covenant is a promise that limits the borrower. A debenture is a bond with no particular asset pledged.

* * *

17.2 OWNERS' FUNDS AS LONG-TERM FINANCE

Debt is not the only long-term money. Owners supply it too, in two ways. The first needs no new cash from anyone.

Retained earnings are the profits a business keeps instead of paying them out. Each year, Meghna and Sons may keep part of its profit to buy inventory, fittings or machines. The money stays as long as the business needs it. It is convenient, but it is not free. The owners could have taken the money and used it elsewhere, so they expect the same return on it as on any other money they leave in. For that reason the case file counts retained earnings inside common equity.

The second way is to issue new stock. Common stock gives its owners a share in the profit that rises and falls with the business,

A covenant is a promise in the loan agreement that limits what the borrower may do.

Covenant. a promise in a loan agreement that limits what the borrower may do

The instalment is found by the payment method of Chapter 5, shown in section 17.6.

A debenture is a bond with no particular asset pledged, so its holder trusts the borrower's standing.

Debenture. a bond backed only by the general standing of the borrower, with no particular asset pledged

Collateral and covenants lower the lender's risk, but they do not remove it, and debt must be repaid whatever the profit.

Recap: a term loan is repaid in instalments; collateral and covenants protect the lender; a debenture pledges nothing in particular.

Owners can supply long-term money by leaving profit in the business or by paying in new money.

Retained earnings are profit kept in the business, and the owners expect a return on it.

Recall: retained earnings, assumed from earlier chapters.

New common or preferred stock brings new money from owners who expect a return

WORKED EXAMPLES: FROM COSTS TO A SINGLE RATE

Four examples apply the method, from clean numbers to a combined problem.

Example 1 uses two sources and round numbers so that the method is the only new thing.

Example 2 repeats the method with the three sources in the case file.

The Steps find the after-tax cost, the weights and the products in turn.

Example 3 changes the situation: the new term loan is added to the funds.

Example 4 finds a loan instalment and then tests a project at the cost of capital.

We work four examples. The first uses clean numbers. The second uses the case file. The third changes the situation, and the fourth combines the loan payment with the use of the cost of capital as a discount rate. Each example follows one order: given, find, formula, steps, answer, check.

Example 1: clean numbers. Given. Debt of Tk 400,000 costs 8% after tax. Common equity of Tk 600,000 costs 12%. *Find.* The weighted average cost of capital. *Formula.* $K = (Wd \times Kd) + (We \times Ke)$, where W is a weight and K is a cost. *Steps.* Total funds: $400,000 + 600,000 = \text{Tk } 1,000,000$. Weights: $400,000 \div 1,000,000 = 0.40$, and $600,000 \div 1,000,000 = 0.60$. $K = 0.40 \times 8\% + 0.60 \times 12\% = 3.2\% + 7.2\% = 10.4\%$. *Answer.* The cost of capital is 10.4%. *Check.* The answer lies between 8% and 12%, and nearer to 12%, because equity is the larger source.

Example 2: realistic numbers. Given. The case file: bank loans Tk 500,000 at 10%, preferred stock Tk 100,000 at 12%, common equity Tk 2,400,000 at 14%, and a tax rate of 25%. *Find.* The weighted average cost of capital. *Formula.* $K = (Wd \times Kd) + (Wp \times Kp) + (We \times Ke)$.

Steps. After-tax cost of debt: $10\% \times (1 - 0.25) = 7.5\%$. Total funds: $500,000 + 100,000 + 2,400,000 = \text{Tk } 3,000,000$. Weights: 0.1667 , 0.0333 and 0.8000 . Products: $0.1667 \times 7.5\% = 1.25\%$. $0.0333 \times 12\% = 0.40\%$. $0.80 \times 14\% = 11.20\%$. Sum: $1.25\% + 0.40\% + 11.20\% = 12.85\%$. *Answer.* The cost of capital is 12.85%. *Check.* The answer lies between 7.5% and 14%, and close to 14%, because common equity is 80% of the funds. A common slip is to use 10% for the debt and forget the tax: that would give 13.27%, which is too high.

Example 3: a reshaped case. This is an exercise built on the case file, not part of it. *Given.* Suppose the Tk 600,000 term loan is added to the existing funds, which are unchanged. The loan costs 7.5% after tax, and the owners' required return stays at 14%. *Find.* The new weighted average cost of capital. *Formula.* The same as in Example 2. *Steps.* Debt: $500,000 + 600,000 = \text{Tk } 1,100,000$. Total funds: $1,100,000 + 100,000 + 2,400,000 = \text{Tk } 3,600,000$. Weights: 0.3056 , 0.0278 and 0.6667 . Products: $0.3056 \times 7.5\% = 2.292\%$. $0.0278 \times 12\% = 0.334\%$. $0.6667 \times 14\% = 9.334\%$. Sum: $2.292\% + 0.334\% + 9.334\% = 11.96\%$. *Answer.* The cost of capital falls to 11.96%. *Check.* The debt share rose from 16.67% to 30.56%, and debt is the cheapest source, so the average falls. In practice owners may ask for more than 14% as debt rises, and then the fall would be smaller.

Example 4: a combined problem. Given. The term loan: Tk 600,000 over 5 years at 10%, repaid in equal instalments at each year end. A project costs Tk 200,000 now and returns Tk 225,000 in one year. The discount rate is 12.85%. *Find.* The yearly instalment, and whether the project passes. *Formula.* Instalment = amount $\times$ interest rate $\div (1 - 1 \div (1 + \text{interest rate})^n)$ multiplied by itself n times, where n is the number of years. Net present value = present value of the return - cost.

The two issues compare as follows. A public issue reaches many new buyers and can raise large sums, but it brings fees, disclosure and new owners. A rights issue is cheaper and keeps control with the old owners, but it raises only as much as they are willing to pay.

The rules for issuing shares differ by country. In Bangladesh, the Bangladesh Securities and Exchange Commission regulates the capital market, as the next section shows.

Before moving on. The capital market is where long-term loans and shares are issued and traded. An initial public offering is a first sale to the public. A rights issue offers new shares to existing owners in proportion to their holdings.

* * *

17.4 LONG-TERM LENDERS AND INVESTORS IN BANGLADESH

Where could Meghna and Sons find five years of money in Dhaka? Several kinds of lender and market venue matter. The names below were checked on 4 October 2026, and names can change, so confirm them before relying on them. Other countries have different names, but the same kinds of supplier appear in most of them.

The table lists the institutions. Read each row as one name, the kind of body it is, and what it offers to a business that needs long-term funds.

SUPPLIERS OF LONG-TERM FUNDS AND MARKET VENUES IN BANGLADESH

NAME	KIND	WHAT IT SUPPLIES
Commercial banks	Banks	Term loans, usually against collateral
Bangladesh Development Bank Limited	Development bank	Long-term loans to industrial projects
Investment Corporation of Bangladesh	State-owned investment institution	Money invested in shares and other securities
Infrastructure Development Company Limited (IDCOL)	Government-owned finance company	Long-term loans for infrastructure and renewable energy
Dhaka Stock Exchange and Chittagong Stock Exchange	Stock exchanges	Places where shares are issued and traded
Bangladesh Securities and Exchange Commission	Regulator	No funds; rules for issuing and trading securities

For a small family business, the nearest source is a commercial bank. Its term loan looks like the one in the case file: instalments, collateral and covenants. A bank decides by the borrower's record and the collateral on offer. For a new business with few assets, this can be difficult.

A public issue brings in new owners; a rights issue keeps control with the old ones.

Rules for issuing shares differ by country, and Bangladesh has its own regulator.

Recap: the capital market issues and trades long-term funds; an initial public offering is the first sale to the public; a rights issue goes to existing owners first.

Bangladesh has commercial banks, development institutions, an investment institution and two stock exchanges.

The table lists each institution, its kind, and what it supplies.

The commercial bank is the nearest source for a small business, and it lends against collateral.

The investment corporation, the exchanges and the regulator make up the capital market side.

A business matches its need to the supplier: a bank for a machine loan, the exchange for a large public issue.

Recap: banks and development institutions lend; the investment corporation invests; the exchanges are the venue; the Commission regulates.

* * *

17.5 THE COST OF CAPITAL

Meghna and Sons raised money from three sources, and each one asks for a different return. How much must the business earn, on average, to keep all three satisfied? That average has a name.

The **cost of capital** is the average return a business must earn on its funds to satisfy those who supplied them. It is a rate, such as 12.85%, and not an amount of money. We find it in three steps: the cost of each source, the weight of each source, and the sum.

Start with the cost of each source. The bank charges 10% a year. But interest is paid before tax is counted, so it lowers the profit that is taxed. At the assumed tax rate of 25%, the state in effect bears a quarter of the interest.¹ The after-tax cost of the loan is $10\% \times (1 - 0.25) = 10\% \times 0.75 = 7.5\%$. Preferred stock costs its dividend rate, 12%, because dividends are not taken off before tax. Common equity costs the return its owners require, 14%, and this includes retained earnings.

Case file, illustrative. Meghna and Sons Limited, 15 November 2025. Long-term funds: bank loans outstanding Tk 500,000 at 10% (after tax $10\% \times 0.75 = 7.5\%$); preferred stock Tk 100,000 at 12%; common equity including retained earnings Tk 2,400,000 with a required return of 14%. Total Tk 3,000,000; weights 16.67%, 3.33% and 80.00%.

The sources differ in size, so they cannot count equally. The weight of a source is its share of the total. Debt: $500,000 \div$

¹ The tax rate of 25% is assumed for illustration only. Tax rules differ by country and change over time.

with probabilities 0.25, 0.50 and 0.25 give an expected value of 12%. (*Chapter IX*)

Factor table. A printed table of ready-made multipliers for given rates and numbers of periods. A reader finds the row for the number of periods, reads the column needed, and multiplies the factor by the payment. (*Chapter V*)

Finance. The work of finding money and deciding how to use it, so that a business or a person can reach chosen aims. It has two halves: finding money, and choosing its use. A stall owner who borrows Tk 800 and spends it on onions to sell is doing finance. (*Chapter I*)

Financial control. Checking that money is spent and recorded as planned, and acting when it is not. If cloth planned at Tk 30,000 costs Tk 36,000, control finds the Tk 6,000 difference, asks why, and changes something. (*Chapter II*)

Financial lease. A long lease that cannot be cancelled, in which the lessee in effect carries the cost and risk of owning the asset. The payments usually cover most of the price plus the lessor's return. Example: a six-year oven lease with no right to leave early. (*Chapter XVIII*)

Financial leverage. The effect of fixed financing charges in making earnings per share change by more than EBIT. With no fixed charges the two change by the same percentage. With interest of Tk 2,000, a 20% change in EBIT of Tk 10,000 moves the profit for each share by 25%. (*Chapter XII*)

Financial manager. The person who plans, raises and controls the money of a business, whatever the job title. In a small shop one brother may hold the duty; in a large business a team may. The manager answers to owners, lenders and employees. (*Chapter II*)

Financial market. A place or network where buyers and sellers trade money, loans and ownership parts of businesses. A bank where one borrower meets one lender is a very small example; large markets let a business find many sources at once. (*Chapter I*)

Financial planning. Deciding in advance how much money a business will need and when. A tailor who knows she must pay Tk 38,000 for cloth and rent in the first week of March can arrange the money before the day arrives. (*Chapter II*)

Financial risk. The extra risk to owners that comes from funding part of a business by borrowing. Interest must be paid in poor years as well as good ones, so what is left for owners varies more. A bookshop with a loan of Tk 600,000 at 10% a year faces it. (*Chapter IX*)

Financing decision. Choosing how to raise the money for purchases, from owners or lenders. It asks how to pay, not what to buy. An oven may be paid for from owners' funds, a bank loan, or both. (*Chapter II*)

Firm-specific risk. Risk that affects one business only and can be reduced by diversifying. A fire at one factory or a failed new product are examples. An investor who holds many businesses feels such a loss as a small part of the whole portfolio. (*Chapter XI*)

What is finance, and how are funds raised, risked and put to work? This treatise answers from the beginning. It sets out the nature and aims of finance, then follows it through the time value of money, the sources of capital, the weighing of risk against return, and the choice of investments. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.

oven for five years while the oven stays the property of its owner. (*Chapter XVIII*)

Lessee. The party who pays to use an asset under a lease. The lessee gains the use of the asset but does not own it. Example: a clinic that pays each month to use a scanner owned by another business. (*Chapter XVIII*)

Lessor. The owner who lets an asset out under a lease. The lessor receives the lease payments and keeps the asset, together with any resale value at the end. Example: a business that owns machines in order to let other businesses use them. (*Chapter XVIII*)

Liability. An amount a business owes to someone else. Debt is one kind of liability; an unpaid bill to a supplier is another. Assets less liabilities gives the owners' wealth in the business. (*Chapter I*)

Liquidity. How easily a business can pay its bills on time from cash it has or can quickly raise. Books on shelves are not liquid until sold, while a bank balance is. Weak liquidity can delay wages and stop deliveries. (*Chapter II*)

Long-term financing. Money borrowed or raised for more than five years. It suits needs that never end or assets that last a long time. Money that owners put into a business as equity, and a bond, are examples. (*Chapter XVI*)

Market risk. Risk that affects all investments in a market and cannot be removed by diversifying. A fall in demand across a whole country is an example. Because it cannot be removed, it is the only risk that earns a premium in the pricing model. (*Chapter X*)

Market value. The price at which something could be sold today in a market. A used bicycle that a buyer would pay Tk 6,000 for today has a market value of Tk 6,000. (*Chapter III*)

Matching principle. The rule that short-term needs are financed with short-term funds and long-term needs with long-term funds. It protects the business from having to renew money in a hurry and from paying interest on idle cash. It is a guide, not a law. (*Chapter XVI*)

Maturity. The date on which a bond is repaid. A bond with five years to maturity pays its par value, with its last coupon interest, at the end of year 5. (*Chapter XIII*)

Mixed stream. A series of payments or receipts that differ in size from one period to the next. Receipts of Tk 1,000, Tk 2,000 and Tk 3,000 at the ends of three years form a mixed stream. An annuity is the special case in which all sizes are equal. (*Chapter VI*)

Mutually exclusive projects. Projects where accepting one means rejecting the others. A machine and a branch fit-out that cannot be run together are an example, as are two scanners when only one will be bought. (*Chapter VII*)

Net present value. The present value of a project's cash inflows minus its initial investment. It shows, in taka today, how much worth the project adds. Accept a project when the value is zero or positive; among exclusive projects, choose the highest. (*Chapter VIII*)

Permanent working capital, 155, 158

Perpetuity, 45–46, 49–52, 135

Portfolio, 91–92, 96–99, 101

Preferred stock, 6, 14, 115–117, 119, 134–136, 138–140, 162–164, 166–170

Premium bond, 128–132

Present value, 36–41, 44, 46, 48–52, 54–61, 70, 76–81, 84, 94, 99, ...

Principal, 25–28

Probability, 86–91

Probability distribution, 86–87, 91

Profit maximisation, 22–23, 25, 28, 30

Profitability, 16–17, 19, 22, 77–81, 124

Profitability index, 77–78, 80–81, 124

R

Rate of return, 76–81, 84–85, 91–92, 124, 130

Relevant cash flow, 67, 69–71

Required return, 74, 76–81, 84–85, 90, 94–95, 97–102, 126–132, 135–140, 166–168

Residual claim, 134–135, 137, 139

Retained earnings, 14–18, 22, 163–164, 166, 169

Return, 14, 24, 27, 30, 32, 48–49, 52, 54, 64–67, 69, ...

Rights issue, 164–165, 170

Risk, 23–25, 28–30, 32, 75–76, 80, 85–87, 89–92, 94–102, 104, 109–111, ...

Risk premium, 94–95, 98–101

Risk-free rate, 94–95, 98–102

S

Sale and leaseback, 174

Seasonal working capital, 155, 158–159

Security market line, 98–99, 101–102

Semi-strong-form efficiency, 100

Short-term financing, 147, 152–153, 162

Simple interest, 34–35, 39, 41

Single sum, 33–36, 39, 44, 46, 48, 54–57, 60, 124, 127

Sinking fund, 50–51

Source of funds, 5, 7, 9, 14–15, 152, 167

Stakeholder, 24–25, 28

Standard deviation, 88–92, 94–95, 98, 101

Strong-form efficiency, 100

T

Term loan, 156–158, 162–165, 168–170

Premium bond. A bond that sells for more than its par value. This happens when the required return is below the coupon rate. The case bond at 7% sells for Tk 1,082.00. (*Chapter XIII*)

Present value. What an amount to be received later is worth today at a stated rate. At 10%, Tk 12,100 due in two years has a present value of Tk 10,000. (*Chapter IV*)

Principal. A person who hires another person to act for them. The owners of a bookshop are principals when they hire someone to run a branch for them. (*Chapter III*)

Probability. A number from 0 to 1 showing how likely an outcome is. Zero means the outcome cannot happen and one means it is certain. A fair coin landing heads has a probability of 0.5. Heavy traffic on 25 days in 100 has a probability of 0.25. (*Chapter IX*)

Probability distribution. A list of possible outcomes with the probability of each, adding up to one. Exactly one outcome must happen, so the probabilities total one. Route times of 10, 25 and 40 minutes with probabilities of 0.25, 0.50 and 0.25 form a probability distribution. (*Chapter IX*)

Profit maximisation. The aim of earning the largest profit possible in a period, usually one year. A shop that always picks the plan with the largest profit for the year follows this aim, even if the plan harms later years. (*Chapter III*)

Profitability. How much profit a business earns compared with the money invested or the sales made. A profit of Tk 20,000 on Tk 200,000 invested is 10% of the money invested; compare ratios, not amounts. (*Chapter II*)

Profitability index. The present value of a project's inflows divided by its initial investment. A value above 1 means each taka invested returns more than a taka in present value. For Press, $1,097,055 \div 900,000$ gives 1.22. (*Chapter VIII*)

Rate of return. The return shown as a percentage of the amount first invested, found by dividing the return by that amount and multiplying by 100. It lets investments of different size be compared. A return of Tk 10,000 on Tk 100,000 is 10%. (*Chapter IX*)

Relevant cash flow. Cash flow that changes only because a project is accepted. It must be cash, lie in the future and differ with the decision. A sunk cost, such as a survey already paid, is not relevant. (*Chapter VII*)

Required return. The lowest return a project must earn to be worth accepting. Also called the required rate of return or hurdle rate. It is used as the discount rate, and it reflects the risk of the project and the cost of the funds. Meghna and Sons uses 12%. (*Chapter VIII*)

Residual claim. The right to whatever is left of a business's income and assets after lenders, suppliers, staff and preferred stockholders are paid. Common stockholders hold it. When the business does well the remainder can be large, and when it does badly it can be nothing. (*Chapter XIV*)

Retained earnings. Profit kept in the business instead of being

D

Debenture, 163, 169

Debt, 6–9, 12, 14, 24, 29, 32–33, 35, 119–120, 124, 132, ...

Degree of financial leverage, 116–121

Degree of operating leverage, 108–111, 114, 117–121

Degree of total leverage, 117–121

Depreciation, 68–71, 80, 105, 174–178

Discount bond, 129–132

Discount rate, 36, 38, 40–41, 48–49, 55, 57–60, 68, 71, 76–78, 80,

...

Discounted cash flow, 55, 57, 60, 81

Discounting, 36–38, 40, 52, 55, 60, 77, 80–81, 128, 131, 137, ...

Diversification, 95–96, 98, 101–102

Dividend, 14–15, 17–19, 22, 114–118, 120, 134–140, 163–164, 166

Dividend decision, 14–15, 17, 19

Dividend growth model, 137, 140

E

Earnings per share, 115–121

EBIT, 68–71, 104–111, 114–121

Effective annual rate, 38–41

Efficient market, 99–102

Equity, 6–9, 12, 14–15, 32, 115, 119, 152, 162–164, 166–170

Expected value, 87–92, 94–95

F

Factor table, 49–50

Finance, 2–6, 8–9, 12–14, 16, 24, 27, 29–30, 33, 36, 41, ...

Financial control, 13, 18

Financial lease, 173–174, 177–178

Financial leverage, 115–121

Financial manager, 4, 12–13, 16–18, 22, 45, 55, 64–66, 75, 85, 95,

...

Financial market, 6, 164

Financial planning, 12, 18

Financial risk, 86, 92, 114, 116, 118–119

Financing decision, 14–15, 17–19

Firm-specific risk, 96–98, 101

Fixed financing charge, 86, 114–117, 120, 134

Fixed operating cost, 86, 105–106, 108, 110–111, 114, 116

Float, 146–150

Funds, 2–9, 12, 14–15, 17, 32–33, 64–66, 68, 71, 76, 79, ...

Future value, 33–37, 39–41, 44, 46–48, 50–52, 55–58, 60–61

source supplies part of the total money that the business has obtained. (*Chapter I*)

Stakeholder. Any person or group affected by a business, such as workers, customers, lenders and owners. A bookshop's stakeholders include its staff, the schools that buy from it and the bank that lent it money. (*Chapter III*)

Standard deviation. A measure of how widely outcomes spread around the expected value. Find each deviation from the expected value, square it, weight it by its probability, add the results and take the square root. A small figure means outcomes lie close to the expected value. (*Chapter IX*)

Strong-form efficiency. The condition in which prices reflect all information, public and private. If it holds, even secret knowledge held by managers is already in the price. Few people believe it holds, and trading on secret information is a crime in many countries. (*Chapter X*)

Term loan. A loan repaid in instalments over a fixed period of more than one year. Each instalment usually holds some interest and some repayment of the amount borrowed. A bank loan of Tk 600,000 repaid yearly over 5 years is an example. (*Chapter XVII*)

Terminal cash flow. Cash received at the end of a project life, such as the resale value of equipment. It also includes working capital that returns, and is reduced by tax on any gain from the sale. (*Chapter VII*)

Time value of money. The idea that money received sooner is worth more than the same amount received later, because money in hand can be put to work. Tk 100,000 today is worth more than Tk 100,000 in a year. (*Chapter IV*)

Trade credit. A supplier allowing a business to pay for goods some days after delivery. It is a loan that appears with each purchase and is the most common source of short-term financing. A bookshop that receives books on the 1st and pays on the 30th is using trade credit. (*Chapter XVI*)

Use of funds. A purpose a business spends money on, such as stock or equipment. Keeping money at the bank for later is also a use. Each use of funds usually gives the business an asset. (*Chapter I*)

Valuation. The work of finding what an asset is worth from the cash it is expected to bring. It needs the expected cash and the required return of the investor. (*Chapter XIII*)

Variable operating cost. An operating cost that rises and falls with the number of units sold. If each notebook costs Tk 12 to buy, then 100 notebooks cost Tk 1,200 and 200 notebooks cost Tk 2,400. (*Chapter XI*)

Voting right. The right of a shareholder to vote on matters put to the owners, such as choosing the directors who guide the business. Voting rights usually come with common stock, often one vote for each share, and usually not with preferred stock. (*Chapter XIV*)

Weak-form efficiency. The condition in which prices already

ANSWERS TO QUESTIONS

CHAPTER I · MONEY, FINANCE AND THE FIRM

1.p1 No. The tailor has accepted an order, but no money has moved today. The cash flow into the business will come next week, when the customer pays.

1.p2 The Tk 300,000 of savings is equity, and the Tk 100,000 loan is debt. The bank must be repaid on agreed dates; the owner shares any profit or loss.

1.t1 Ticket sales of Tk 40,000 are a cash flow into the business today. The film hire of Tk 25,000 is owed next month, so it is not a cash flow today. The Tk 10,000 loan instalment is a cash flow out today.

1.t2 The net cash flow today is Tk 40,000 less Tk 10,000, which is Tk 30,000 in. The film hire of Tk 25,000 is a liability; it will become a cash flow out next month, when it is paid.

1.1 For her, finance is finding money and deciding how to use it. She might find Tk 800 by asking her cousin, and decide to spend it on onions to sell.

1.2 Recording the money received is accounting, because it records what has happened. Deciding whether to borrow for the machine is finance, because it looks forward and asks where to find money and how to use it.

1.3 The two sources are the owner's savings, which are equity, and the loan, which is debt. Two uses might be ovens and other equipment, and ingredients to bake with, or money kept at the bank.

1.4 The sale is counted in June, when the goods are taken. The money moves only on 28 July, so the cash flow into the business falls in July.

1.5 Cash flow in is Tk 3,000, and cash flow out is Tk 2,000. The net movement is Tk 3,000 less Tk 2,000, which is Tk 1,000 in.

1.6 The Tk 800 is a source of funds, and it is debt, because it must be repaid. Afterwards she owes her cousin Tk 800, which is a liability, plus any extra charge they agree.

1.7 A source might be the school fund; a use might be buying ingredients for the canteen. The canteen's daily sales bring cash flow into it.

1.8 The sources are the owners' savings, which are equity, and the bank loan, which is debt. Two assets might be medical equipment and a cash and bank balance. Finance is one business function beside the clinic's other work; it finds funds and judges the money side of the other functions' decisions.

CHAPTER II · THE FINANCIAL MANAGER'S WORK

2.p1 The decision to buy the oven for Tk 200,000 is the investment decision. Paying Tk 80,000 from owners' funds and borrowing Tk 120,000 is the financing decision.

2.p2 The dividend is Tk 25,000, the part of the profit paid out. The

business would have to sell them or borrow again. Long-term funds stay for as long as the assets take to pay for themselves.

17.1 Any three. Collateral: the lender can take an asset if the loan is not repaid, which lowers its risk. A covenant, such as a minimum level of owners' equity: it protects the cushion behind the loan. A limit on new borrowing or on profit paid out: it stops money leaving the business. A schedule of instalments: it returns the money along the way.

17.2 A public issue offers shares to the general public. It reaches many buyers and can raise large sums, but it brings fees, disclosure and new owners. A rights issue offers new shares only to existing shareholders in proportion to their holdings. It is cheaper and keeps control with the old owners, but it raises only what they are willing to pay.

17.3 Any four. Commercial banks: term loans. Bangladesh Development Bank Limited: long-term loans to industrial projects. Investment Corporation of Bangladesh: money invested in shares and securities. IDCOL: long-term loans for infrastructure and renewable energy. Dhaka Stock Exchange and Chittagong Stock Exchange: places where shares are issued and traded.

17.4 $11\% \times (1 - 0.25) = 11\% \times 0.75 = 8.25\%$.

17.5 $0.25 \times 7.5\% = 1.875\%$. $0.05 \times 12\% = 0.60\%$. $0.70 \times 14\% = 9.80\%$. Sum: 12.275%, or about 12.28%.

CHAPTER XVIII · LEASING

18.p1 $60,000 \times (1 - 0.25) = \text{Tk } 45,000$. The payment saves Tk 15,000 of tax.

18.t1 Lease P: $60,000 \times 0.75 = \text{Tk } 45,000$ a year; $45,000 \times 2.600527 = \text{Tk } 117,024$. Lease Q: $45,000 \times 0.75 = \text{Tk } 33,750$ in years 1 and 2, worth $33,750 \times 1.795566 = \text{Tk } 60,600$; year 3 hire $70,000 \times 0.75 = \text{Tk } 52,500$, worth $52,500 \times 0.804961 = \text{Tk } 42,260$; total about Tk 102,861. Lease Q is cheaper by about Tk 14,163.

18.t2 The minibus hired for one week is an operating lease. The six-year oven lease that cannot be cancelled is a financial lease. The press sold and leased back is a sale and leaseback.

18.1 The owner is the lessor and the puller is the lessee. A lease is a contract in which the owner of an asset lets another party use it in return for regular payments.

18.2 An operating lease is short or cancellable and the lessor keeps most risks, for example a scanner hired by the month. A financial lease is long and cannot be cancelled, and the lessee in effect carries the cost and risk of owning, for example a six-year oven lease.

18.3 $80,000 \times (1 - 0.25) = \text{Tk } 60,000$. The payment lowers taxable profit by Tk 80,000, which saves Tk 20,000 of tax.

18.4 First, find the after-tax cash outflow of leasing each year. Second, find the after-tax cash outflow of buying each year. Third, convert each outflow to a present value at the after-tax cost of debt and add. Fourth, compare the totals and choose the lower.

18.5 Discount rate = $8\% \times 0.75 = 6\%$. Lease: $100,000 \times 0.75 = 75,000$

perhaps driving hard and wearing out the car. Their rewards differ, so their choices differ.

3.4 Examples: pay the manager by long-term value (cost: a harder pay rule to design); give the manager a stake in the business (cost: a share of the owners' return); check records and require reports (cost: time and money for the owners).

CHAPTER IV · THE TIME VALUE OF MONEY: SINGLE SUMS

4.p1 Year 1: Tk 11,000. Year 2: Tk 12,100. Year 3: Tk 12,100 $\times$ 1.10 = Tk 13,310. Check: Tk 10,000 $\times$ 1.331 = Tk 13,310.

4.p2 Year by year: Tk 13,310 $\div$ 1.10 = Tk 12,100, then $\div$ 1.10 = Tk 11,000, then $\div$ 1.10 = Tk 10,000. The present value is Tk 10,000.

4.t1 PV = Tk 130,000 $\div$ 1.07 to the power of 5 = Tk 130,000 $\div$ 1.402552 = about Tk 92,688. This is above Tk 90,000, so waiting is better. Check: Tk 90,000 $\times$ 1.402552 = about Tk 126,230, which is below Tk 130,000.

4.t2 Factor = 21,277.79 $\div$ 15,000 = 1.418519. In the 6% column of the future value table this factor is in the row for 6 years, so $n = 6$ years.

4.1 Factor = 1.09 to the power of 6 = 1.677100. FV = Tk 15,000 $\times$ 1.677100 = Tk 25,156.50.

4.2 Factor = 1 $\div$ 1.11 to the power of 4 = 1 $\div$ 1.518070 = 0.658731. PV = Tk 80,000 $\times$ 0.658731 = Tk 52,698.48.

4.3 Simple: Tk 25,000 $\times$ (1 + 0.06 $\times$ 8) = Tk 37,000. Compound: Tk 25,000 $\times$ 1.06 to the power of 8 = Tk 25,000 $\times$ 1.593848 = Tk 39,846.20. Compound interest is Tk 2,846.20 higher.

4.4 Rate per quarter = 12% $\div$ 4 = 3%. Effective annual rate = 1.03 to the power of 4 - 1 = 1.125509 - 1 = 12.55%.

4.5 Opportunity cost of idle cash is the interest it could have earned. Example: Tk 50,000 kept in a drawer for a year loses the Tk 4,000 that a bank paying 8% would have given.

CHAPTER V · ANNUITIES

5.p1 Factor = (1.10⁴ - 1) $\div$ 0.10 = (1.4641 - 1) $\div$ 0.10 = 4.641. Future value = Tk 1,000 $\times$ 4.641 = Tk 4,641.

5.p2 Tk 3,310 $\times$ 1.10 = Tk 3,641, or Tk 1,331 + Tk 1,210 + Tk 1,100 = Tk 3,641.

5.t1 Present value of the plan = Tk 90,000 $\times$ 3.790787 = about Tk 341,171. This is less than Tk 400,000, so the payment plan costs less today. The delay in paying is worth something at 10%.

5.t2 Each payment of an annuity due is made one period earlier, so it earns one more period of interest, or is discounted one period less. Every payment is therefore worth 1 + i times as much.

5.1 Factor = (1.09⁶ - 1) $\div$ 0.09 = 7.523335. Ordinary: Tk 12,000 $\times$ 7.523335 = about Tk 90,280. Due: Tk 90,280 $\times$ 1.09 = about Tk 98,405.

5.2 Factor = (1 - 1.10⁻⁵) $\div$ 0.10 = 3.790787. Present value = Tk 45,000 $\times$ 3.790787 = about Tk 170,585.

5.3 Factor = (1.07⁴ - 1) $\div$ 0.07 = 4.439943. Deposit = Tk 300,000 $\div$

13.1 C = Tk 80, $a = 3.169865$, $f = 0.683013$. Coupons: 80 $\times$ 3.169865 = 253.589. Par value: 1,000 $\times$ 0.683013 = 683.013. Value = Tk 936.60.

13.2 Discount bond: the required return is above the coupon rate, so the fixed coupon is less attractive and the price falls below par. Par bond: the two rates are equal, so the price equals par. Premium bond: the required return is below the coupon rate, so the coupon is more attractive and the price rises above par.

13.3 Coupon Tk 30, 4 periods, 4% per half-year: $a = 3.629895$, $f = 0.854804$. Coupons: 30 $\times$ 3.629895 = 108.897. Par value: 1,000 $\times$ 0.854804 = 854.804. Value = Tk 963.70.

13.4 At 10% the value is Tk 962.09; at 11% it is Tk 926.08. The fraction is (962.09 - 950) $\div$ (962.09 - 926.08) = 12.09 $\div$ 36.01 = 0.336. The yield to maturity is about 10.3%.

13.5 The coupons and the par value are fixed. A higher required return means each future payment is discounted more heavily, so its present value is smaller and the total falls.

CHAPTER XIV · VALUING COMMON AND PREFERRED STOCK

14.p1 Value = 10 $\div$ (0.15 - 0.05) = 10 $\div$ 0.10 = Tk 100.00.

14.t1 The bond suits her better. It pays fixed interest by contract, and a lender is paid before owners if the business ends. Common stock pays dividends the directors may cut, and its holder has only a residual claim, paid last, so the risk of loss is higher.

14.1 P = 9 $\div$ 0.12 = Tk 75.00.

14.2 D₁ = 3.50 $\times$ 1.05 = 3.675. Then $r - g = 0.11 - 0.05 = 0.06$, so P₀ = 3.675 $\div$ 0.06 = Tk 61.25.

14.3 If g equals r , the bottom of the fraction is zero and no value exists. If g is above r , the value is negative, which has no meaning. A business also cannot raise dividends faster than the required return for ever.

14.4 Any three: a bond holder lends and a stockholder owns; bond interest is fixed while dividends are decided by the directors; a bond gives no vote while common stock does; a lender is paid before owners; the stockholder carries more risk.

14.5 At 12%: 6 $\div$ 0.12 = Tk 50.00. At 15%: 6 $\div$ 0.15 = Tk 40.00. The value falls by Tk 10.00, because a higher required return lowers the present value.

14.6 A: D₁ = 4.16, so P₀ = 4.16 $\div$ 0.08 = Tk 52.00. B: D₁ = 4.28, so P₀ = 4.28 $\div$ 0.05 = Tk 85.60. The gap of Tk 33.60 arises because higher growth shrinks $r - g$ from 0.08 to 0.05.

14.7 Bond: 80 $\times$ 2.48685 + 1,000 $\times$ 0.75131 = 198.95 + 751.31 = Tk 950.26. Stock: D₁ = 2.00 $\times$ 1.05 = 2.10, so P₀ = 2.10 $\div$ 0.05 = Tk 42.00; ten shares are Tk 420.00. Total = Tk 1,370.26.

CHAPTER XV · WORKING CAPITAL AND THE CASH CYCLE

15.p1 40 + 15 = 55 days.

15.p2 70 - 30 = 40 days. For 40 days the business must use its own money.

15.t1 Each item rises by one fifth, so the net investment is Tk

implement (buy, install, train the staff); review (compare real cash flows with the forecast).

7.3 Independent projects do not affect each other, such as a new oven and a new delivery bicycle. Mutually exclusive projects allow only one, such as two different ovens for the same space, or a machine against a branch fit-out that cannot be run together.

7.4 Relevant: (a) the price, (c) the extra electricity and (d) the extra stock, because each changes only if the freezer is bought. Not relevant: (b) the report, a sunk cost, and (e) the rent, which is the same either way.

7.5 EBIT = 240,000 – 120,000 – 40,000 = Tk 80,000. Tax = $0.25 \times 80,000$ = Tk 20,000. Operating cash flow = 80,000 – 20,000 + 40,000 = Tk 100,000. Check: 240,000 – 120,000 – 20,000 = 100,000.

7.6 Gain = 70,000 – 50,000 = 20,000; tax = $0.25 \times 20,000$ = 5,000; sale after tax = 65,000. Add working capital of 20,000: terminal cash flow = Tk 85,000.

CHAPTER VIII · CAPITAL BUDGETING: EVALUATION TECHNIQUES

8.p1 Project A: $60,000 \div 30,000 = 2$ years. Project B: 3 years ($3 \times 20,000 = 60,000$). Payback misses the cash after the payback point: A returns only its cost, while B returns Tk 120,000 in all.

8.p2 Present values: $12,000 \div 1.10 = 10,909$ and $12,000 \div 1.21 = 9,917$, total 20,826. NPV = $20,826 - 20,000 = \text{Tk } 826$. It is positive, so accept.

8.t1 Payback: X = 2.00 years; $Y = 3 + 50,000 \div 400,000 = 3.13$ years. NPV at 10%: $X = 272,727 + 165,289 + 75,131 + 68,301 - 500,000 = \text{Tk } 81,449$; $Y = 90,909 + 123,967 + 150,263 + 273,205 - 500,000 = \text{Tk } 138,344$. Payback prefers X, net present value prefers Y. Choose Y, because it adds Tk 56,895 more value and net present value uses all the cash and its timing.

8.t2 Net present value is the present value of the inflows minus the initial investment. At the internal rate of return the discounted inflows are exactly equal to the initial investment, so the difference is zero.

8.1 After 2 years the inflows total 270,000; unrecovered 130,000. Year 3 inflow is 180,000: $130,000 \div 180,000 = 0.72$. Payback period = 2.72 years.

8.2 Present values: $109,091 + 123,967 + 135,237 + 136,603 = \text{about } 504,897$. NPV = about Tk 104,897. It is positive, so accept.

8.3 Payback: accept if no longer than the limit set by managers. Net present value: accept if zero or more. Internal rate of return: accept if at least the required return. Profitability index: accept if at least 1.

8.4 NPV at 20% = +4,784; at 25% = –33,920. IRR = $20 + 5 \times 4,784 \div (4,784 + 33,920) = 20 + 0.62 = \text{about } 20.6\%$.

8.5 M: index $330,000 \div 300,000 = 1.10$; NPV = Tk 30,000. N: index $540,000 \div 500,000 = 1.08$; NPV = Tk 40,000. M ranks first by index; N ranks first by net present value.

CHAPTER IX · RETURN AND RISK

9.p1 No. The probabilities add up to $0.2 + 0.5 + 0.4 = 1.1$, and a probability distribution must add up to exactly one.

9.p2 Deviations below the expected value are negative and those above are positive, so they would cancel to zero. Squaring makes every deviation positive, so the spread shows.

9.t1 Coefficient of variation: Project X = $10,000 \div 50,000 = 0.20$; Project Y = $24,000 \div 80,000 = 0.30$. Project Y carries more risk for each unit of return.

9.1 Return = $5,600 + (82,000 - 80,000) = \text{Tk } 7,600$. Rate of return = $7,600 \div 80,000 = 0.095$, or 9.5%.

9.2 Business risk comes from the trade: sales, costs and trade conditions, for example a rise in the price of paper. Financial risk comes from borrowing: interest of Tk 60,000 a year on a loan of Tk 600,000 at 10% must be paid in poor years too, so what is left for the owners varies more.

9.3 Expected value = $0.25 \times 6 + 0.50 \times 10 + 0.25 \times 14 = 1.5 + 5.0 + 3.5 = 10\%$. Deviations –4, 0, +4; squares 16, 0, 16; weighted $4 + 0 + 4 = 8$; standard deviation = $\sqrt{8} = 2.83\%$.

9.4 Asset M: $3 \div 10 = 0.30$. Asset N: $9 \div 18 = 0.50$. Asset N carries more risk for each unit of return, although its expected return is higher.

CHAPTER X · RISK, REQUIRED RETURN AND MARKET EFFICIENCY

10.p1 About 10%. A beta of 2 means the investment moves about twice as far as the market, so $2 \times 5\% = 10\%$.

10.p2 Market risk premium = $10\% - 6\% = 4\%$. Required return = $6\% + 0.5 \times 4\% = 8\%$.

10.t1 Premium = $14\% - 8\% = 6\%$. Required return = $8\% + 1.0 \times 6\% = 14\%$. The expected return of 13% is below 14%, so the line lies below the security market line. Reject it.

10.1 Market risk premium = $13\% - 7\% = 6\%$. Required return = $7\% + 1.2 \times 6\% = 7\% + 7.2\% = 14.2\%$.

10.2 Firm-specific risk hits one business only, so losses at some businesses are offset by gains at others when many are held. Market risk hits all businesses at once, so holding more of them cannot remove it.

10.3 The line starts at 6% at beta 0 and rises 4 points for each whole step of beta, reaching 10% at beta 1 and 14% at beta 2. At beta 0.5 the required return is 8%; at beta 1.5 it is 12%.

10.4 Weak form: studying past price charts does not help. Semi-strong form: reading a public report after it appears does not help. Strong form: using secret information known only to managers does not help.

10.5 No. An expected return below the line is less than the required return for that much market risk, so the investment does not pay enough for the risk. The investor should reject it, or accept only if its expected return rises to the line.